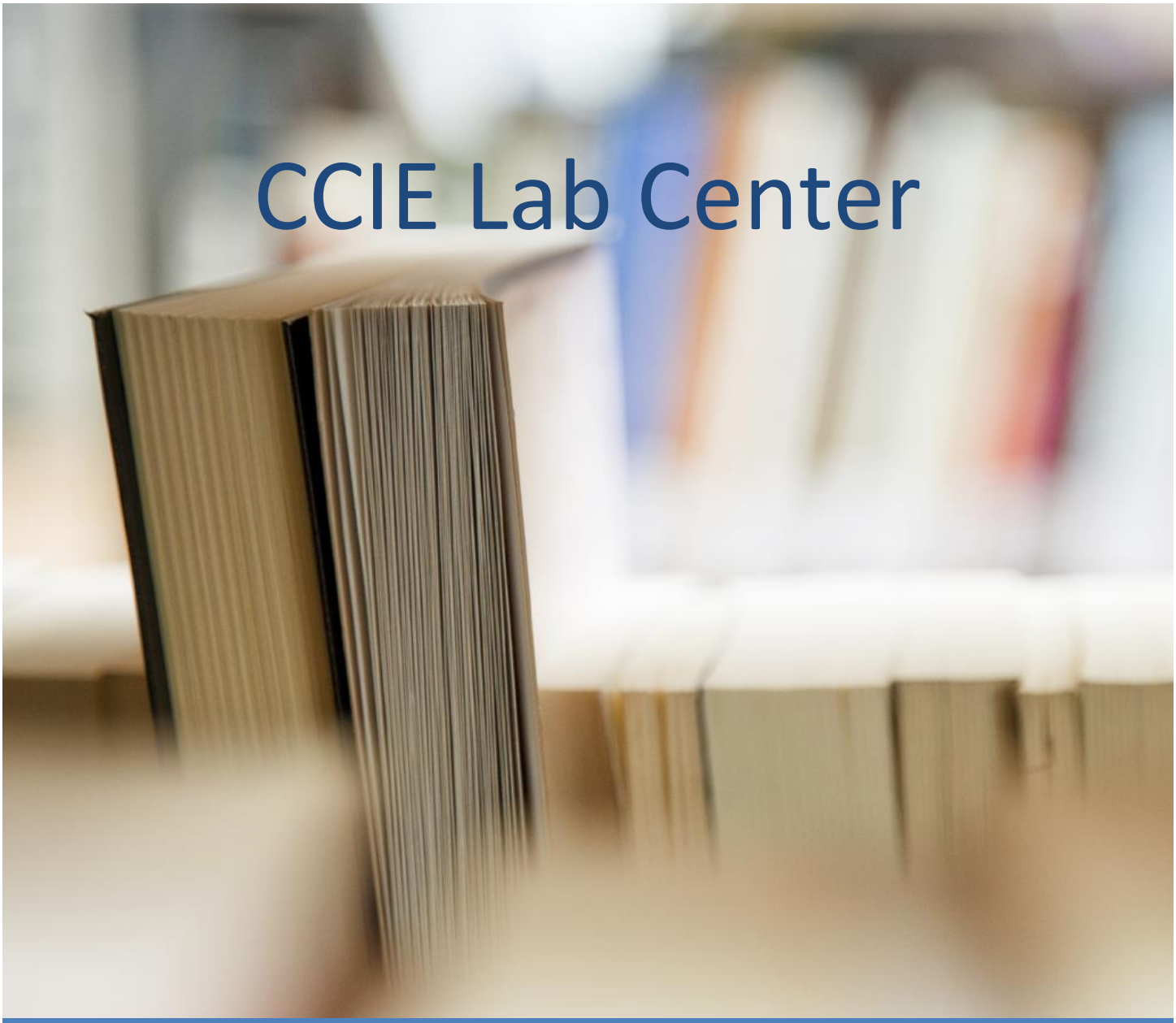
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CLC - CCIE Enterprise Infrastructure v1.0: Real Lab v1.0

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CCIE Enterprise Infrastructure Real Lab v1.0



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Workbook Description

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Table of Contents

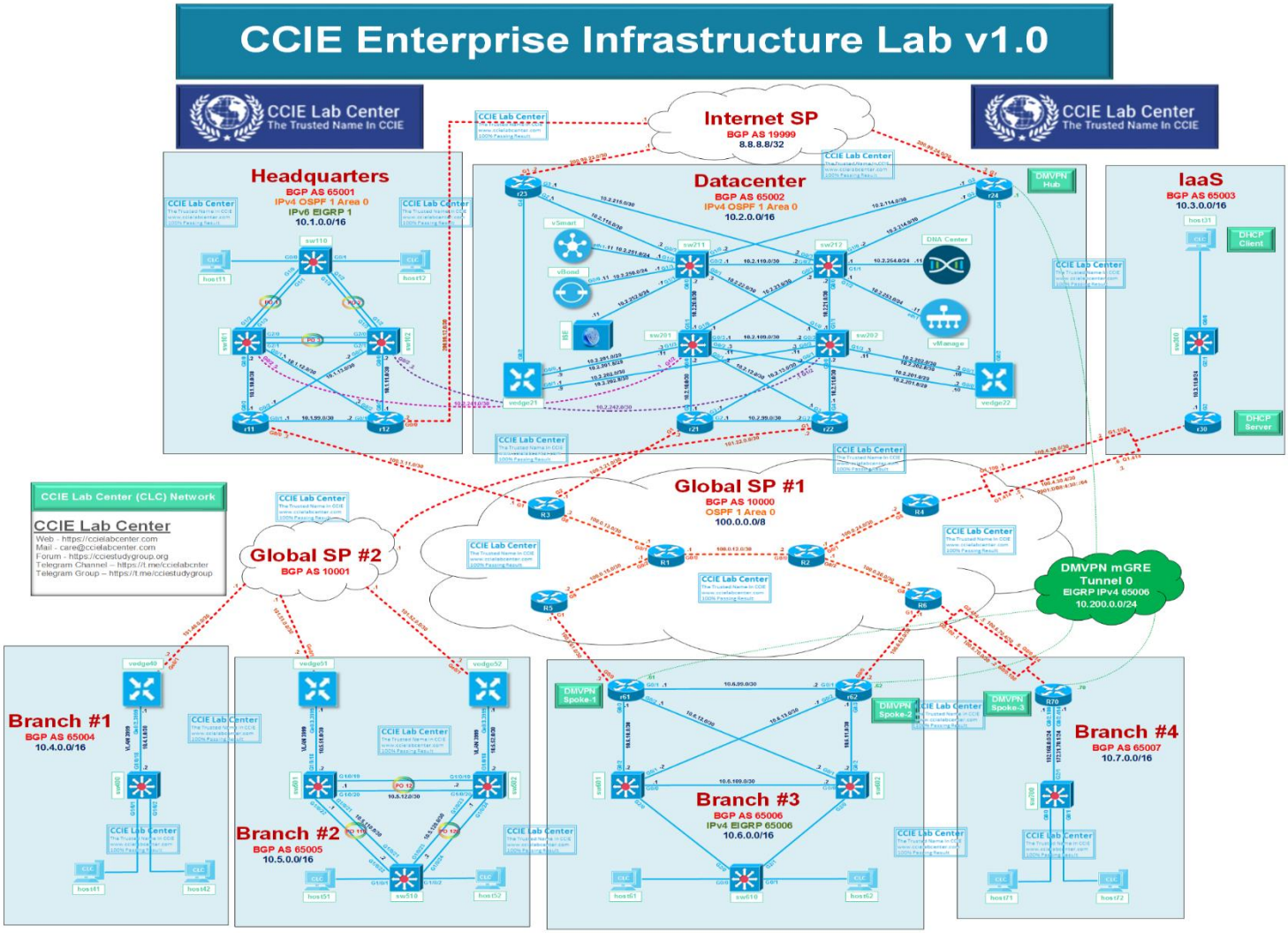
0. Lab Details
    - Lab Topology
    - Device Initial Configuration
1. Section 1: Existing Network Review & Tuning
    - 1.1 Introduction
    - 1.2 Layer 2 Technologies in HQ
    - 1.3 First Hop Redundancy Protocol in HQ
    - 1.4 OSPFv2 Between HQ & DC
    - 1.5 DHCP IPv4 Services for HQ
    - 1.6 IPv6 in HQ
    - 1.7 IPv6 EIGRP in HQ
    - 1.8 OSPFv2 in DC
    - 1.9 BGP Between HQ & DC and Service Providers
    - 1.10 Bringing Up VPNv4/VPNv6 in SP1
    - 1.11 Fixing DMVPN Network Between DC and Branches 3 & 4
    - 1.12 Tuning EIGRP on DMVPN and DMVPN Enabled Sites
    - 1.13 IPv4 Networks on Legacy Branches
    - 1.14 Multicast in FADB2
    - 1.15 Extending Connectivity to IAAS
    - 1.16 Enabling Internet access to FADB2
2. Section 2: Implementing Proof of Concept SDX Branches
    - 2.1 Correcting the IP Adresses of Managed Switches in DNA Center
    - 2.2 Completing VN Configuration in DNA Center
    - 2.3 Mapping SDA VNs to SD-WAN VPNs
    - 2.4 Configuring SD-WAN VPN Route Leaking
    - 2.5 Handling Guest Traffic
    - 2.6 Support for Silent Hosts in Branch #2
3. Section 3: Making use of Programmability
    - 3.1 Enabling CLI Access to R30
    - 3.2 Using Guest Shell and Python on r30
    - 3.3 Automated Configuration Backup Script
    - 3.4 The End

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SECTION 0: Lab Details

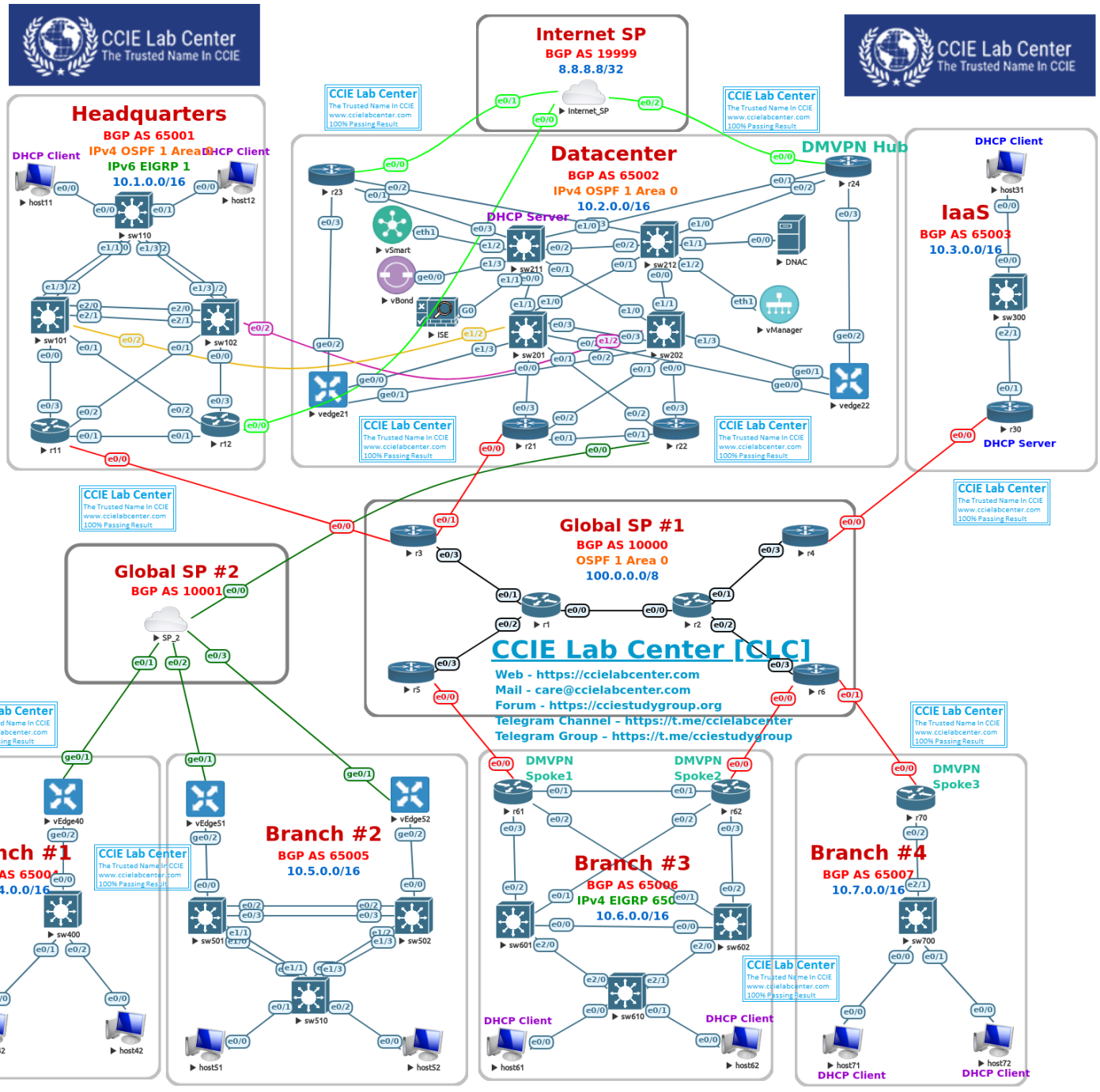
SECTION 0.1: Lab Topoloy

CLC CCIE Enterprise Infrastructure Real Lab 1.0: Diagram 1.1# Main Topology



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CLC CCIE Enterprise Infrastructure Real Lab 1.0: Diagram 1.2# Eve-ng Topology



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Section 1: Existing Network Review & Tuning

SECTION 1.1: Introduction

The topology you will be working with will be similar, but not necessarily identical to the network that was designed in the previous module and may include technologies and feature sets not touched upon.

The best of success!

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SECTION 1.2: Layer 2 Technologies in HQ (2 Points)

QUESTION

Complete and correct the EtherChannel configuration between switches sw101, sw102, sw110 according to these requirements:

1. At the end of task, all EtherChannels between switches sw101, sw102, sw110 must be up and operational including all their physical member links.
2. Do not create new Port-Channel interfaces, reuse those that already exist on the switches.
3. When resolving existing issues, do not change the preconfigured negotiation protocol (if any).
4. On EtherChannels that use a negotiation protocol, tune its mode of operation for the shortest link bundling time possible.

Configure Spanning Tree Protocol on switches sw101, sw102, sw110 according to these requirements:

1. The STP root for VLAN 2000 must be sw101
2. The STP root for VLAN 2001 must be sw102
3. The roots must be elected based on bridge priority
4. On the three switches, have STP perform cost calculations in 32-bit arithmetic
5. On the three switches, use the Rapid STP version and ensure that it can achieve rapid convergence on all interconnections between the switches.
6. On sw110, prevent all current and future access mode interface from being affected by the Proposal/Agreement process.

Solution

On sw101

```
sw101(config)#interface range gigabitEthernet 1/2-3
sw101(config-if-range)#Channel-group 1 mode on
sw101(config-if-range)#exit
sw101(config)#

sw101(config)# interface range gigabitEthernet 2/0-3
sw101(config-if-range)#channel-group 1 mode passive
sw101(config-if-range)#exit
sw101(config)#

sw101(config)#spanning-tree mode rapid-pvst
sw101(config)#spanning-tree vlan 2000 priority 0
```

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```
sw101(config)#spanning-tree pathcost method long
```

On sw102

```
sw102(config)#interface range gigabitEthernet 1/2-3
sw102(config-if-range)#channel-group 2 mode active
sw102(config-if-range)#exit
sw102(config)#

sw102(config)#Spanning-tree mode rapid-pvst
sw102(config)#spanning-tree vlan 2001 priority 0
sw102(config)#spanning-tree pathcost method long
```

On sw110

```
sw110(config)#spanning-tree mode rapid-pvst
sw110(config)#spanning-tree portfast edge default
sw110(config)#spanning-tree pathcost method long
sw110(config)#
```

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Verification

On sw101

sw101

```
sw101#show etherchannel summary
Flags:  D - down        P - bundled in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      N - not in use, no aggregation
        f - failed to allocate aggregator

        M - not in use, minimum links not met
        m - not in use, port not aggregated due to minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

        A - formed by Auto LAG


Number of channel-groups in use: 2
Number of aggregators:           2

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------
1      Po1(SU)          -       Gi1/2(P)    Gi1/3(P)
3      Po3(SU)        LACP      Gi2/0(P)    Gi2/1(P)

sw101#
```

sw101

```
sw101#show spanning-tree summary
Switch is in rapid-pvst mode
Root bridge for: VLAN2000
Extended system ID                        is enabled
Portfast Default                          is disabled
Portfast Edge BPDU Guard Default          is disabled
Portfast Edge BPDU Filter Default         is disabled
Loopguard Default                         is disabled
PVST Simulation Default                   is enabled but inactive in rapid-pvst mode
Bridge Assurance                          is enabled
EtherChannel misconfig guard              is enabled
Configured Pathcost method used is long
UplinkFast                                is disabled
BackboneFast                              is disabled

Name                   Blocking Listening Learning Forwarding STP Active
---------------------- -------- --------- -------- ---------- ----------
VLAN0001                     1         0        0          6          7
VLAN2000                     0         0        0          2          2
VLAN2001                     1         0        0          1          2
---------------------- -------- --------- -------- ---------- ----------
3 vlans                      2         0        0          9         11
sw101#
```

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On sw102

sw102

```
sw102#show etherchannel summary
Flags:  D - down        P - bundled in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      N - not in use, no aggregation
        f - failed to allocate aggregator

        M - not in use, minimum links not met
        m - not in use, port not aggregated due to minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

        A - formed by Auto LAG


Number of channel-groups in use: 2
Number of aggregators:           2

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------
2      Po2(SU)         LACP      Gi1/2(P)    Gi1/3(P)
3      Po3(SU)         LACP      Gi2/0(P)    Gi2/1(P)

sw102#
```

sw102

```
sw102#show spanning-tree summary
Switch is in rapid-pvst mode
Root bridge for: VLAN0001, VLAN2001
Extended system ID                        is enabled
Portfast Default                          is disabled
Portfast Edge BPDU Guard Default          is disabled
Portfast Edge BPDU Filter Default         is disabled
Loopguard Default                         is disabled
PVST Simulation Default                   is enabled but inactive in rapid-pvst mode
Bridge Assurance                          is enabled
EtherChannel misconfig guard              is enabled
Configured Pathcost method used is long
UplinkFast                                is disabled
BackboneFast                              is disabled

Name                   Blocking Listening Learning Forwarding STP Active
---------------------- -------- --------- -------- ---------- ----------
VLAN0001                      0         0        0          7          7
VLAN2000                      0         0        0          2          2
VLAN2001                      0         0        0          2          2
---------------------- -------- --------- -------- ---------- ----------
3 vlans                       0         0        0         11         11
sw102#
```

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On sw110

sw110

```
sw110#show etherchannel summary
Flags:  D - down        P - bundled in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      N - not in use, no aggregation
        f - failed to allocate aggregator

        M - not in use, minimum links not met
        m - not in use, port not aggregated due to minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

        A - formed by Auto LAG


Number of channel-groups in use: 2
Number of aggregators:           2

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------
1      Po1(SU)           -      Gi1/0(P)    Gi1/1(P)
2      Po2(SU)         LACP     Gi1/2(P)    Gi1/3(P)

sw110#
```

sw110

```
sw110#show spanning-tree summary
Switch is in rapid-pvst mode
Root bridge for: none
Extended system ID                          is enabled
Portfast Default                            is edge
Portfast Edge BPDU Guard Default            is disabled
Portfast Edge BPDU Filter Default           is disabled
Loopguard Default                           is disabled
PVST Simulation Default                     is enabled but inactive in rapid-pvst mode
Bridge Assurance                            is enabled
EtherChannel misconfig guard                is enabled
Configured Pathcost method used is long
UplinkFast                                  is disabled
BackboneFast                                is disabled

Name                   Blocking Listening Learning Forwarding STP Active
---------------------- -------- --------- -------- ---------- ----------
VLAN0001                      0         0        0          8          8
VLAN2000                      1         0        0          2          3
VLAN2001                      0         0        0          3          3
---------------------- -------- --------- -------- ---------- ----------
3 vlans                       1         0        0         13         14
sw110#
```

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SECTION 1.3: First Hop Redundancy Protocol in HQ (2 Points)

QUESTION

For IPv4, implement an FHRP mechanism on sw101 and sw102 for VLANs 2000 and 2001 according to these requirements:

1. Use group number 100 for VLAN 2000 and group number 101 for VLAN 2001.
2. Use the first available IPv4 address in the subnet for the address of the virtual router.
3. For VLAN 2000 - SW101 must be the preferred gateway & for VLAN 2001 - SW102 must be the preferred gateway. Do not rely on the IPv4 addresses of the switches as role tiebreakers. The role must be determined by an explicit configuration solely on the intended preferred gateway.
4. Each preferred gateway must monitor the reachability of both routers r11 and r12 using the loopback IPv4 address of the routers by an ICMP Echo. The reachability is to be verified every 5 seconds with a timeout of 400 msec. A router must be declared unreachable as soon as it does not respond to three probes in a row. If both r11 and r12 are declared unreachable from a preferred gateway, the other switch must be allowed to assume the gateway role.
5. Use the FHRP protocol that allows the virtual IPv4 address to match the IPv4 address of a member router.

Solution

On sw101

```
sw101(config)#ip sla 1
sw101(config-ip-sla)#icmp-echo 10.1.255.11 source-interface Vlan2000
sw101(config-ip-sla-echo)#threshold 400
sw101(config-ip-sla-echo)#timeout 400
sw101(config-ip-sla-echo)#frequency 5
sw101(config-ip-sla-echo)#exit
sw101(config)#

sw101(config)#ip sla schedule 1 start-time now life forever

sw101(config)#ip sla 2
sw101(config-ip-sla)#icmp-echo 10.1.255.12 source-interface Vlan2000
sw101(config-ip-sla-echo)#threshold 400
sw101(config-ip-sla-echo)#timeout 400
sw101(config-ip-sla-echo)#frequency 5
sw101(config-ip-sla-echo)#exit
sw101(config)#
```

CCIE Lab Center

```
sw101(config)#ip sla schedule 2 start-time now  life forever

sw101(config)#track 1 ip sla 1 reachability
sw101(config-track)#delay down 10 up 10
sw101(config-track)#exit
sw101(config)#

sw101(config-track)#track 2 ip sla 2 reachability
sw101(config-track)#delay down 10 up 10
sw101(config-track)#exit
sw101(config)#

sw101(config)#interface Vlan2000
sw101(config-if)# ip address 10.1.100.2 255.255.255.0
sw101(config-if)# vrrp 100 ip 10.1.100.1
sw101(config-if)# vrrp 100 priority 120
sw101(config-if)# vrrp 100 track 1 decrement 11
sw101(config-if)# vrrp 100 track 2 decrement 11
sw101(config-if)#exit
sw101(config)#

sw101(config)#interface Vlan2001
sw101(config-if)# ip address 10.1.101.2 255.255.255.0
sw101(config-if)# vrrp 101 ip 10.1.101.1
sw101(config-if)#exit
sw101(config)#
```

On sw102

```
sw102(config)#ip sla 1
sw102(config-ip-sla)#icmp-echo 10.1.255.11 source-interface Vlan2001
sw102(config-ip-sla-echo)#threshold 400
sw102(config-ip-sla-echo)#timeout 400
sw102(config-ip-sla-echo)#frequency 5
sw102(config-ip-sla-echo)#exit
sw102(config)#

sw102(config)#ip sla schedule 1 start-time now life forever

sw102(config)#ip sla 2
sw102(config-ip-sla)#icmp-echo 10.1.255.12 source-interface Vlan2001
sw102(config-ip-sla-echo)#threshold 400
sw102(config-ip-sla-echo)#timeout 400
sw102(config-ip-sla-echo)#frequency 5
sw102(config-ip-sla-echo)#exit
```

CCIE Lab Center

```
sw102(config)#

sw102(config)#ip sla schedule 2 start-time now  life forever

sw102(config)#track 1 ip sla 1 reachability
sw102(config-track)#delay down 10 up 10
sw102(config-track)#exit
sw102(config)#

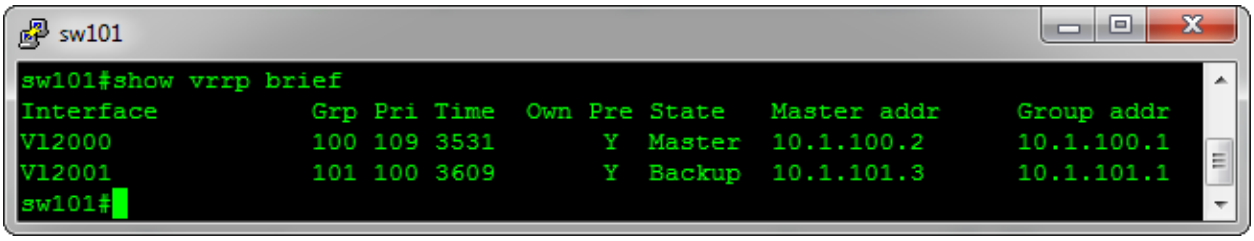
sw102(config-track)#track 2 ip sla 2 reachability
sw102(config-track)#delay down 10 up 10
sw102(config-track)#exit
sw102(config)#

sw102(config)#interface Vlan2000
sw102(config-if)#ip address 10.1.100.3 255.255.255.0
sw102(config-if)#vrrp 100 ip 10.1.100.1
sw102(config-if)#exit
sw102(config)#

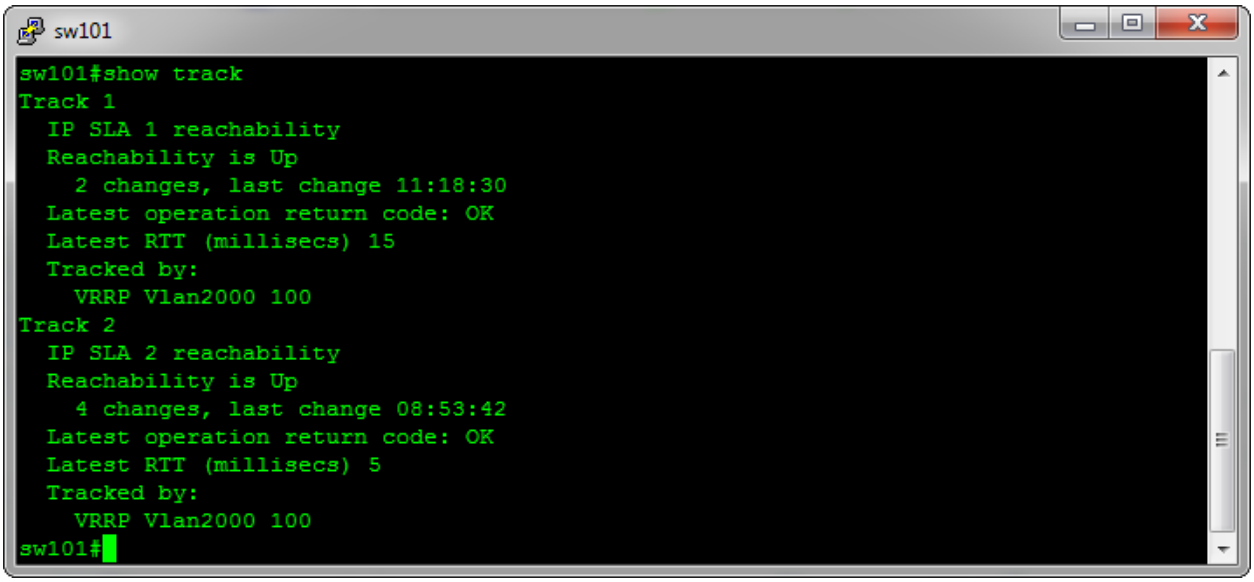
sw102(config)#interface Vlan2001
sw102(config-if)#ip address 10.1.101.3 255.255.255.0
sw102(config-if)#vrrp 101 ip 10.1.101.1
sw102(config-if)#vrrp 101 priority 120
sw102(config-if)#vrrp 101 track 1 decrement 11
sw102(config-if)#vrrp 101 track 2 decrement 11
sw102(config-if)#exit
sw102(config)#
```

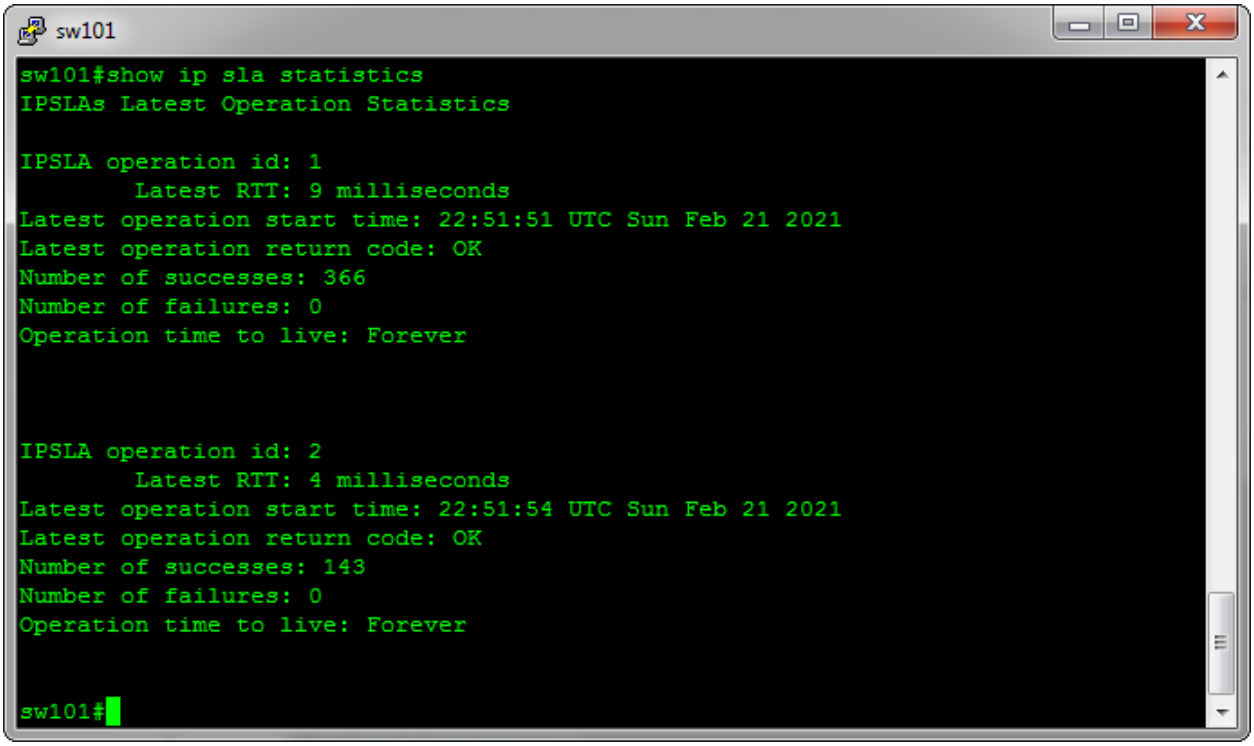
Verification

On sw101



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sw101

```
sw101#show track 1
Track 1
  IP SLA 1 reachability
  Reachability is Up
    2 changes, last change 00:09:03
  Delay up 10 secs, down 10 secs
  Latest operation return code: OK
  Latest RTT (millisecs) 8
  Tracked by:
    VRRP Vlan2000 100
sw101#
sw101#show track 2
Track 2
  IP SLA 2 reachability
  Reachability is Up
    2 changes, last change 00:09:52
  Delay up 10 secs, down 10 secs
  Latest operation return code: OK
  Latest RTT (millisecs) 6
  Tracked by:
    VRRP Vlan2000 100
sw101#
```

On SW102

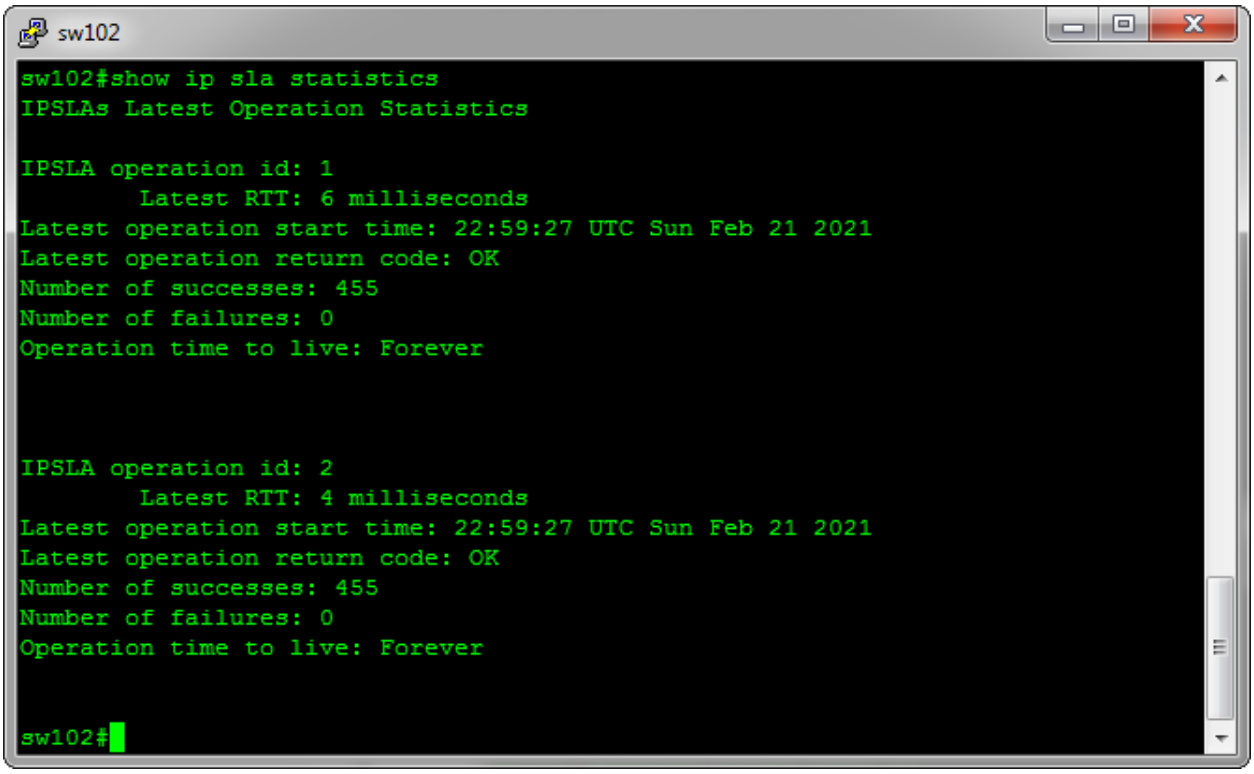
sw102

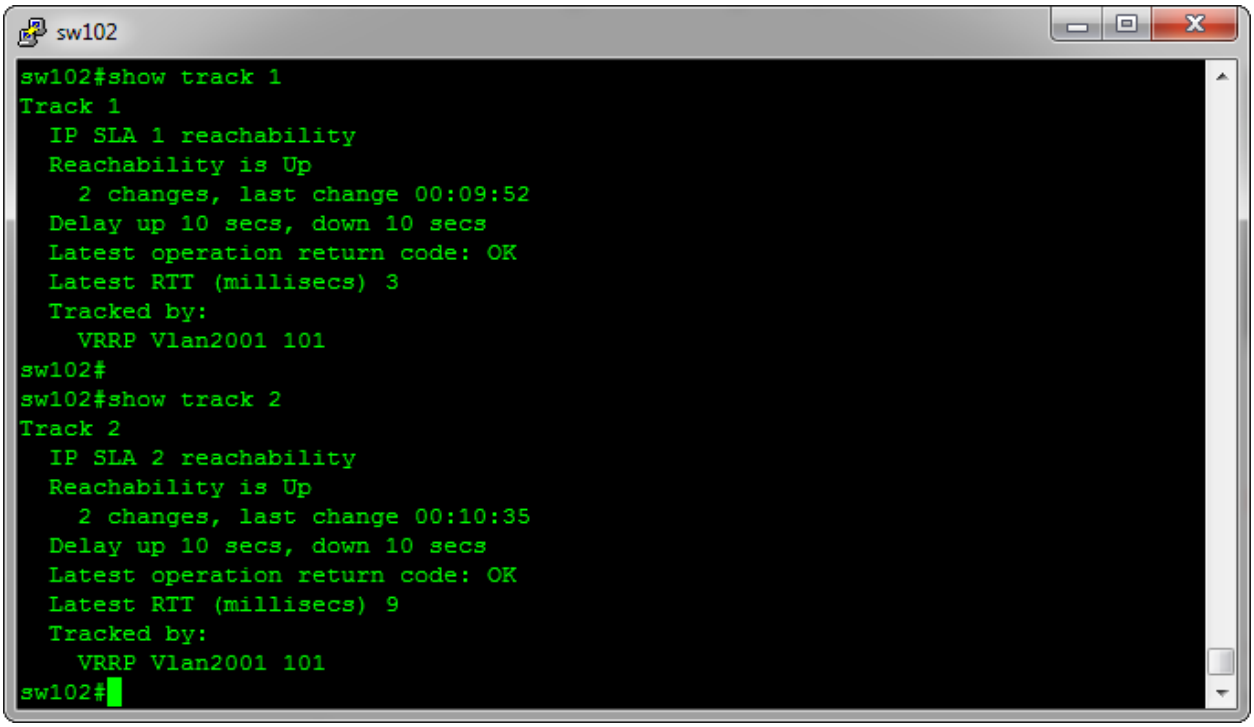
```
sw102# show vrrp brief
Interface          Grp Pri Time  Own Pre State   Master addr     Group addr
Vl2000             100 100 3609      Y  Backup  10.1.100.2      10.1.100.1
Vl2001             101 109 3531      Y  Master  10.1.101.3      10.1.101.1
sw102#
```

sw102

```
sw102#show track
Track 1
  IP SLA 1 reachability
  Reachability is Up
    2 changes, last change 11:25:44
  Latest operation return code: OK
  Latest RTT (millisecs) 6
  Tracked by:
    VRRP Vlan2001 101
Track 2
  IP SLA 2 reachability
  Reachability is Up
    4 changes, last change 08:59:32
  Latest operation return code: OK
  Latest RTT (millisecs) 1
  Tracked by:
    VRRP Vlan2001 101
sw102#
```

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You will get the exact above outputs after completing section 1.4

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SECTION 1.4: OSPFv2 between HQ and DC (3 Points)

QUESTION

Complete and correct the OSPF configuration on the switches sw101, sw102, sw201 and sw202 according to these requirements:

1. Enable OSPFv2 on the redundant interconnections between the DC and HQ sites. Make sure that OSPF establishes adjacencies on these interconnections and exchange routing information between the DC and HQ sites
2. Protect the authencity and integrity of the OSPFv2 sessions on the redundant interconnections between DC and HQ with the SHA-384 mechanism. Use key ID 1 and a shared secret of “cci3” (without quotes)
3. Improve the detection of unreachable OSPFv2 neighbors on the redundant interconnections between DC and HQ so that OSPF can detect the loss of a neighbor within 300 msec, with the probes being sent every 100 msec. It is not allowed to modify OSPF timers to accomplish this requirements.

Solution

On sw101

```
sw101(config)#key chain ccie
sw101(config-keychain)#key 1
sw101(config-keychain-key)#key-string cci3
sw101(config-keychain-key)#cryptographic-algorithm hmac-sha-384
sw101(config-keychain-key)#exit
sw101(config-keychain)#exit
sw101(config)#

sw101(config)#interface gigabitEthernet 0/2
sw101(config-if)#ip ospf authentication key-chain ccie
sw101(config-if)#ip ospf bfd
sw101(config-if)#bfd interval 100 min_rx 100 multiplier 3
sw101(config-if)#exit
sw101(config)#

sw101(config)#router ospf 1
sw101(config-router)#network 10.1.0.0 0.0.255.255 area 0
sw101(config-router)#passive-interface vlan 2000
sw101(config-router)#passive-interface vlan 2001
```

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```
sw101(config-router)#exit
sw101(config)#
```

On sw102

```
sw102(config)#key chain ccie
sw102(config-keychain)#key 1
sw102(config-keychain-key)#key-string cci3
sw102(config-keychain-key)#cryptographic-algorithm hmac-sha-384
sw102(config-keychain-key)#exit
sw102(config-keychain)#exit
sw102(config)#

sw102(config)#interface gigabitEthernet 0/2
sw102(config-if)#ip ospf authentication key-chain ccie
sw102(config-if)#ip ospf bfd
sw102(config-if)#bfd interval 100 min_rx 100 multiplier 3
sw102(config-if)#exit
sw102(config)#

sw102(config)#router ospf 1
sw102(config-router)#network 10.1.0.0 0.0.255.255 area 0
sw102(config-router)#passive-interface vlan 2000
sw102(config-router)#passive-interface vlan 2001
sw102(config-router)#exit
sw102(config)#
```

On sw201

```
sw201(config)#key chain ccie
sw201(config-keychain)#key 1
sw201(config-keychain-key)#key-string cci3
sw201(config-keychain-key)#cryptographic-algorithm hmac-sha-384
sw201(config-keychain-key)#exit
sw201(config-keychain)#exit
sw201(config)#

sw201(config)#interface gigabitEthernet 1/2
sw201(config-if)#ip ospf authentication key-chain ccie
sw201(config-if)#ip ospf bfd
sw201(config-if)#bfd interval 100 min_rx 100 multiplier 3
sw201(config-if)#exit
sw201(config)#

sw201(config)#router ospf 1
```

CCIE Lab Center

```
sw201(config-router)#no passive-interface GigabitEthernet1/2
sw201(config-router)#exit
sw201(config)#
```

On sw202

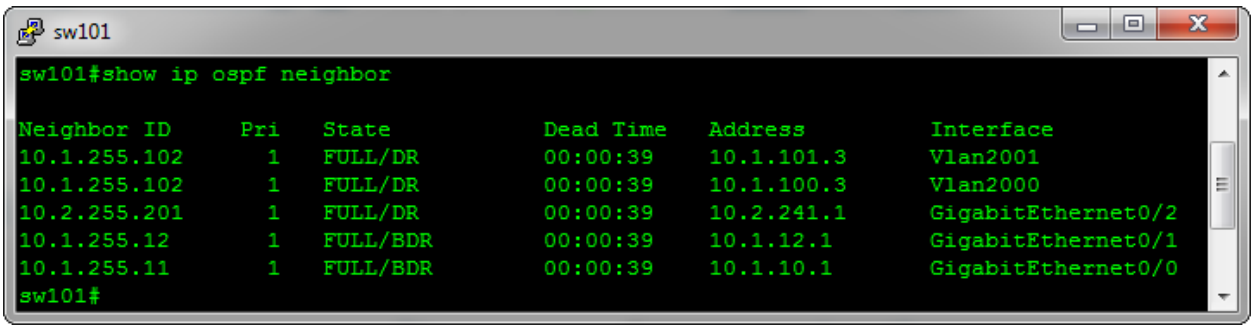
```
sw202(config)#key chain ccie
sw202(config-keychain)#key 1
sw202(config-keychain-key)#key-string cci3
sw202(config-keychain-key)#cryptographic-algorithm hmac-sha-384
sw202(config-keychain-key)#exit
sw202(config-keychain)#exit
sw202(config)#

sw202(config)#interface gigabitEthernet 1/2
sw202(config-if)#ip ospf authentication key-chain ccie
sw202(config-if)#ip ospf bfd
sw202(config-if)#bfd interval 100 min_rx 100 multiplier 3
sw202(config-if)#exit
sw202(config)#

sw202(config)#router ospf 1
sw202(config-router)#no passive-interface GigabitEthernet1/2
sw202(config-router)#exit
sw202(config)#
```

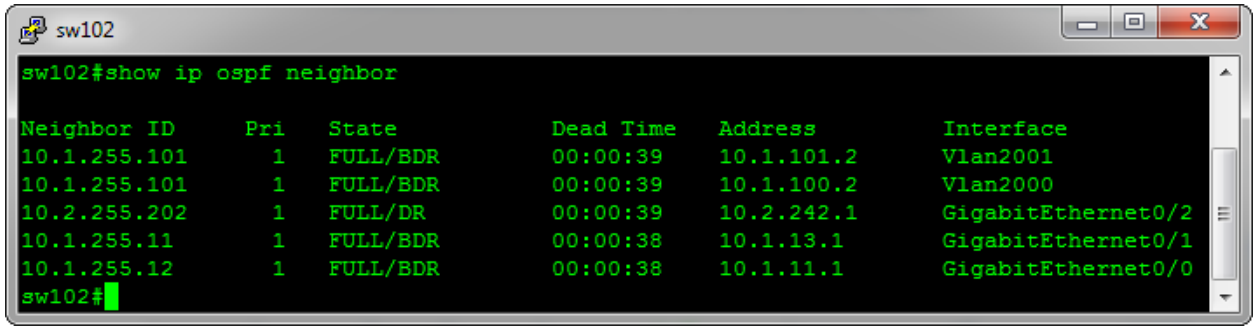
Verification

On sw101



CCIE Lab Center

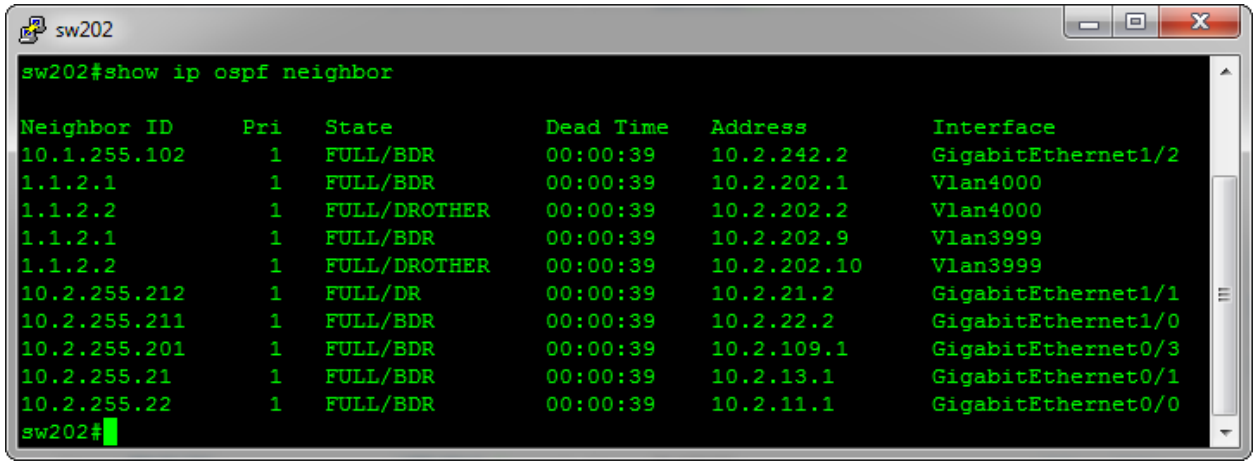
On sw102



On sw201



On sw202



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SECTION 1.5: DHCP IPv4 Service for HQ (2 Points)

QUESTION

Enable hosts in HQ VLAN 2000 and VLAN 2001 to obtain their IP configuration via DHCP according to these requirements.

1. On SW211, create IPv4 DHCP pools named hq_2000 and hq_2001 for HQ VLANs 2000 and 2001 respectively. In each subnet assign addresses from .101 up to .254 inclusively and the appropriate gateway to clients.
2. Enable DHCP Snooping on sw110 in VLANs 2000 and 2001 to protect against DHCP related attacks.
3. Place host11 into VLAN 2000
4. Place host12 into VLAN 2001
5. Perform the necessary configuration on switches sw101, sw102, sw110 to enable hosts in VLANs 2000 and 2001 to obtain IPv4 configuration through DHCP. The DHCP server running at sw211 in the DC must be referred to by its loopback IPv4 address 10.2.255.211. Do not disable the Option 82 insertion, and do not enable DHCP Snooping on other switches.
6. Verify that host11 and host12 have the IP connectivity to the Cisco DNA Center, vManage, ISE running in the DC using their internal (in Band Connectivity) address.

Solution

On sw211

```
sw211(config)#ip dhcp pool hq_2000
sw211(dhcp-config)#network 10.1.100.1 255.255.255.0
sw211(dhcp-config)#default-router 10.1.100.1
sw211(dhcp-config)#exit
sw211(config)#

sw211(config)#ip dhcp pool hq_2001
sw211(dhcp-config)#network 10.1.101.1 255.255.255.0
sw211(dhcp-config)#default-router 10.1.101.1
sw211(dhcp-config)#exit
sw211(config)#

sw211(config)#ip dhcp excluded-address 10.1.100.1 10.1.100.100
sw211(config)#ip dhcp excluded-address 10.1.100.255
sw211(config)#ip dhcp excluded-address 10.1.101.1 10.1.101.100
sw211(config)#ip dhcp excluded-address 10.1.101.255
```

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On sw101

```
sw101(config)#interface range vlan 2000-2001
sw101(config-if-range)#ip helper-address 10.2.255.211
sw101(config-if-range)#exit
sw101(config)#

sw101(config)#interface vlan 2000
sw101(config-if)#ip dhcp relay information trusted
sw101(config-if)#exit
sw101(config)#

sw101(config)#interface vlan 2001
sw101(config-if)#ip dhcp relay information trusted
sw101(config-if)#exit
sw101(config)#
```

On sw102

```
sw102(config)#interface range vlan 2000-2001
sw102(config-if-range)#ip helper-address 10.2.255.211
sw102(config-if-range)#exit
sw102(config)#

sw102(config)#interface vlan 2000
sw102(config-if)#ip dhcp relay information trusted
sw102(config-if)#exit
sw102(config)#

sw102(config)#interface vlan 2001
sw102(config-if)#ip dhcp relay information trusted
sw102(config-if)#exit
sw102(config)#
```

On sw110

```
sw110(config)#ip dhcp snooping
sw110(config)#ip dhcp snooping vlan 2000-2001

sw110(config)#interface Port-channel1
sw110(config-if)#ip dhcp snooping trust
sw110(config-if)#exit
sw110(config)#

sw110(config)#interface Port-channel2
```

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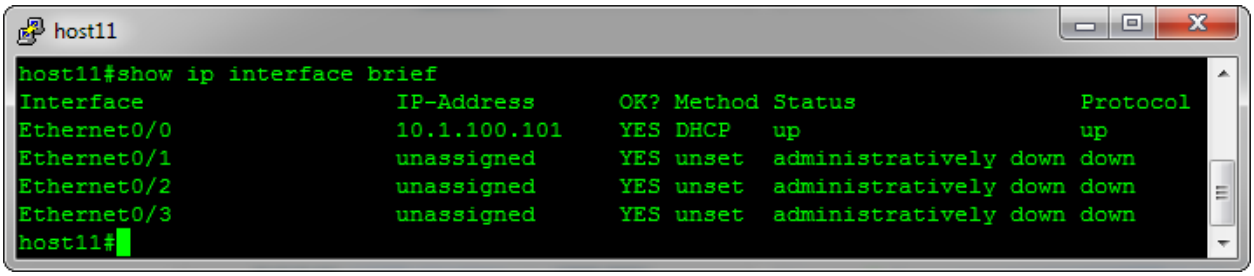
```
sw110(config-if)#ip dhcp snooping trust
sw110(config-if)#exit
sw110(config)#

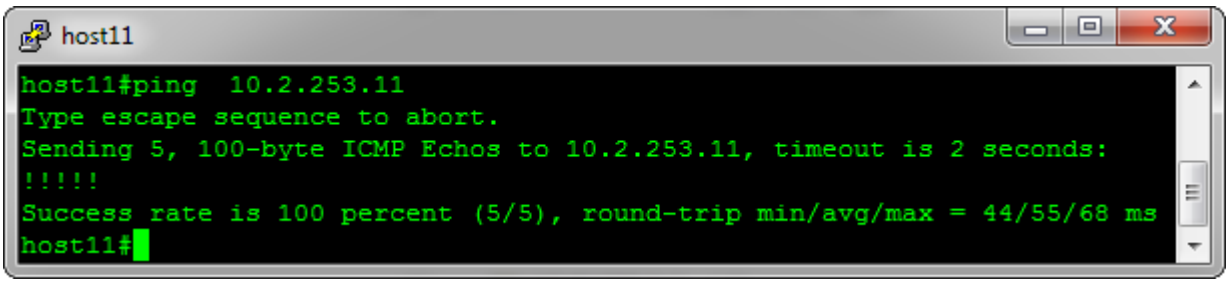
sw110(config)#interface gigabitEthernet 0/0
sw110(config-if)#switchport mode access
sw110(config-if)#switchport access vlan 2000
sw110(config-if)#exit
sw110(config)#

sw110(config)#interface gigabitEthernet 0/1
sw110(config-if)#switchport mode access
sw110(config-if)#switchport access vlan 2001
sw110(config-if)#exit
sw110(config)#
```

Verification

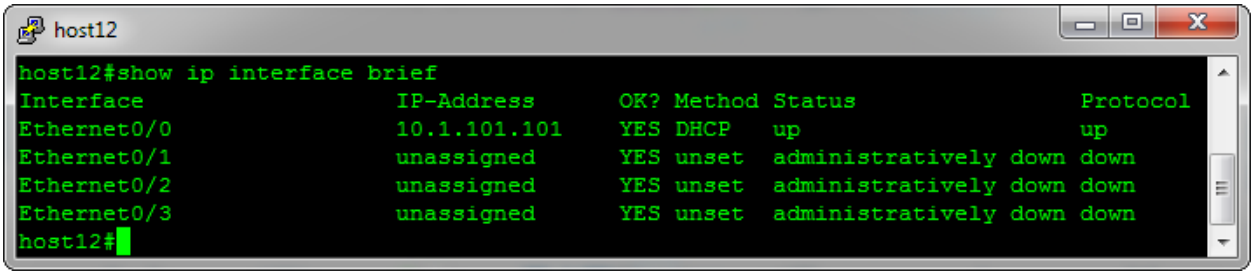
On host11

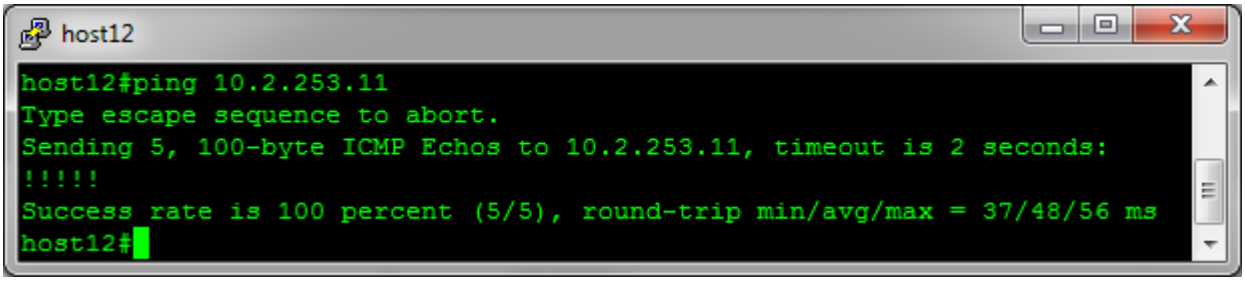




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On host12





CCIE Lab Center

SECTION 1.6: IPv6 in HQ (1 Points)

QUESTION

Implement IPv6 on sw101 and sw102 for switch virtual interfaces (SVI's) Vlan2000 and Vlan2001 according to these requirements:

1. SW101
   - Interface VLAN 2000: 2001:DB8:1:100::1/64
   - Interface VLAN 2001: 2001:DB8:1:101::1/64
2. SW102
   - Interface VLAN 2000: 2001:DB8:1:100::2/64
   - Interface VLAN 2001: 2001:DB8:1:101::2/64
3. The configuration must enable hosts in these VLANs to obtain their IPv6 configuration via SLAAC and keep a stable connectivity with other IPv6 networks
4. Use native IPv6 means to provide gateway redundancy with sw101 being the preferred gateway in VLAN 2000 and sw102 being the preferred gateway in VLAN 2001. The role must be detrmined by an explicit configuration solely on the intended preferred gateway.
5. Hosts must be able to detect the failure of the preferred gateway in as little as 3 seconds.

Solution

On sw101

```
sw101(config)#interface vlan 2000
sw101(config-if)#ipv6 address 2001:DB8:1:100::1/64
sw101(config-if)#ipv6 nd router-preference high
sw101(config-if)#ipv6 nd ra interval msec 1000
sw101(config-if)#ipv6 nd ra lifetime 3
sw101(config-if)#exit
sw101(config)#

sw101(config)#interface vlan 2001
sw101(config-if)#ipv6 address 2001:DB8:1:101::1/64
sw101(config-if)#ipv6 nd ra interval msec 1000
sw101(config-if)#ipv6 nd ra lifetime 3
sw101(config-if)#exit
sw101(config)#
```

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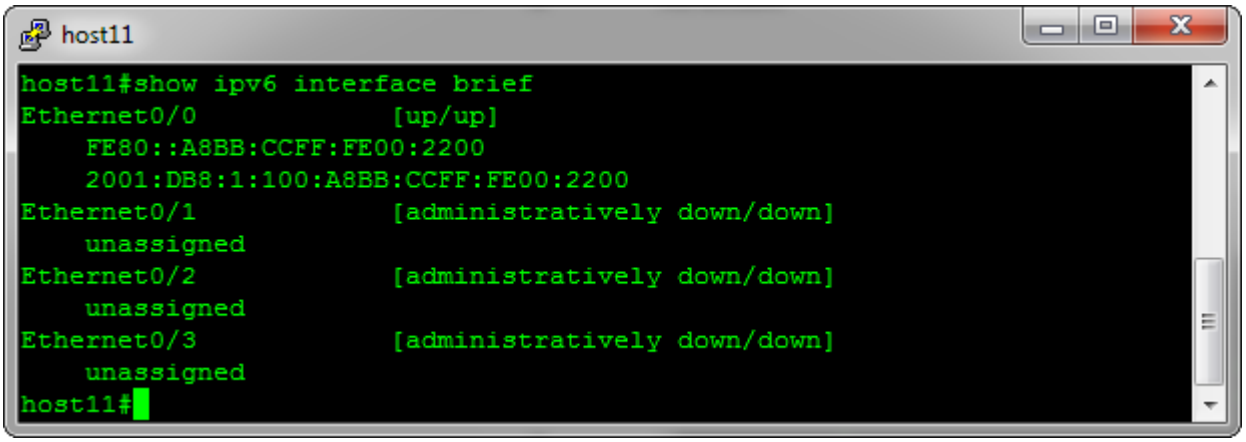
On sw102

```
sw102(config)#interface Vlan2000
sw102(config-if)#ipv6 address 2001:DB8:1:100::2/64
sw102(config-if)#ipv6 nd ra interval msec 1000
sw102(config-if)#ipv6 nd ra lifetime 3
sw102(config-if)#exit
sw102(config)#

sw102(config)#interface Vlan2001
sw102(config-if)#ipv6 address 2001:DB8:1:101::2/64
sw102(config-if)#ipv6 nd router-preference high
sw102(config-if)#ipv6 nd ra interval msec 1000
sw102(config-if)#ipv6 nd ra lifetime 3
sw102(config-if)#exit
sw102(config)#
```

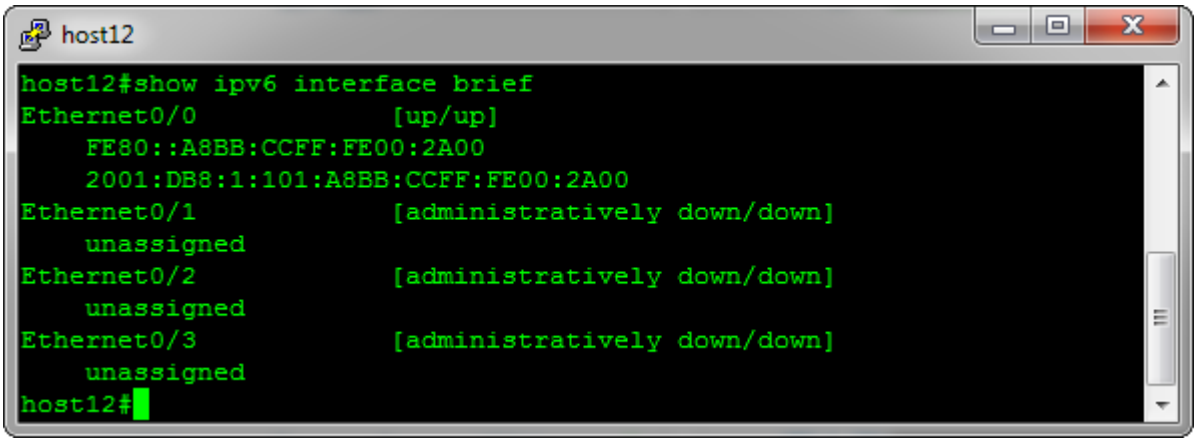
Verification

On host11



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On host12



CCIE Lab Center

SECTION 1.7: IPv6 EIGRP in HQ (2 Points)

QUESTION

In HQ enable EIGRP for IPv6 on r11, r12, sw101 and sw102 according to these requirements:

1. Use process name “ccie” (without the quotes) and AS number 65001.
2. Do not configure any additional IPv6 addresses
3. IPv6 EIGRP may form adjacencies only over the physical Layer3 interface between r11, r12, sw101 and sw102.
4. Prevent IPv6 EIGRP from automatically running on, or advertising attached prefixes from new IPv6-enabled interfaces in the future unless allowed explicitly.
5. Ensure that the attached IPv6 prefixes on SVI’s Vlan2000 and Vlan2001 on SW101 and sw102 are advertised in IPv6 EIGRP and learned on r11 and r12.
6. No route filtering is allowed to accomplish this entire task.

Solution

On r11

```
r11(config)#router eigrp ccie
r11(config-router)#address-family ipv6 unicast autonomous-system 65001

r11(config-router-af)#af-interface default
r11(config-router-af-interface)#passive-interface
r11(config-router-af-interface)#exit-af-interface
r11(config-router-af)#

r11(config-router-af)#af-interface gigabitEthernet 0/1
r11(config-router-af-interface)#no passive-interface
r11(config-router-af-interface)#exit-af-interface
r11(config-router-af)#

r11(config-router-af)#af-interface gigabitEthernet 0/2
r11(config-router-af-interface)#no passive-interface
r11(config-router-af-interface)#exit
r11(config-router-af)#

r11(config-router-af)#af-interface gigabitEthernet 0/3
r11(config-router-af-interface)#no passive-interface
```

```
r11(config-router-af-interface)#exit
r11(config-router-af)#exit-address-family
r11(config-router)#exit
r11(config)#
```

On r12

```
r12(config)#router eigrp ccie
r12(config-router)#address-family ipv6 unicast autonomous-system 65001

r12(config-router-af)#af-interface default
r12(config-router-af-interface)#passive-interface
r12(config-router-af-interface)#exit-af-interface
r12(config-router-af)#

r12(config-router-af)#af-interface gigabitEthernet 0/1
r12(config-router-af-interface)#no passive-interface
r12(config-router-af-interface)#exit-af-interface
r12(config-router-af)#

r12(config-router-af)#af-interface gigabitEthernet 0/2
r12(config-router-af-interface)#no passive-interface
r12(config-router-af-interface)#exit
r12(config-router-af)#

r12(config-router-af)#af-interface gigabitEthernet 0/3
r12(config-router-af-interface)#no passive-interface
r12(config-router-af-interface)#exit
r12(config-router-af)#exit-address-family
r12(config-router)#exit
r12(config)#
```

On sw101

```
sw101(config)#router eigrp ccie
sw101(config-router)#address-family ipv6 unicast autonomous-system 65001

sw101(config-router-af)#af-interface default
sw101(config-router-af-interface)#passive-interface
sw101(config-router-af-interface)#exit-af-interface
sw101(config-router-af)#

sw101(config-router-af)#af-interface gigabitEthernet 0/0
sw101(config-router-af-interface)#no passive-interface
```

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```
sw101(config-router-af-interface)#exit-af-interface
sw101(config-router-af)#

sw101(config-router-af)#af-interface gigabitEthernet 0/1
sw101(config-router-af-interface)#no passive-interface
sw101(config-router-af-interface)#exit
sw101(config-router-af)#exit-address-family
sw101(config-router)#exit
sw101(config)#
```

On sw102

```
sw102(config)#router eigrp ccie
sw102(config-router)#address-family ipv6 unicast autonomous-system 65001

sw102(config-router-af)#af-interface default
sw102(config-router-af-interface)#passive-interface
sw102(config-router-af-interface)#exit-af-interface
sw102(config-router-af)#

sw102(config-router-af)#af-interface gigabitEthernet 0/0
sw102(config-router-af-interface)#no passive-interface
sw102(config-router-af-interface)#exit-af-interface
sw102(config-router-af)#

sw102(config-router-af)#af-interface gigabitEthernet 0/1
sw102(config-router-af-interface)#no passive-interface
sw102(config-router-af-interface)#exit
sw102(config-router-af)#exit-address-family
sw102(config-router)#exit
sw102(config)#
```

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Verification

On r11

r11

```
r11#show ipv6 eigrp neighbors
EIGRP-IPv6 VR(ccie) Address-Family Neighbors for AS(65001)
H   Address                 Interface              Hold Uptime   SRTT   RTO  Q  Seq
                                                   (sec)         (ms)       Cnt Num
2   Link-local address:     Gi0/2                    12 00:01:50  102   612  0  13
    FE80::102
1   Link-local address:     Gi0/3                     9 00:04:24   78   468  0  12
    FE80::101
0   Link-local address:     Gi0/1                    10 00:07:10   33   198  0  24
    FE80::12
r11#
```

r11

```
r11#show ipv6 route
IPv6 Routing Table - default - 5 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, HA - Home Agent, MR - Mobile Router, R - RIP
       H - NHRP, I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea
       IS - ISIS summary, D - EIGRP, EX - EIGRP external, NM - NEMO
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       RL - RPL, O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1
       OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       la - LISP alt, lr - LISP site-registrations, ld - LISP dyn-eid
       a - Application
D   2001:DB8:1:100::/64 [90/15360]
     via FE80::101, GigabitEthernet0/3
     via FE80::102, GigabitEthernet0/2
D   2001:DB8:1:101::/64 [90/15360]
     via FE80::101, GigabitEthernet0/3
     via FE80::102, GigabitEthernet0/2
LC  2001:DB8:1:255::11/128 [0/0]
     via Loopback0, receive
D   2001:DB8:1:255::12/128 [90/10880]
     via FE80::12, GigabitEthernet0/1
L   FF00::/8 [0/0]
     via Null0, receive
r11#
```

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r11

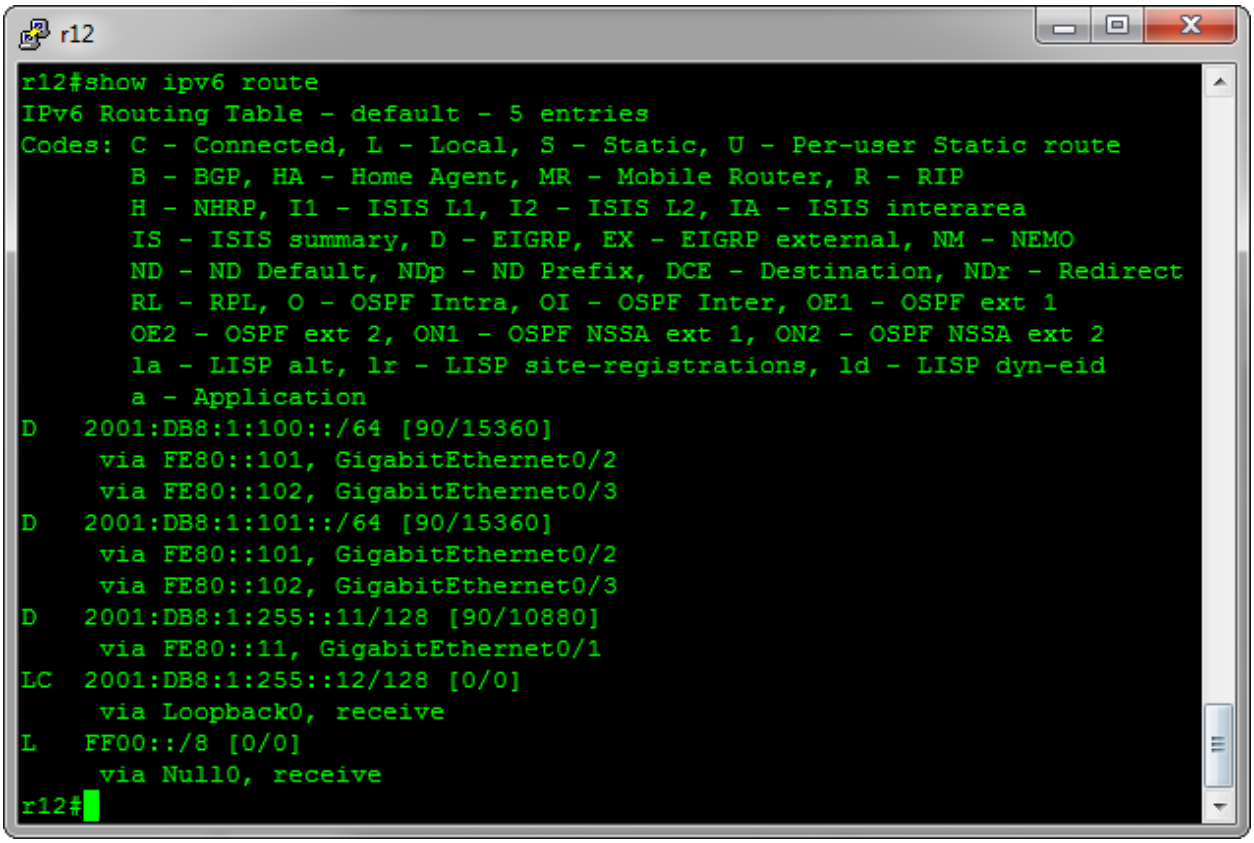
```
r11#ping 2001:DB8:1:100::1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:1:100::1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 12/20/32 ms
r11#
r11#ping 2001:DB8:1:100::2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:1:100::2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 14/18/25 ms
r11#
r11#ping 2001:DB8:1:101::1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:1:101::1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 16/22/32 ms
r11#
r11#ping 2001:DB8:1:101::2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:1:101::2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 13/23/37 ms
r11#
```

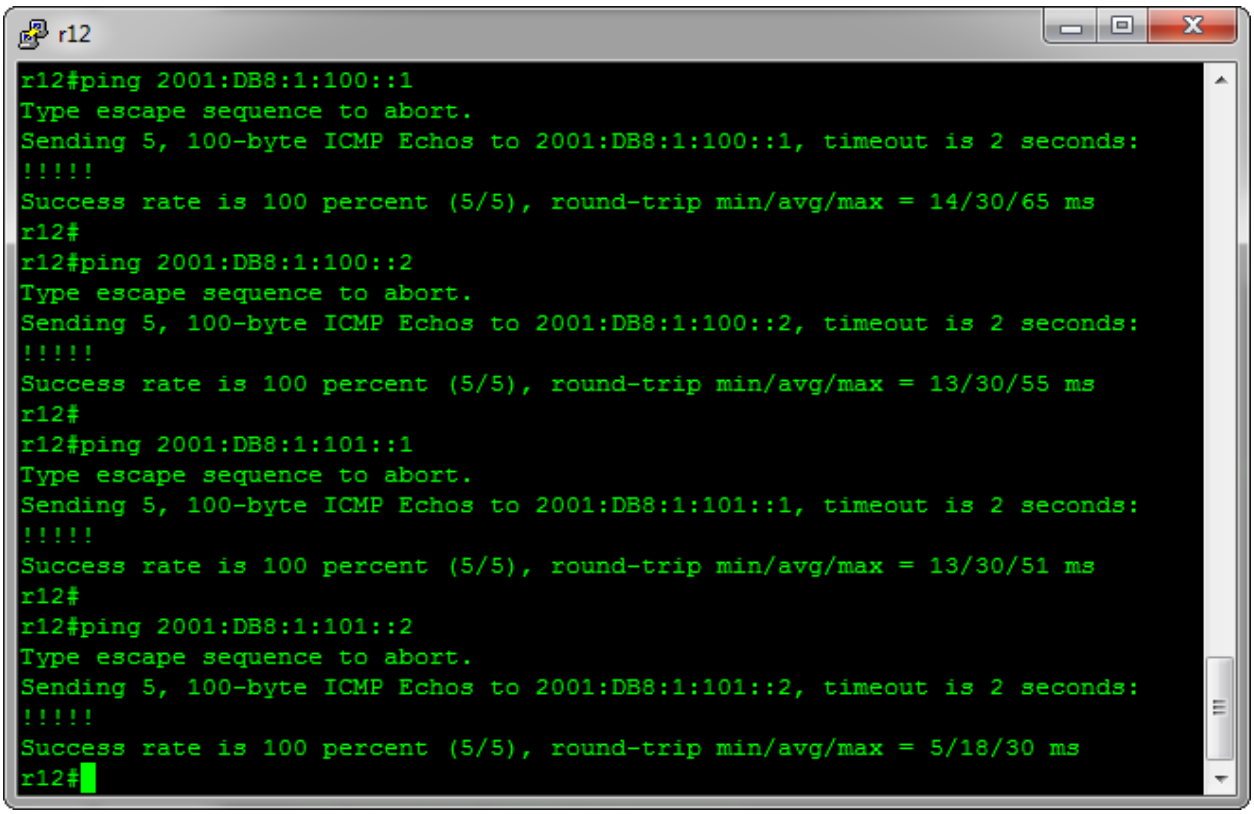
On r12

r12

```
r12#show ipv6 eigrp neighbors
EIGRP-IPv6 VR(ccie) Address-Family Neighbors for AS(65001)
H   Address                 Interface              Hold Uptime   SRTT   RTO  Q  Seq
                                                   (sec)         (ms)       Cnt Num
2   Link-local address:     Gi0/3                    13 00:04:42   43   258  0  12
    FE80::102
1   Link-local address:     Gi0/2                    10 00:07:14   72   432  0  13
    FE80::101
0   Link-local address:     Gi0/1                    10 00:10:02   31   186  0  23
    FE80::11
r12#
```

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On sw101

sw101

```
sw101#show ipv6 eigrp neighbors
EIGRP-IPv6 VR(ccie) Address-Family Neighbors for AS(65001)
H   Address                 Interface              Hold Uptime   SRTT   RTO  Q  Seq
                                                   (sec)         (ms)       Cnt Num
1   Link-local address:     Gi0/1                    11 00:09:36 1393   5000  0  23
    FE80::12
0   Link-local address:     Gi0/0                    13 00:09:37   38    228  0  22
    FE80::11
sw101#
```

sw101

```
sw101#show ipv6 route
IPv6 Routing Table - default - 7 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2
       IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX - EIGRP external
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       RL - RPL, a - Application
C   2001:DB8:1:100::/64 [0/0]
     via Vlan2000, directly connected
L   2001:DB8:1:100::1/128 [0/0]
     via Vlan2000, receive
C   2001:DB8:1:101::/64 [0/0]
     via Vlan2001, directly connected
L   2001:DB8:1:101::1/128 [0/0]
     via Vlan2001, receive
D   2001:DB8:1:255::11/128 [90/10880]
     via FE80::11, GigabitEthernet0/0
D   2001:DB8:1:255::12/128 [90/10880]
     via FE80::12, GigabitEthernet0/1
L   FF00::/8 [0/0]
     via Null0, receive
sw101#
```

sw101

```
r12#ping 2001:DB8:1:255::11
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:1:255::11, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 9/13/22 ms
r12#
r12#ping 2001:DB8:1:255::12
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:1:255::12, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 4/5/7 ms
r12#
```

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On sw102

sw102

```
sw102#show ipv6 eigrp neighbors
EIGRP-IPv6 VR(ccie) Address-Family Neighbors for AS(65001)
H   Address                 Interface              Hold Uptime   SRTT   RTO  Q  Seq
                                                   (sec)         (ms)       Cnt Num
1   Link-local address:     Gi0/1                    13 00:12:57   40   240  0  24
    FE80::11
0   Link-local address:     Gi0/0                    13 00:12:58   49   294  0  25
    FE80::12
sw102#
```

sw102

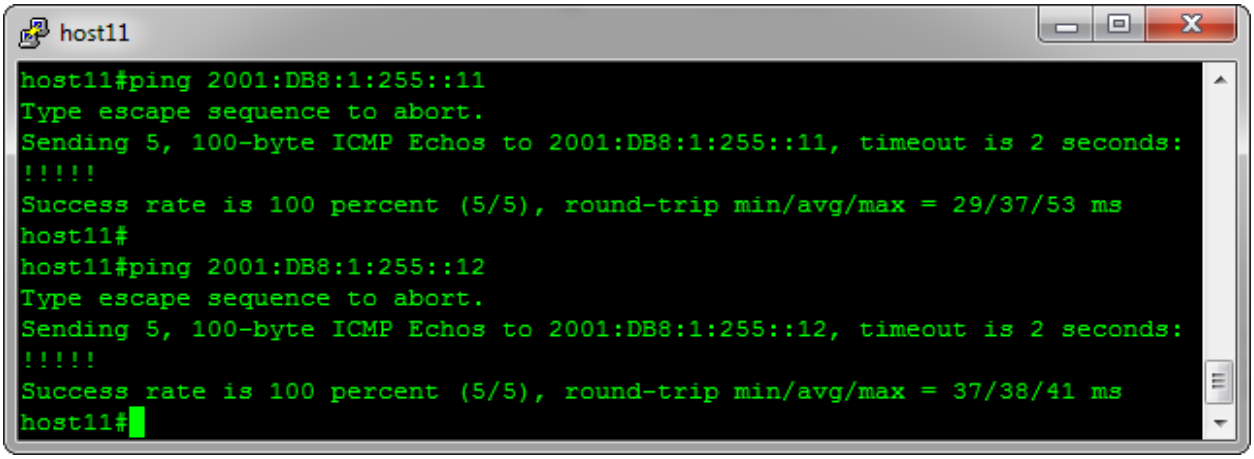
```
sw102#show ipv6 route
IPv6 Routing Table - default - 7 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2
       IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX - EIGRP external
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       RL - RPL, a - Application
C   2001:DB8:1:100::/64 [0/0]
     via Vlan2000, directly connected
L   2001:DB8:1:100::2/128 [0/0]
     via Vlan2000, receive
C   2001:DB8:1:101::/64 [0/0]
     via Vlan2001, directly connected
L   2001:DB8:1:101::2/128 [0/0]
     via Vlan2001, receive
D   2001:DB8:1:255::11/128 [90/10880]
     via FE80::11, GigabitEthernet0/1
D   2001:DB8:1:255::12/128 [90/10880]
     via FE80::12, GigabitEthernet0/0
L   FF00::/8 [0/0]
     via Null0, receive
sw102#
```

sw102

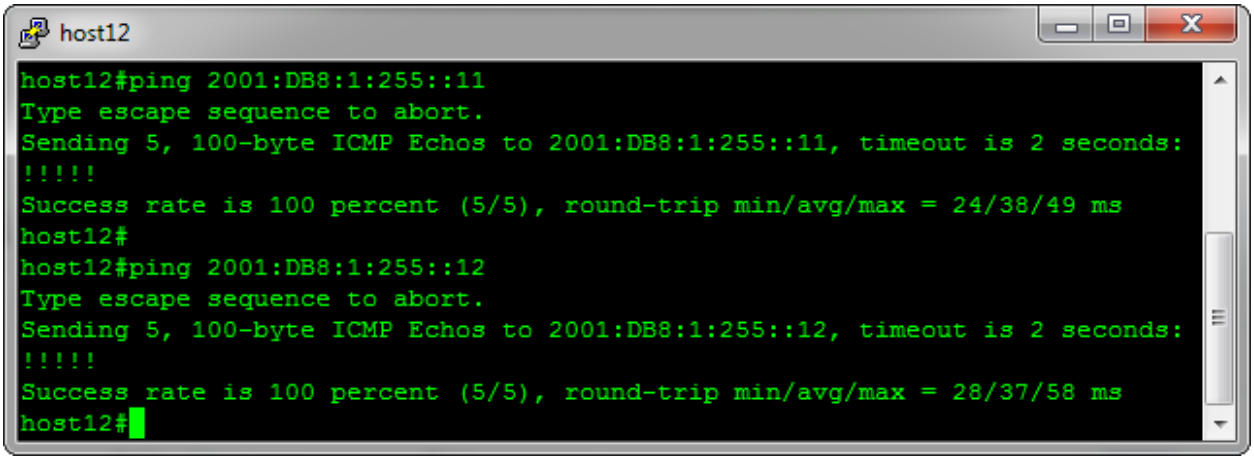
```
sw102#ping 2001:DB8:1:255::11
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:1:255::11, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 9/15/20 ms
sw102#
sw102#ping 2001:DB8:1:255::12
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:1:255::12, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 4/13/23 ms
sw102#
```

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On host11



On host12



CCIE Lab Center

SECTION 1.8: OSPFv2 in DC (4 Points)

QUESTION

Configure devices in the DC according to these requirements:

1. Switches sw201 and sw202 must establish a stable OSPF adjacency in the Full state with vedge21 and vedge22 on interface Vlan3999. Any configuration changes and corrections necessary to meet this requirement may be performed only on the switches and any mismatched parameters causing the issue must be changed to exactly match the configuration of the vedges.
2. All OSPF speakers in the DC running Cisco IOS and IOS-XE software must be configured to keep the number of advertised internal routes to an absolute minimum while not impacting the reachability of the services. This includes the reachability of ISE, DNA Center, vManage, vBond, vSmart on their internal (In Band Connectivity) addresses as well as any existing and future devices in VLAN4000 on sw201 and SW202. The configuration of this requirement must be completed exclusively within the “router ospf” and “interface Vlan” context without causing any impact to existing OSPF adjacencies.
3. Router r24 must advertise two prefixes, 10.6.0.0/16 and 10.200.0.0/24 as Type-5 LSAs in OSPFv2 to provide HQ and DC with the reachability to the DMVPN tunnel and Branches#3 and #4. The configuration of this requirement must be completed exclusively within the “router ospf” context.
4. Any route from 10.2.0.0/16 range that keeps being advertised in OSPF must continue being advertised as an intra-area route.
5. It is not allowed to modify existing areas to accomplish this entire task.

Solution

On sw201

```
sw201(config)#interface vlan 3999
sw201(config-if)#mtu 1496
sw201(config-if)#exit
sw201(config)#

sw201(config)#router ospf 1
sw201(config-router)#prefix-suppression
sw201(config-router)#exit
sw201(config)#
```

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On sw202

```
sw202(config)#interface vlan 3999
sw202(config-if)#mtu 1496
sw202(config-if)#exit
sw202(config)#

sw202(config)#router ospf 1
sw202(config-router)#prefix-suppression
sw202(config-router)#exit
sw202(config)#
```

On sw211

```
sw211(config)#router ospf 1
sw211(config-router)#prefix-suppression
sw211(config-router)#passive-interface Ethernet1/1
sw211(config-router)#passive-interface Ethernet1/2
sw211(config-router)#passive-interface Ethernet1/3
sw211(config-router)#exit
sw211(config)#
```

On sw212

```
sw212(config)#router ospf 1
sw212(config-router)#prefix-suppression
sw212(config-router)#passive-interface Ethernet1/1
sw212(config-router)#passive-interface Ethernet1/2
sw212(config-router)#exit
sw212(config)#
```

On r21

```
r21(config)#router ospf 1
r21(config-router)#prefix-suppression
r21(config-router)#exit
r21(config)#
```

On r22

```
r22(config)#router ospf 1
r22(config-router)#prefix-suppression
r22(config-router)#exit
```

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r22(config)#

On r23

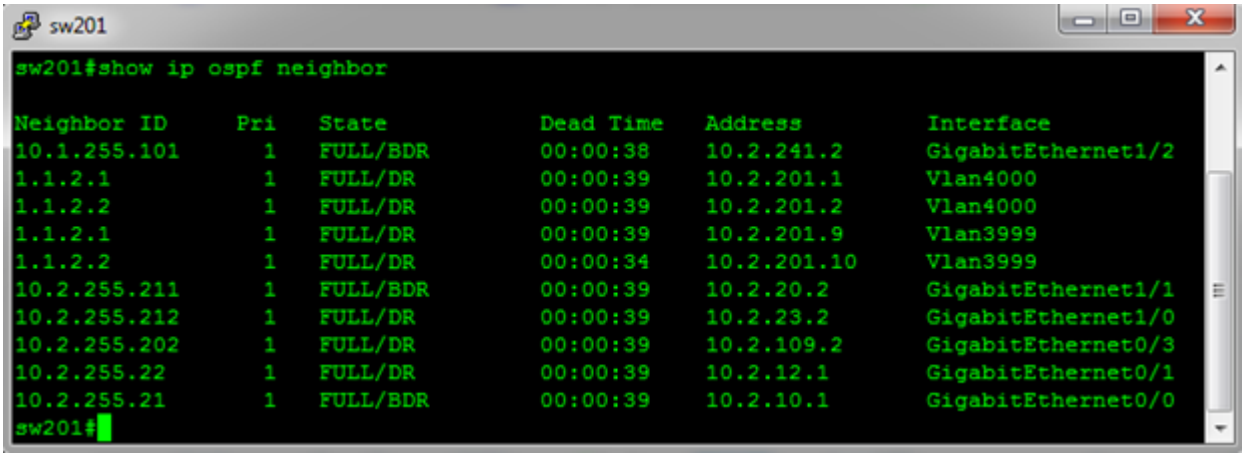
r23(config)#router ospf 1
r23(config-router)#prefix-suppression
r23(config-router)#exit
r23(config)#

On r24

r24(config)#router ospf 1
r24(config-router)#redistribute eigrp 65006 subnets tag 150
r24(config-router)#summary-address 10.6.0.0 255.255.0.0
r24(config-router)#prefix-suppression
r24(config-router)#exit
r24(config)#

Verification

On sw201



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On sw202



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SECTION 1.9: BGP between HQ - DC and Service Providers

(3 Points)

QUESTION

Configure the BGP peering between HQ/DC and Global SP #1 and Global SP #2 according to these requirements:

1. Bring up the BGP peering between HQ r11 and SP#1 r3
2. Bring up the BGP peering between DC r21 and SP#1 r3
3. Bring up the BGP peering between DC r22 and SP#2
4. Ensure that the routes learned over eBGP sessions and further advertised in iBGP will be considered reachable even if the networks on inter-AS links are not advertised in OSPF. The configuration of this requirement must be completed exclusively with the “router bgp” context.
5. On r11, r21, r22 perform mutual redistribution between OSPFv2 and BGP. However, prevent routes that were injected into OSPF from BGP to be reinjected back into BGP. This requirement must be solved on r11, r21, r22 using only a single route-map on each of the routers and without any reference to ACLs, prefix lists, or route-types.
6. Prevent HQ and DC from ever communicating through SP#1 r3. All communication between HQ and DC must occur only over the direct sw101/sw201 and sw102/sw202 interconnections. Any other communication must remain unaffected. This requirement must be solved on r21 and r22 by route filtering based on a well-known mandatory attribute without the use of route-maps.
7. No command may be removed from configuration on r11 to accomplish this entire task.
8. It is not allowed to modify existing configuration commands on r21 and r22 to accomplish this entire task.

Solution

On r11

```
r11(config)#route-map TAG deny 10
r11(config-route-map)#match tag 150
r11(config-route-map)#exit

r11(config)#route-map TAG permit 20
r11(config-route-map)#exit
r11(config)#

r11(config)#router bgp 65001
r11(config-router)#neighbor 100.3.11.1 remote-as 10000
r11(config-router)#address-family ipv4
```

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```
r11(config-router-af)#neighbor 100.3.11.1 activate
r11(config-router-af)#redistribute ospf 1 match external 1 external 2 route-map TAG
r11(config-router-af)#exit
r11(config-router)#exit
r11(config)#

r11(config)#router ospf 1
r11(config-router)#redistribute bgp 65001 subnets tag 150
r11(config-router)#exit
r11(config)#
```

On r21

```
r21(config)#ip as-path access-list 1 deny _65001_
r21(config)#ip as-path access-list 1 permit .*

r21(config)#ip prefix-list 1 deny 10.1.0.0/16 le 32
r21(config)#ip prefix-list 1 permit 0.0.0.0/0 le 32

r21(config)#route-map TAG deny 10
r21(config-route-map)#match tag 150
r21(config-route-map)#exit

r21(config)#route-map TAG permit 20
r21(config-route-map)#exit
r21(config)#

r21(config)#router bgp 65002
r21(config-router)#neighbor 10.2.255.22 next-hop-self
r21(config-router)#neighbor 100.3.21.1 remote-as 10000
r21(config-router)#neighbor 100.3.21.1 filter-list 1 in
r21(config-router)#neighbor 100.3.21.1 prefix-list 1 out
r21(config-router)#redistribute ospf 1 match external 1 external 2 route-map TAG
r21(config-router)#exit
r21(config)#

r21(config)#router ospf 1
r21(config-router)#redistribute bgp 65002 subnets tag 150
r21(config-router)#exit
r21(config)#
```

On r22

```
r22(config)#ip as-path access-list 1 deny _65001_
r22(config)#ip as-path access-list 1 permit .*
```

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```
r22(config)#ip prefix-list 1 deny 10.1.0.0/16 le 32
r22(config)#ip prefix-list 1 permit 0.0.0.0/0 le 32


r22(config)#route-map TAG deny 10
r22(config-route-map)#match tag 150
r22(config-route-map)#exit

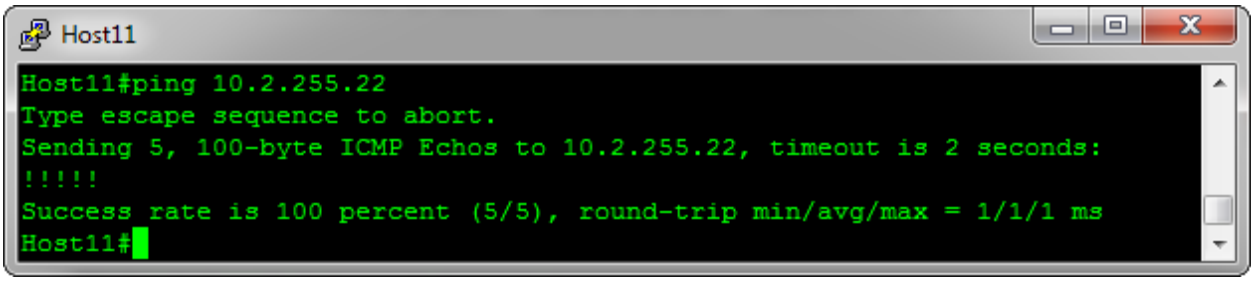
r22(config)#route-map TAG permit 20
r22(config-route-map)#exit
r22(config)#

r22(config)#router bgp 65002
r22(config-router)#neighbor 101.22.0.1 remote-as 10001
r22(config-router)#neighbor 101.22.0.1 filter-list 1 in
r22(config-router)#neighbor 101.22.0.1 prefix-list 1 out
r22(config-router)#redistribute ospf 1 match external 1 external 2 route-map TAG
r22(config-router)#exit
r22(config)#

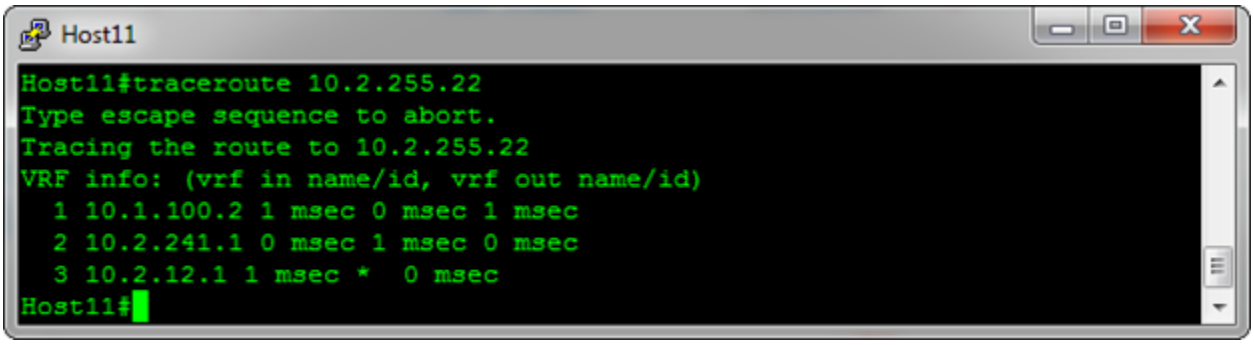
r22(config)#router ospf 1
r22(config-router)#redistribute bgp 65002 subnets tag 150
r22(config-router)#exit
r22(config)#
```

Verification

On host11



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More testings can be done by shutting down links of r21 (e0/1-3)

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SECTION 1.10: Bringing up VPNv4/VPNv6 in SP#1 (3 Points)

QUESTION

Configure routers r3, r4, r5 and r6 in SP#1 accoding to these requirements:

1. Configure r3 through r6 for mutual VPNv4 and VPNv6 route exchange without the use of a route-reflector. Use Lo0 IPv4 addresses for peerings.
2. Configure r3 through r6 to assign (allocate/bind) as few unique MPLS labels to all existing and future VPNv4 and VPNv6 routes as possible.
3. On routers r3 through r6, prevent any existing and future customer from discovering details about the inner topology of SP#1, It is not allowed to use ACLs to accomplish this requirement.

Solution

On r3

```
r3(config)#router bgp 10000
r3(config-router)#neighbor 100.255.254.4 remote-as 10000
r3(config-router)#neighbor 100.255.254.4 update-source Loopback0
r3(config-router)#neighbor 100.255.254.5 remote-as 10000
r3(config-router)#neighbor 100.255.254.5 update-source Loopback0
r3(config-router)#neighbor 100.255.254.6 remote-as 10000
r3(config-router)#neighbor 100.255.254.6 update-source Loopback0

r3(config-router)#address-family vpnv4
r3(config-router-af)#neighbor 100.255.254.4 activate
r3(config-router-af)#neighbor 100.255.254.5 activate
r3(config-router-af)#neighbor 100.255.254.6 activate
r3(config-router-af)#exit-address-family

r3(config-router)#address-family vpnv6
r3(config-router-af)#neighbor 100.255.254.4 activate
r3(config-router-af)#neighbor 100.255.254.5 activate
r3(config-router-af)#neighbor 100.255.254.6 activate
r3(config-router-af)#exit-address-family
r3(config-router)#exit
r3(config)#

r3(config)#no mpls ip propagate-ttl
r3(config)#mpls label mode all-vrfs protocol bgp-vpnv4 per-vrf
r3(config)#mpls label mode all-vrfs protocol bgp-vpnv6 per-vrf
```

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On r4

```
r4(config)#router bgp 10000
r4(config-router)#neighbor 100.255.254.3 remote-as 10000
r4(config-router)#neighbor 100.255.254.3 update-source Loopback0
r4(config-router)#neighbor 100.255.254.5 remote-as 10000
r4(config-router)#neighbor 100.255.254.5 update-source Loopback0
r4(config-router)#neighbor 100.255.254.6 remote-as 10000
r4(config-router)#neighbor 100.255.254.6 update-source Loopback0

r4(config-router)#address-family vpnv4
r4(config-router-af)#neighbor 100.255.254.3 activate
r4(config-router-af)#neighbor 100.255.254.5 activate
r4(config-router-af)#neighbor 100.255.254.6 activate
r4(config-router-af)#exit-address-family

r4(config-router)#address-family vpnv6
r4(config-router-af)#neighbor 100.255.254.3 activate
r4(config-router-af)#neighbor 100.255.254.5 activate
r4(config-router-af)#neighbor 100.255.254.6 activate
r4(config-router-af)#exit-address-family
r4(config-router)#exit
r4(config)#

r4(config)#no mpls ip propagate-ttl
r4(config)#mpls label mode all-vrfs protocol bgp-vpnv4 per-vrf
r4(config)#mpls label mode all-vrfs protocol bgp-vpnv6 per-vrf
```

On r5

```
r5(config)#router bgp 10000
r5(config-router)#neighbor 100.255.254.3 remote-as 10000
r5(config-router)#neighbor 100.255.254.3 update-source Loopback0
r5(config-router)#neighbor 100.255.254.4 remote-as 10000
r5(config-router)#neighbor 100.255.254.4 update-source Loopback0
r5(config-router)#neighbor 100.255.254.6 remote-as 10000
r5(config-router)#neighbor 100.255.254.6 update-source Loopback0

r5(config-router)#address-family vpnv4
r5(config-router-af)#neighbor 100.255.254.3 activate
r5(config-router-af)#neighbor 100.255.254.4 activate
r5(config-router-af)#neighbor 100.255.254.6 activate
r5(config-router-af)#exit-address-family

r5(config-router)#address-family vpnv6
r5(config-router-af)#neighbor 100.255.254.3 activate
```

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```
r5(config-router-af)#neighbor 100.255.254.4 activate
r5(config-router-af)#neighbor 100.255.254.6 activate
r5(config-router-af)#exit-address-family
r5(config-router)#exit
r5(config)#

r5(config)#no mpls ip propagate-ttl
r5(config)#mpls label mode all-vrfs protocol bgp-vpnv4 per-vrf
r5(config)#mpls label mode all-vrfs protocol bgp-vpnv6 per-vrf
```

On r6

```
r6(config)#router bgp 10000
r6(config-router)#bgp router-id 10.255.254.6
r6(config-router)#bgp log-neighbor-changes
r6(config-router)#neighbor 100.255.254.3 remote-as 10000
r6(config-router)#neighbor 100.255.254.3 update-source Loopback0
r6(config-router)#neighbor 100.255.254.4 remote-as 10000
r6(config-router)#neighbor 100.255.254.4 update-source Loopback0
r6(config-router)#neighbor 100.255.254.5 remote-as 10000
r6(config-router)#neighbor 100.255.254.5 update-source Loopback0

r6(config-router)#address-family vpnv4
r6(config-router-af)#neighbor 100.255.254.3 activate
r6(config-router-af)#neighbor 100.255.254.4 activate
r6(config-router-af)#neighbor 100.255.254.5 activate
r6(config-router-af)#exit-address-family

r6(config-router)#address-family vpnv6
r6(config-router-af)#neighbor 100.255.254.3 activate
r6(config-router-af)#neighbor 100.255.254.4 activate
r6(config-router-af)#neighbor 100.255.254.5 activate
r6(config-router-af)#exit-address-family
r6(config-router)#exit
r6(config)#

r6(config)#no mpls ip propagate-ttl
r6(config)#mpls label mode all-vrfs protocol bgp-vpnv4 per-vrf
r6(config)#mpls label mode all-vrfs protocol bgp-vpnv6 per-vrf
```

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Verification

On r3

```
r3
r3#show bgp vpnv4 unicast all summary
BGP router identifier 100.255.254.3, local AS number 10000
BGP table version is 2423, main routing table version 2423
64 network entries using 16384 bytes of memory
82 path entries using 11152 bytes of memory
26/14 BGP path/bestpath attribute entries using 7904 bytes of memory
5 BGP AS-PATH entries using 136 bytes of memory
1 BGP extended community entries using 24 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 35600 total bytes of memory
BGP activity 853/789 prefixes, 1546/1464 paths, scan interval 60 secs
64 networks peaked at 07:33:51 Feb 22 2021 UTC (02:32:01.885 ago)

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
100.3.11.2      4        65001    1427    1598     2423    0    0 22:29:15        18
100.3.21.2      4        65002     422     264     2423    0    0 02:53:20        54
100.255.254.4   4        10000    7290    8129     2423    0    0 4d14h            0
100.255.254.5   4        10000    7298    8133     2423    0    0 4d14h            2
100.255.254.6   4        10000    7297    8128     2423    0    0 4d14h            2

r3#
```

```
r3
r3#show bgp vpnv6 unicast all summary
BGP router identifier 100.255.254.3, local AS number 10000
BGP table version is 1, main routing table version 1

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
100.255.254.4   4        10000    7290    8130        1    0    0 4d14h            0
100.255.254.5   4        10000    7299    8134        1    0    0 4d14h            0
100.255.254.6   4        10000    7298    8129        1    0    0 4d14h            0

r3#
```

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On r5

r5

```
r5#show bgp vpnv4 unicast all summary
BGP router identifier 100.255.254.5, local AS number 10000
BGP table version is 4565, main routing table version 4565
118 network entries using 30208 bytes of memory
118 path entries using 16048 bytes of memory
17/16 BGP path/bestpath attribute entries using 5168 bytes of memory
5 BGP AS-PATH entries using 136 bytes of memory
1 BGP extended community entries using 24 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 51584 total bytes of memory
BGP activity 1558/1440 prefixes, 1999/1881 paths, scan interval 60 secs
118 networks peaked at 07:33:51 Feb 22 2021 UTC (02:35:15.519 ago)

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
100.5.61.2      4        65006    3991    4387     4565    0    0 2d15h           1
100.255.254.3   4        10000    8137    7302     4565    0    0 4d14h          56
100.255.254.4   4        10000       3       7     4565    0    0 00:00:50        0
100.255.254.6   4        10000    7292    7296     4565    0    0 4d14h           2

r5#
```

r5

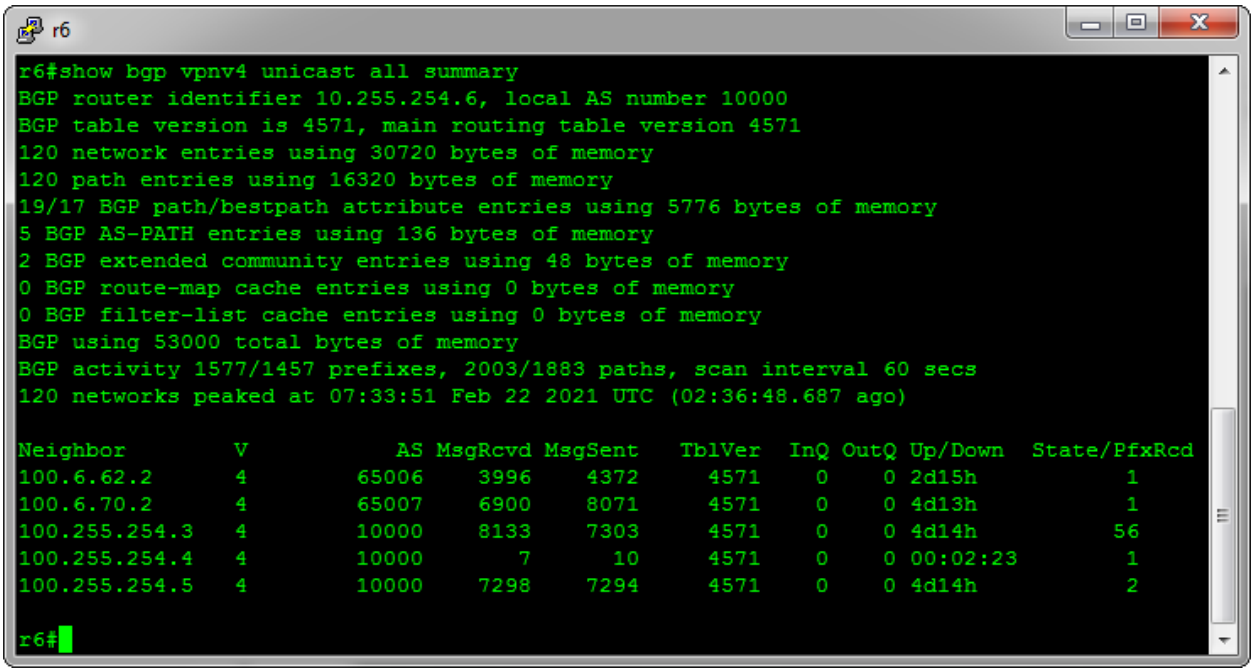
```
r5#show bgp vpnv6 unicast all summary
BGP router identifier 100.255.254.5, local AS number 10000
BGP table version is 1, main routing table version 1

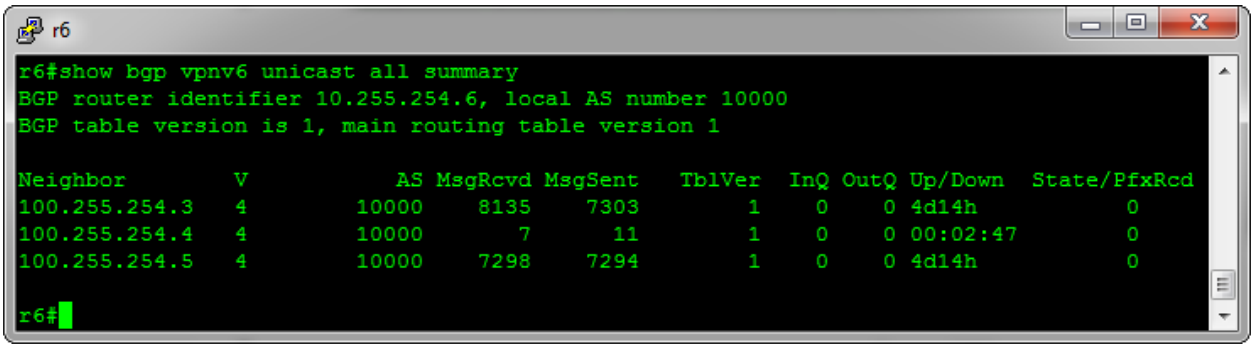
Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
100.255.254.3   4        10000    8137    7302        1    0    0 4d14h           0
100.255.254.4   4        10000       4       8        1    0    0 00:01:16        0
100.255.254.6   4        10000    7293    7296        1    0    0 4d14h           0

r5#
```

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On r6





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SECTION 1.11: Fixing Broken DMVPN between DC and Branches #3 & #4 (3 Points)

QUESTION

Correct the configuration issues resulting in broken DMVPN tunnel connectivity between DC, Branch3 and Branch4 according to these requirements:

1. The DMVPN must operate in IPsec-protected Phase 3 mode.
2. Using the FVRF approach, safeguard the DMVPN operation against any potential recursive routing issues involving the tunnel.
3. Do not create any new VRFs.
4. Do not change the tunnel source commands on tunnel interfaces.
5. On spokes do not add new BGP neighbors; reuse those that are currently up while changing their VRF membership as needed.
6. It is not allowed to modify configuration on DC r24 to complete this entire task.

Solution

On r24

```
r24(config)#crypto isakmp policy 10
r24(config-isakmp)#hash md5
r24(config-isakmp)#exit
r24(config)#

r24(config)#interface tunnel0
r24(config-if)#ip nhrp map multicast dynamic (command is already there but hidden)
r24(config-if)#exit
r24(config)#
```

On r61

```
r61(config)#crypto keyring KR vrf WAN
r61(conf-keyring)#  pre-shared-key address 0.0.0.0 0.0.0.0 key cisco
r61(conf-keyring)#  exit
r61(config)#

r61(config)#interface loopback 0
```

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```
r61(config-if)#vrf forwarding WAN
r61(config-if)#ip address 10.6.255.61 255.255.255.255
r61(config-if)#exit
r61(config)#

r61(config)#interface GigabitEthernet0/0
r61(config-if)#vrf forwarding WAN
r61(config-if)#ip address 100.5.61.2 255.255.255.252
r61(config-if)#exit
r61(config)#

r61(config)#interface tunnel 0
r61(config-if)#ip mtu 1440
r61(config-if)#ip nhrp shortcut
r61(config-if)#no ip nhrp map 10.2.255.24 10.200.0.1
r61(config-if)#ip nhrp map 10.200.0.1 10.2.255.24
r61(config-if)#tunnel vrf WAN
r61(config-if)#exit
r61(config)#

r61(config)#router bgp 65006
r61(config-router)#no network 10.6.255.61 mask 255.255.255.255
r61(config-router)#no neighbor 100.5.61.1 remote-as 10000

r61(config-router)#address-family ipv4 vrf WAN
r61(config-router-af)#network 10.6.255.61 mask 255.255.255.255
r61(config-router-af)#neighbor 100.5.61.1 remote-as 10000
r61(config-router-af)#neighbor 100.5.61.1 activate
r61(config-router-af)#exit-address-family
r61(config-router)#exit
r61(config)#
```

On r62

```
r62(config)#crypto keyring KR vrf WAN
r62(conf-keyring)#  pre-shared-key address 0.0.0.0 0.0.0.0 key cisco
r62(conf-keyring)#exit
r62(config)#

r62(config)#interface Loopback0
r62(config-if)# vrf forwarding WAN
r62(config-if)# ip address 10.6.255.62 255.255.255.255
r62(config-if)#exit
r62(config)#

r62(config)#interface GigabitEthernet0/0
```

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```
r62(config-if)# vrf forwarding WAN
r62(config-if)# ip address 100.6.62.2 255.255.255.252
r62(config-if)#exit
r62(config)#

r62(config)#interface tunnel 0
r62(config-if)#ip mtu 1440
r62(config-if)#ip nhrp shortcut
r62(config-if)#ip nhrp network-id 1010
r62(config-if)#tunnel vrf WAN
r62(config-if)#exit
r62(config)#

r62(config)#router bgp 65006
r62(config-router)#no network 10.6.255.62 mask 255.255.255.255
r62(config-router)#no neighbor 100.6.62.1 remote-as 10000

r62(config-router)# address-family ipv4 vrf WAN
r62(config-router-af)#  network 10.6.255.62 mask 255.255.255.255
r62(config-router-af)#  neighbor 100.6.62.1 remote-as 10000
r62(config-router-af)#  neighbor 100.6.62.1 activate
r62(config-router-af)# exit-address-family
r62(config-router)#exit
r62(config)#
```

On r70

```
r70(config)#crypto isakmp key cisco address 0.0.0.0

r70(config)#interface Tunnel0
r70(config-if)# ip mtu 1440
r70(config-if)#ip nhrp shortcut
r70(config-if)#ip nhrp redirect
r70(config-if)#tunnel vrf WAN
r70(config-if)#exit
r70(config)#
```

Verification

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On r24

r24

```
r24#show dmvpn detail
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
        N - NATed, L - Local, X - No Socket
        T1 - Route Installed, T2 - Nexthop-override
        C - CTS Capable, I2 - Temporary
        # Ent --> Number of NHRP entries with same NBMA peer
        NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
        UpDn Time --> Up or Down Time for a Tunnel
==========================================================================

Interface Tunnel0 is up/up, Addr. is 10.200.0.1, VRF ""
   Tunnel Src./Dest. addr: 10.2.255.24/Multipoint, Tunnel VRF ""
   Protocol/Transport: "multi-GRE/IP", Protect "prof"
   Interface State Control: Disabled
   nhrp event-publisher : Disabled
Type:Hub, Total NBMA Peers (v4/v6): 3

# Ent  Peer NBMA Addr Peer Tunnel Add State  UpDn Tm Attrb    Target Network
----- --------------- --------------- ----- -------- ----- -----------------
    1 10.6.255.61         10.200.0.61    UP 02:51:07     D     10.200.0.61/32
    1 10.6.255.62         10.200.0.62    UP 02:39:21     D     10.200.0.62/32
    1 10.7.255.70         10.200.0.70    UP 02:39:22     D     10.200.0.70/32
```

On r61

r61

```
r61#show dmvpn detail
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
        N - NATed, L - Local, X - No Socket
        T1 - Route Installed, T2 - Nexthop-override
        C - CTS Capable
        # Ent --> Number of NHRP entries with same NBMA peer
        NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
        UpDn Time --> Up or Down Time for a Tunnel
==========================================================================

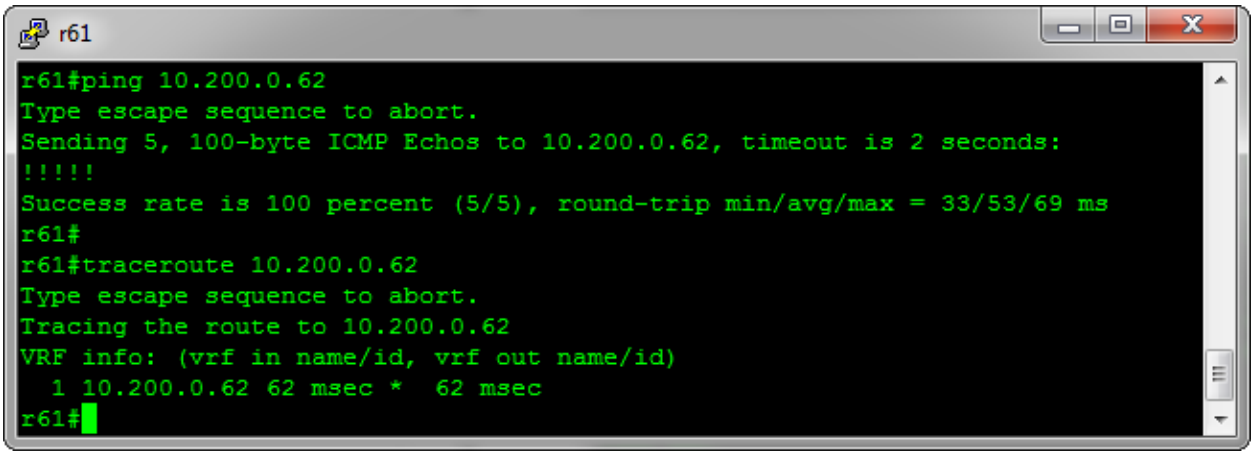
Interface Tunnel0 is up/up, Addr. is 10.200.0.61, VRF ""
   Tunnel Src./Dest. addr: 10.6.255.61/MGRE, Tunnel VRF "WAN"
   Protocol/Transport: "multi-GRE/IP", Protect "prof"
   Interface State Control: Disabled
   nhrp event-publisher : Disabled

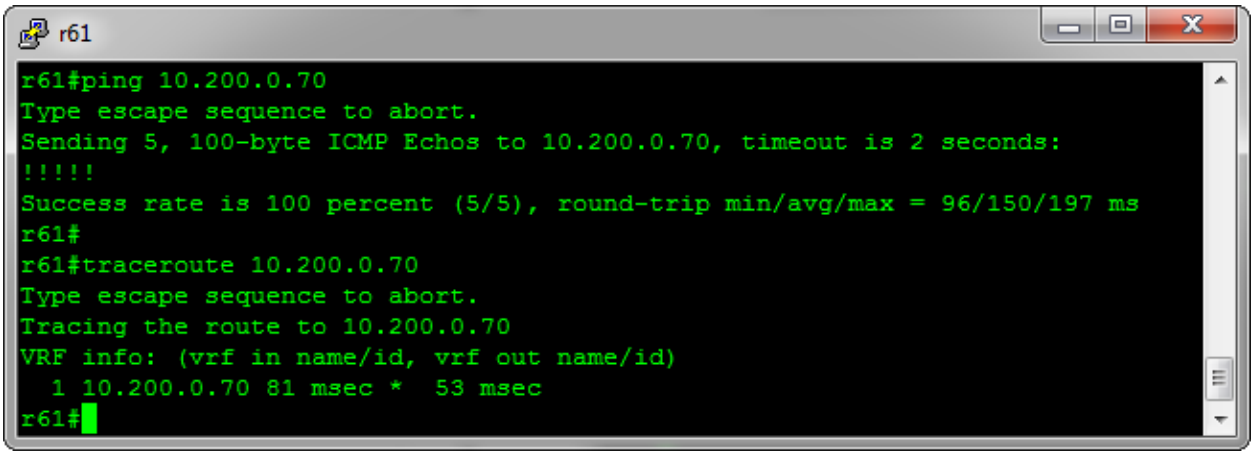
IPv4 NHS:
10.200.0.1  RE priority = 0 cluster = 0
Type:Spoke, Total NBMA Peers (v4/v6): 1

# Ent  Peer NBMA Addr Peer Tunnel Add State  UpDn Tm Attrb    Target Network
----- --------------- --------------- ----- -------- ----- -----------------
    1 10.2.255.24         10.200.0.1     UP 02:32:14     S      10.200.0.1/32


r61#
```

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On r62

r62

```
r62#show dmvpn detail
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
        N - NATed, L - Local, X - No Socket
        T1 - Route Installed, T2 - Nexthop-override
        C - CTS Capable
        # Ent --> Number of NHRP entries with same NBMA peer
        NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
        UpDn Time --> Up or Down Time for a Tunnel
==============================================================================

Interface Tunnel0 is up/up, Addr. is 10.200.0.62, VRF ""
   Tunnel Src./Dest. addr: 10.6.255.62/MGRE, Tunnel VRF "WAN"
   Protocol/Transport: "multi-GRE/IP", Protect "prof"
   Interface State Control: Disabled
   nhrp event-publisher : Disabled

IPv4 NHS:
10.200.0.1  RE priority = 0 cluster = 0
Type:Spoke, Total NBMA Peers (v4/v6): 1

# Ent  Peer NBMA Addr Peer Tunnel Add State  UpDn Tm Attrb    Target Network
----- --------------- --------------- ----- -------- ----- -----------------
    1 10.2.255.24          10.200.0.1    UP 02:33:49     S      10.200.0.1/32


r62#
```

r62

```
r62#ping 10.200.0.61
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.200.0.61, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 43/51/74 ms
r62#
r62#traceroute 10.200.0.61
Type escape sequence to abort.
Tracing the route to 10.200.0.61
VRF info: (vrf in name/id, vrf out name/id)
  1 10.200.0.61 75 msec *  57 msec
r62#
```

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r62

```
r62#ping 10.200.0.70
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.200.0.70, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 105/164/271 ms
r62#
r62#traceroute 10.200.0.70
Type escape sequence to abort.
Tracing the route to 10.200.0.70
VRF info: (vrf in name/id, vrf out name/id)
  1 10.200.0.70 51 msec *  29 msec
r62#
```

On r70

r70

```
r70#show dmvpn detail
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
        N - NATed, L - Local, X - No Socket
        T1 - Route Installed, T2 - Nexthop-override
        C - CTS Capable
        # Ent --> Number of NHRP entries with same NBMA peer
        NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
        UpDn Time --> Up or Down Time for a Tunnel
==============================================================================

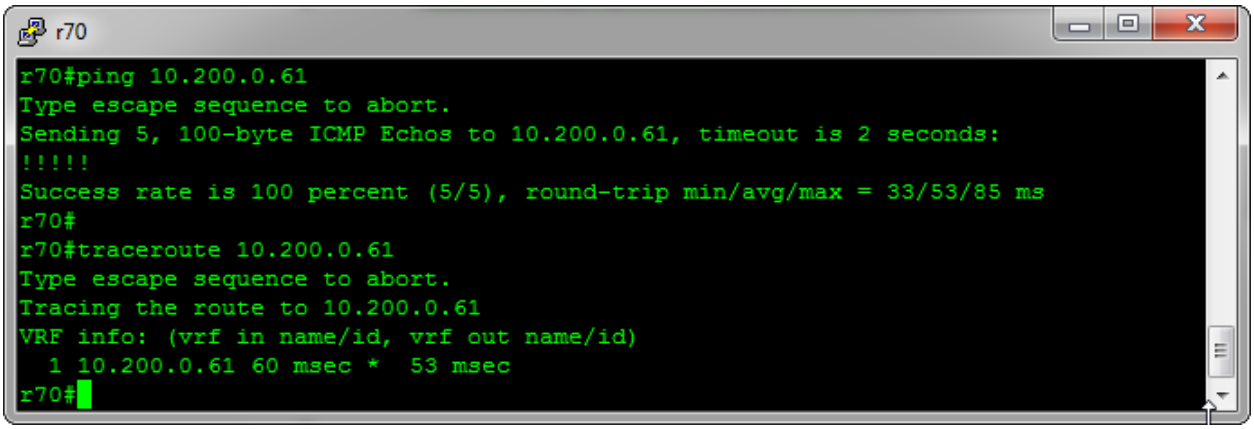
Interface Tunnel0 is up/up, Addr. is 10.200.0.70, VRF ""
   Tunnel Src./Dest. addr: 10.7.255.70/MGRE, Tunnel VRF "WAN"
   Protocol/Transport: "multi-GRE/IP", Protect "prof"
   Interface State Control: Disabled
   nhrp event-publisher : Disabled

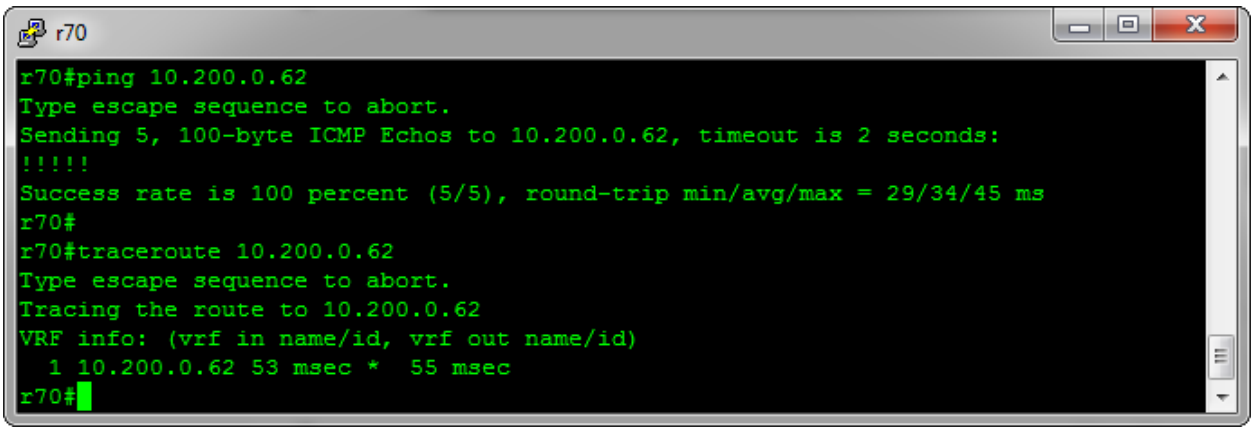
IPv4 NHS:
10.200.0.1  RE priority = 0 cluster = 0
Type:Spoke, Total NBMA Peers (v4/v6): 1

# Ent  Peer NBMA Addr Peer Tunnel Add State  UpDn Tm Attrb    Target Network
----- --------------- --------------- ----- -------- ----- -----------------
    1 10.2.255.24          10.200.0.1    UP 02:34:35     S      10.200.0.1/32


r70#
```

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SECTION 1.12: Tuning EIGRP on DMVPN and DMVPN-enabled Sites (3 Points)

QUESTION

Optimize the DMVPN operation according to these requirements:

1. Ensure that Branch 3 & Branch 4 can receive only a default route over EIGRP in DMVPN.
2. The default route origination must be done on r24 without the use of any static routes, redistribution, or route filtering.
3. It is not allowed to modify the configuration of r61 and r62 in Branch#3 to accomplish this task.
4. It is allowed to add commands to the configuration of r70 in Branch#4 to accomplish this task; none of the existing configuration on r70 may be removed to accomplish this task.

Configure sw601 and sw602 at Branch#3 according to these requirements:

1. Routers r61 and r62 must not send EIGRP queries to sw601 and sw602.
2. Switches sw601 and sw602 must allow advertising any current or future directly connected network to r61 and r62 after the network is added to EIGRP.
3. Switches sw601 and sw602 must continue to propagate the default route received from r61 and r62 to each other. To select the default route, use a prefix list with a “permit” type entry only.
4. Switches sw601 and sw602 must not propagate the default route back to r61 and r62.
5. If the prefix list that allows the propagation of selected EIGRP learned networks between sw601 and sw602 is modified in the future, the same set of networks must be disallowed from being advertised back to r61 and r62 automatically, without any additional configuration.

Solution

On r24

```
r24(config)#router eigrp ccie
r24(config-router)#address-family ipv4 unicast autonomous-system 65006
r24(config-router-af)#af-interface tunnel 0
r24(config-router-af-interface)#no passive-interface
r24(config-router-af-interface)#summary-address 0.0.0.0 0.0.0.0
r24(config-router-af-interface)#exit-af-interface
r24(config-router-af)#exit-address-family
r24(config-router)#exit
r24(config)#
```

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On r70

```
r70(config)#router eigrp ccie
r70(config-router)#address-family ipv4 unicast autonomous-system 65006
r70(config-router-af)#af-interface tunnel 0
r70(config-router-af-interface)#no passive-interface
r70(config-router-af-interface)#exit-af-interface
r70(config-router-af)#exit-address-family
r70(config-router)#exit
r70(config)#
```

On sw601

```
sw601(config)#ip prefix-list PERMIT seq 1 permit 0.0.0.0/0

sw601(config)#route-map PERMIT permit 1
sw601(config-route-map)#match ip address prefix-list PERMIT
sw601(config-route-map)#exit
sw601(config)#

sw601(config)#access-list 1 permit 10.6.0.0 0.0.255.255
sw601(config)#access-list 1 deny   any

sw601(config)#router eigrp ccie
sw601(config-router)# address-family ipv4 unicast autonomous-system 65006
sw601(config-router-af)#eigrp stub connected leak-map PERMIT
sw601(config-router-af)#topology base
sw601(config-router-af-topology)#distribute-list 1 out Ethernet0/1
sw601(config-router-af-topology)#distribute-list 1 out Ethernet0/2
sw601(config-router-af-topology)#exit-af-topology
sw601(config-router-af)#exit-address-family
sw601(config-router)#exit
sw601(config)#
```

On sw602

```
sw602(config)#ip prefix-list PERMIT seq 1 permit 0.0.0.0/0

sw602(config)#route-map PERMIT permit 1
sw602(config-route-map)#match ip address prefix-list PERMIT
sw602(config-route-map)#exit
sw602(config)#

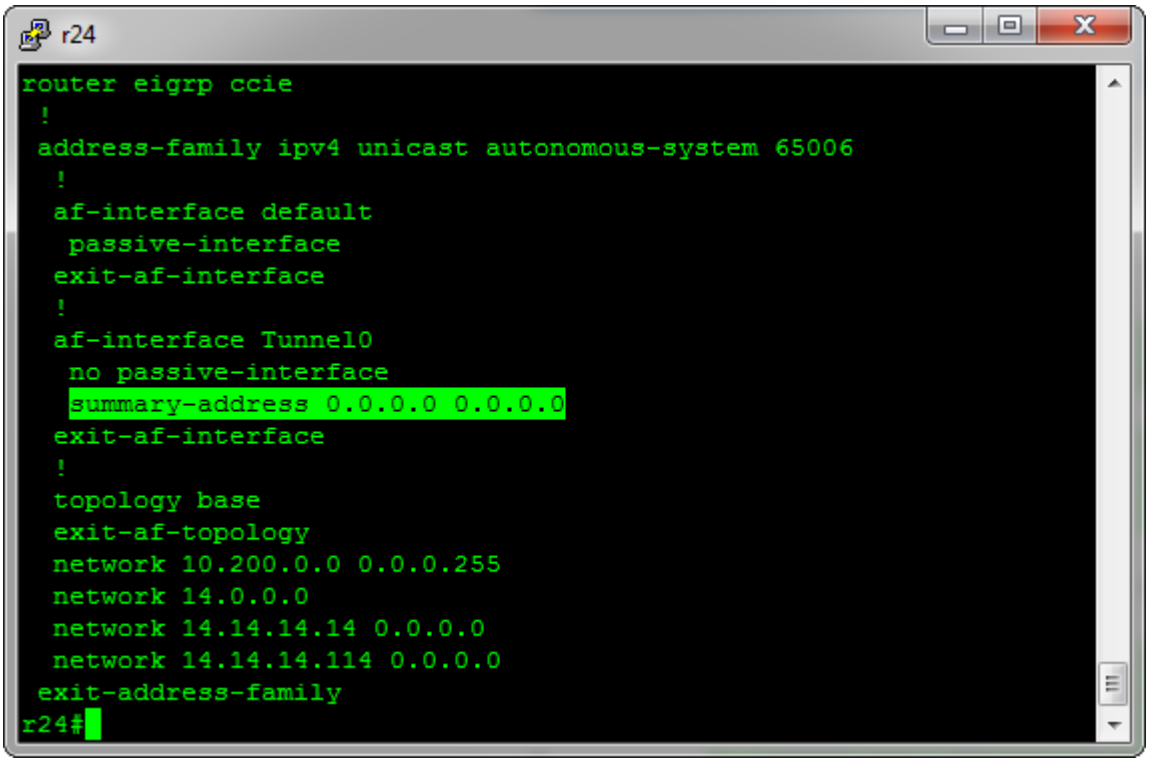
sw602(config)#access-list 1 permit 10.6.0.0 0.0.255.255
sw602(config)#access-list 1 deny   any
```

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```
sw602(config)#router eigrp ccie
sw602(config-router)#address-family ipv4 unicast autonomous-system 65006
sw602(config-router-af)#eigrp stub connected leak-map PERMIT
sw602(config-router-af)#topology base
sw602(config-router-af-topology)#distribute-list 1 out Ethernet0/1
sw602(config-router-af-topology)#distribute-list 1 out Ethernet0/2
sw602(config-router-af-topology)#exit-af-topology
sw602(config-router-af)#exit-address-family
sw602(config-router)#exit
sw602(config)#
```

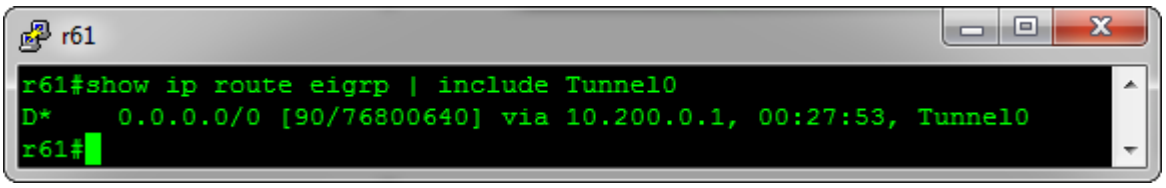
Verification

On r24

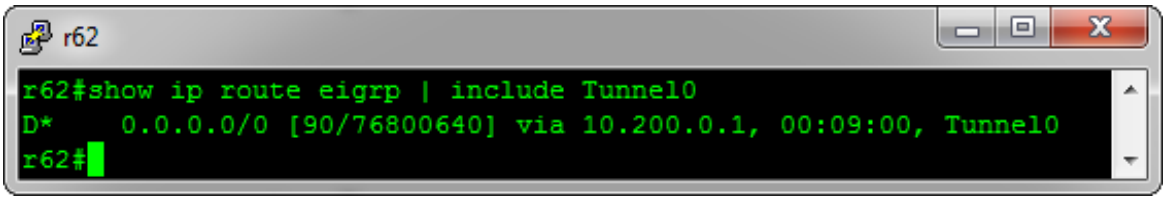


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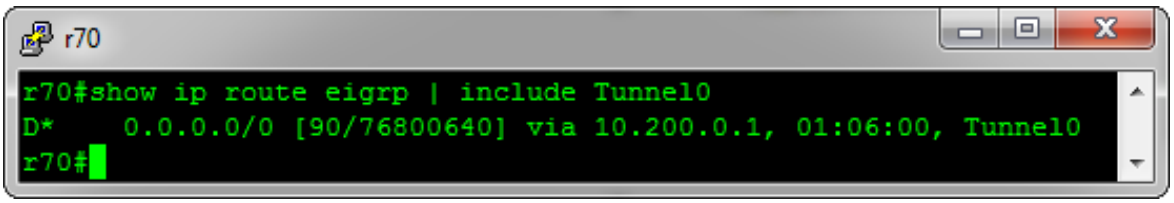
On r61



On r62

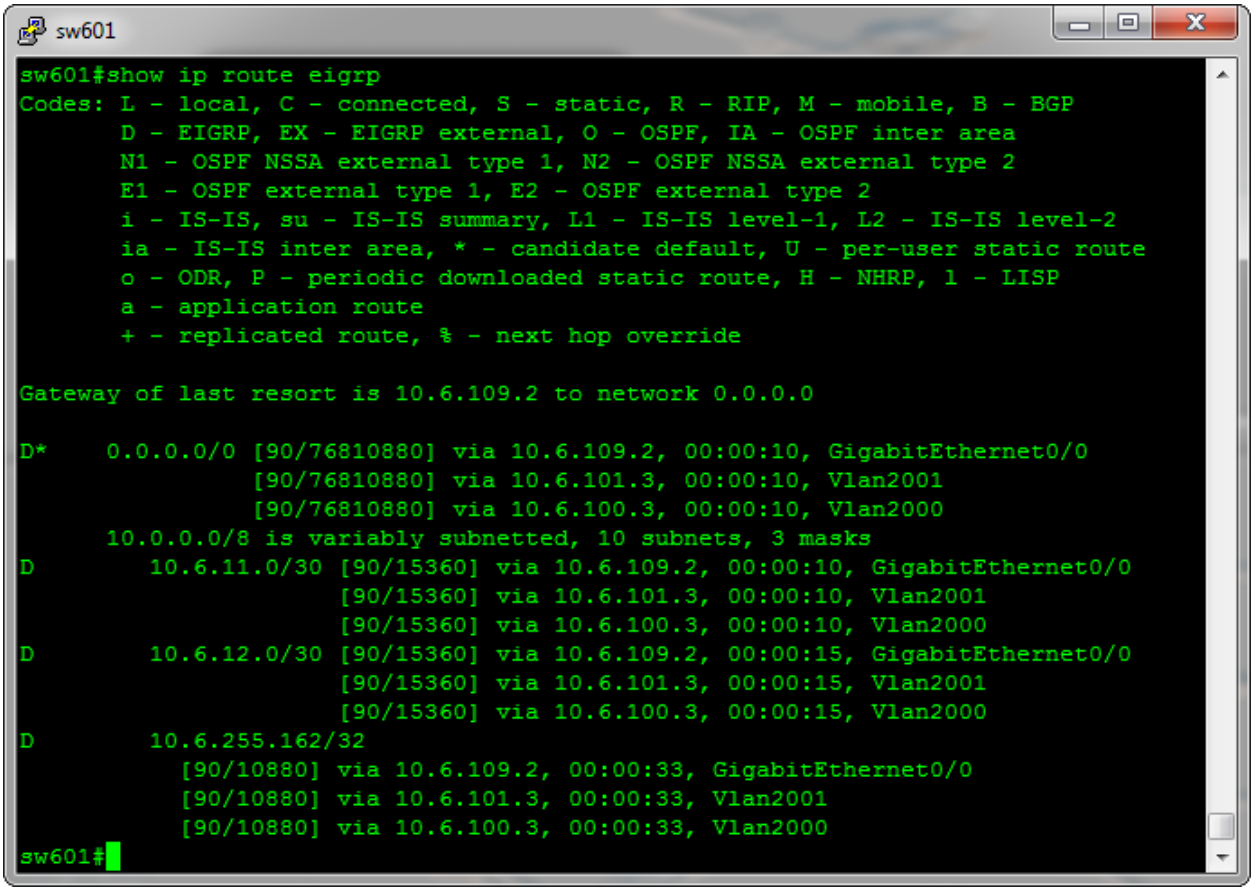


On r70



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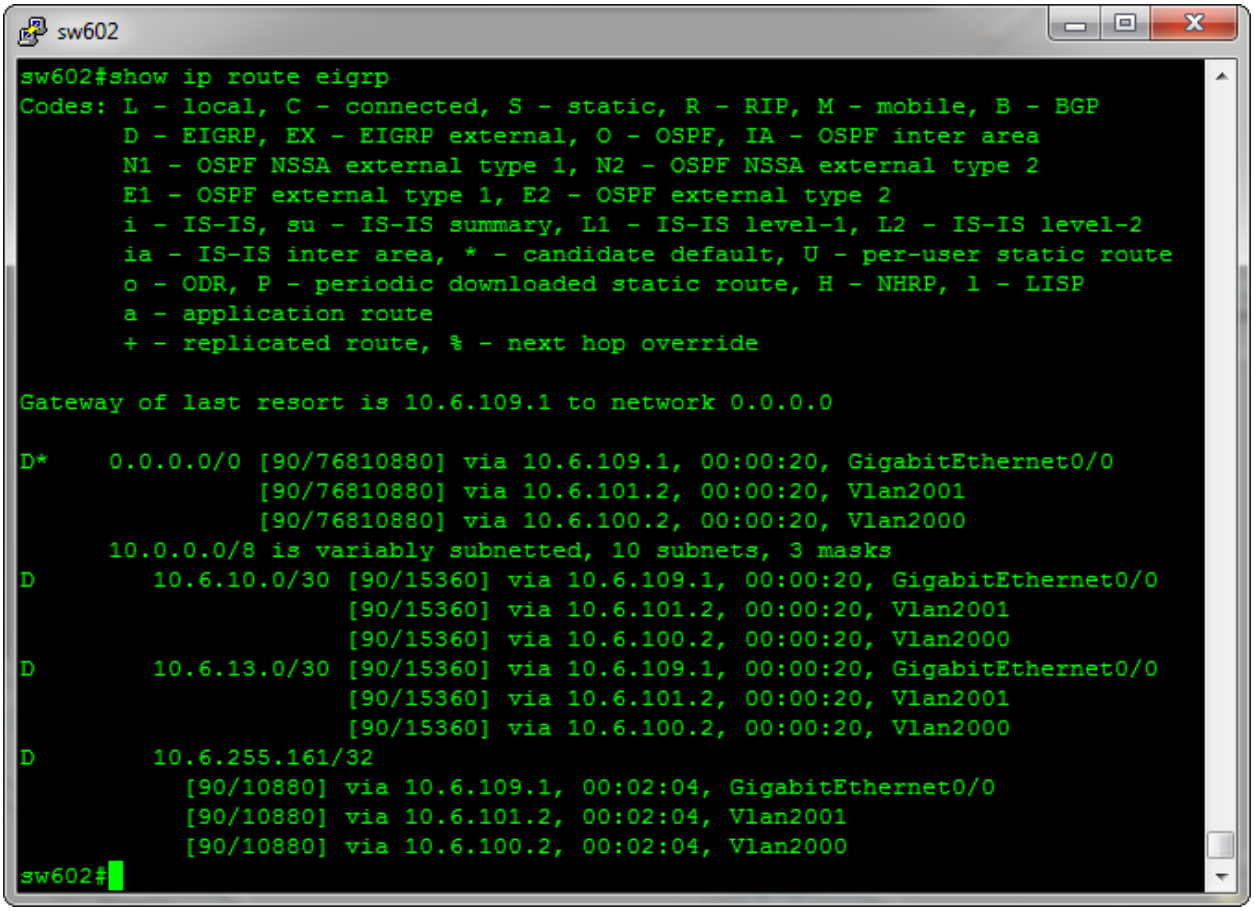
On sw601



The above output can be seen after shutdown of sw601 uplinks (gi0/1 & gi0/2)

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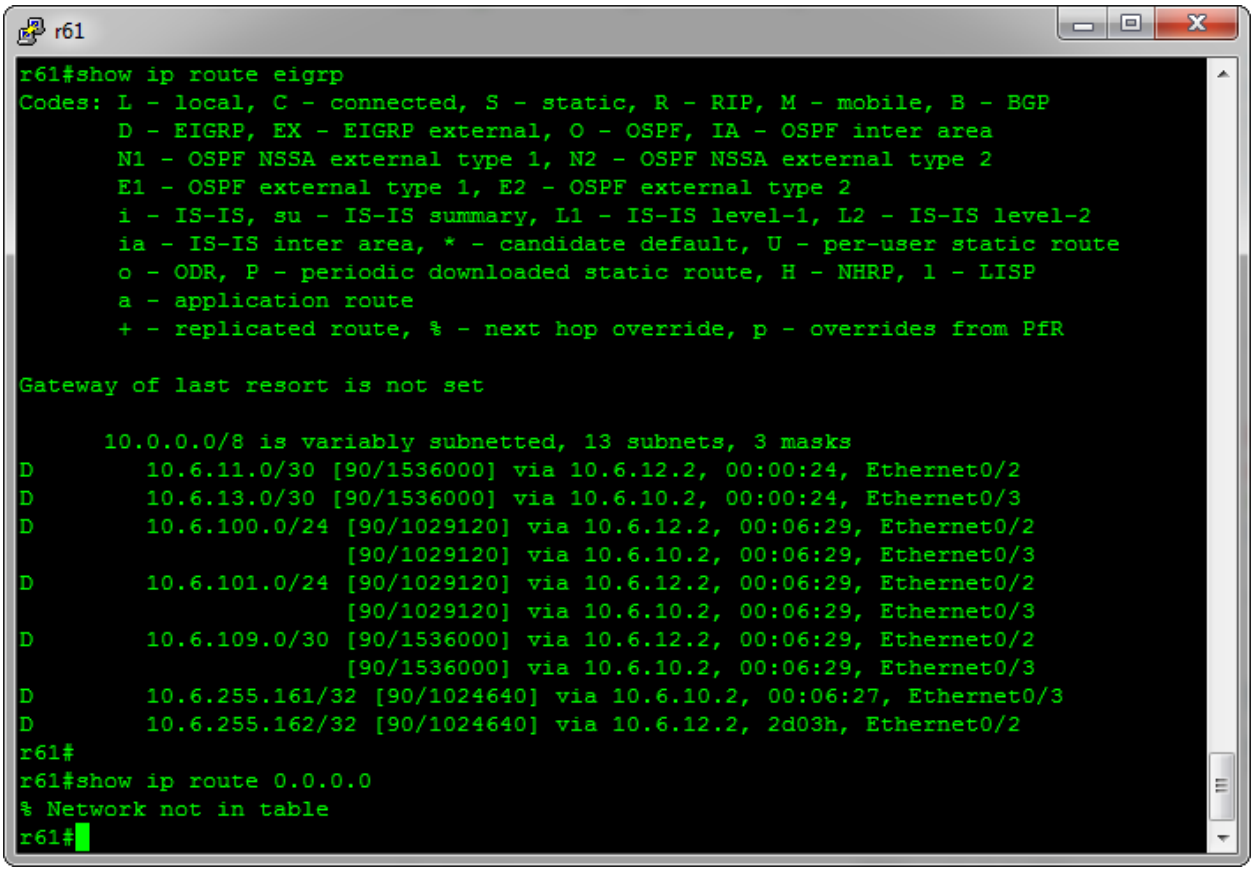
On sw602



The above output can be seen after shutdown of sw602 uplinks (gi0/1 & gi0/2)

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On r61



The above output can be seen after shutdown of r61 links (g0/0 & g0/1). Similary you can test on r62. Even if the prefix-list is modified in furtue, the same set of network will be denied.

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SECTION 1.13: IPv4 Networks on Legacy Branches (2 Points)

QUESTION

On sw211 in DC, complete the DHCP server configuration according to these requirements:

1. Create IPv4 DHCP pools named br3_v2000 and br3_v2001 for Branch#3 VLANs 2000 (10.6.100.0/24) and 2001 (10.6.101.0/24), respectively.
2. Create IPv4 DHCP pool named br_v1 for the subnet 10.7.1.0/24 on Branch#4
3. In each subnet assign addresses from .101 up to .254 inclusively and the appropriate gateway to clients.

On Branch#3 complete and correct the configuration on switches sw601, sw602 and sw610 to allow HSRP and DHCP Relay operation in VLANs 2000 and 2001 according to these requirements:

1. HSRP must implicitly use the vMAC address range of 0000.0c9f.f000 through 0000.0c9f.ffff
2. The group number must be 100 for VLAN 2000 and 101 for VLAN 2001
3. Sw601 must be Active gateway for VLAN 2000 with a priority of 110; the Active role ownership must be deterministic.
4. Sw602 must be Active gateway for VLAN 2001 with a priority of 110; the Active role ownership must be deterministic.
5. Each Active switch must track its uplick interface g0/1 and g0/2. If either of these interface goes down, the Active switch must allow the other switch to become Active. Howeve, it is not allowed for the tracking to modify the HSRP priority to accomplish this requirements.
6. Both sw601 and sw602 must be configured as DHCP relay agents in both VLANs 2000 and 2001, pointing toward the DHCP server 10.2.255.211 at sw211. However, at anytime, only the Active router in the particular VLAN should relay the DHCP messages.
7. Place host61 and host62 into VLANs 2000 and 2001 respectively and make sure they are assigned their correct IPv4 configuration.
8. It is not permitted to use any kind of scripting to complete this task.

On Branch#3 complete the configuration of the router r70 according to these requirements:

1. Assign IP address 10.7.1.1/24 to g0/2
2. Enable DHCP relay on this interface and point it to the DHCP server 10.2.255.211 at sw211
3. It is allowed to add one additional missing command to the r70 configuration to allow clients connected to g0/2 obtain their IPv4 configuration.
4. Make sure that host r71 and host r72 are assigned their correct IPv4 configuration

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Solution

On sw211

```
sw211(config)#ip dhcp pool br3_v2000
sw211(dhcp-config)#network 10.6.100.0 255.255.255.0
sw211(dhcp-config)#default-router 10.6.100.1
sw211(dhcp-config)#exit
sw211(config)#

sw211(config)#ip dhcp excluded-address 10.6.100.1 10.6.100.100
sw211(config)#ip dhcp excluded-address 10.6.100.255

sw211(config)#ip dhcp pool br3_v2001
sw211(dhcp-config)#network 10.6.101.0 255.255.255.0
sw211(dhcp-config)#default-router 10.6.101.1
sw211(dhcp-config)#exit
sw211(config)#

sw211(config)#ip dhcp excluded-address 10.6.101.1 10.6.101.100
sw211(config)#ip dhcp excluded-address 10.6.101.255

sw211(config)#ip dhcp pool br_v1
sw211(dhcp-config)#network 10.7.1.0 255.255.255.0
sw211(dhcp-config)#default-router 10.7.1.1
sw211(dhcp-config)#exit
sw211(config)#

sw211(config)#ip dhcp excluded-address 10.7.1.1 10.7.1.100
sw211(config)#ip dhcp excluded-address 10.7.1.255
sw211(config)#
```

On sw601

```
sw601(config)#interface vlan 2000
sw601(config-if)#standby 100 preempt
sw601(config-if)#ip helper-address 10.2.255.211
sw601(config-if)#exit
sw601(config)#

sw601(config)#interface vlan 2001
sw601(config-if)#standby version 2
sw601(config-if)#exit
sw601(config)#
```

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```
sw601(config)#track 1 interface gigabitEthernet 0/1 line-protocol
sw601(config-track)#exit
sw601(config)#

sw601(config)#track 2 interface gigabitEthernet 0/2 line-protocol
sw601(config-track)#exit
sw601(config)#


sw601(config)#interface vlan 2000
sw601(config-if)#standby 100 track 1 shutdown
sw601(config-if)#standby 100 track 2 shutdown
sw601(config-if)#exit
sw601(config)#
```

On sw602

```
sw602(config)#interface Vlan2000
sw602(config-if)#standby version 2
sw602(config-if)#exit
sw602(config)#

sw602(config)#interface Vlan2001
sw602(config-if)#standby version 2
sw602(config-if)#no standby 0 ip 10.6.101.1
sw602(config-if)#standby 101 ip 10.6.101.1
sw602(config-if)#standby 101 priority 110
sw602(config-if)#standby 101 preempt
sw602(config-if)#ip helper-address 10.2.255.211
sw602(config-if)#exit
sw602(config)#

sw602(config)#track 1 interface gigabitEthernet 0/1 line-protocol
sw602(config-track)#exit
sw602(config)#

sw602(config)#track 2 interface gigabitEthernet 0/2 line-protocol
sw602(config-track)#exit
sw602(config)#

sw602(config)#interface vlan 2001
sw602(config-if)#standby 101 track 1 shutdown
sw602(config-if)#standby 101 track 2 shutdown
sw602(config-if)#exit
sw602(config)#
```

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On sw610

```
sw610(config)#vlan 2000
sw610(config-vlan)#exit
sw610(config)#

sw610(config)#interface gigabitEthernet 0/0
sw610(config-if)#switchport mode access
sw610(config-if)#switchport access vlan 2000
sw610(config-if)#exit
sw610(config)#

sw610(config)#interface gigabitEthernet 0/1
sw610(config-if)#switchport mode access
sw610(config-if)#switchport access vlan 2001
sw610(config-if)#exit
sw610(config)#

sw610(config)#interface GigabitEthernet2/0
sw610(config-if)#no switchport trunk allowed vlan 1,2000
sw610(config-if)#switchport trunk allowed vlan 1,2000-2001
sw610(config-if)#exit
sw610(config)#

sw610(config)#interface GigabitEthernet2/1
sw610(config-if)#no switchport trunk allowed vlan 1,2000
sw610(config-if)#switchport trunk allowed vlan 1,2000-2001
sw610(config-if)#exit
sw610(config)#
```

On r70

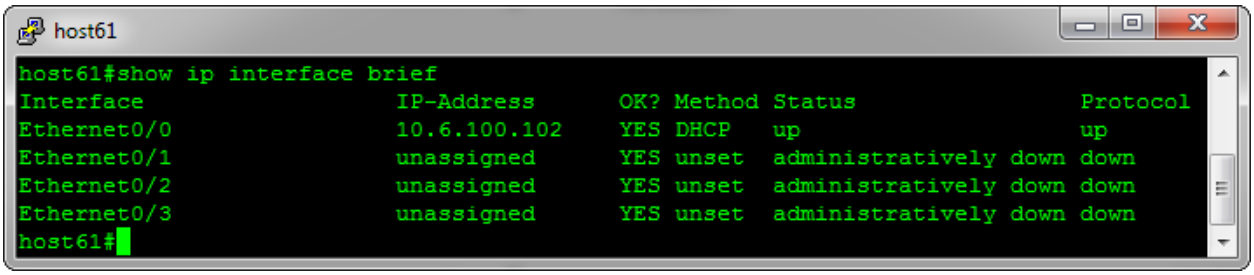
```
r70(config)#interface gigabitEthernet 0/2
r70(config-if)#ip address 10.7.1.1 255.255.255.0
r70(config-if)#ip helper-address 10.2.255.211
r70(config-if)#exit
r70(config)#

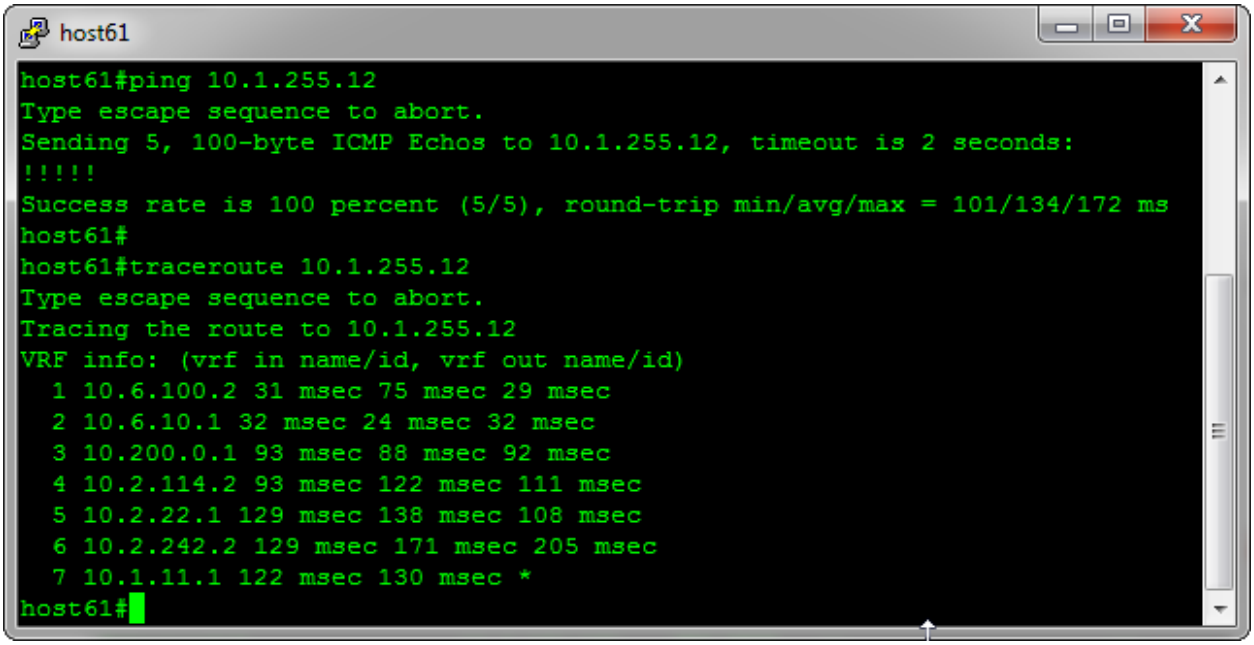
r70(config)#router eigrp ccie
r70(config-router)# address-family ipv4 unicast autonomous-system 65006
r70(config-router-af)#network 10.7.1.0 0.0.0.255
r70(config-router-af)#network 172.31.70.0 0.0.0.255
r70(config-router-af)#network 192.168.0.0 0.0.255.255
r70(config-router-af)#exit-address-family
r70(config-router)#exit
r70(config)#
```

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Verification

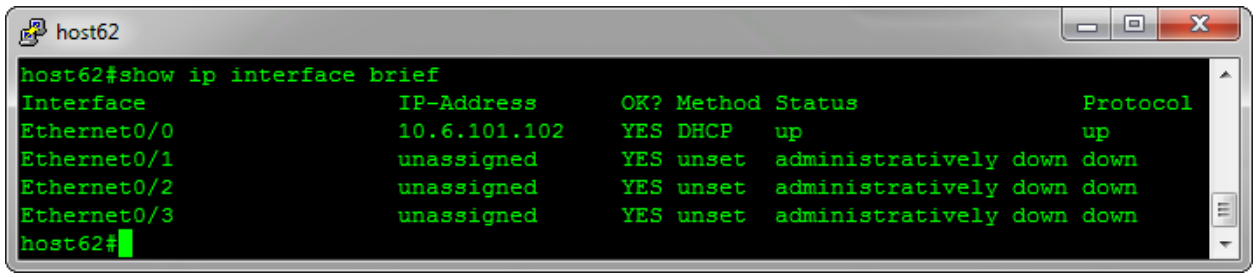
On host61

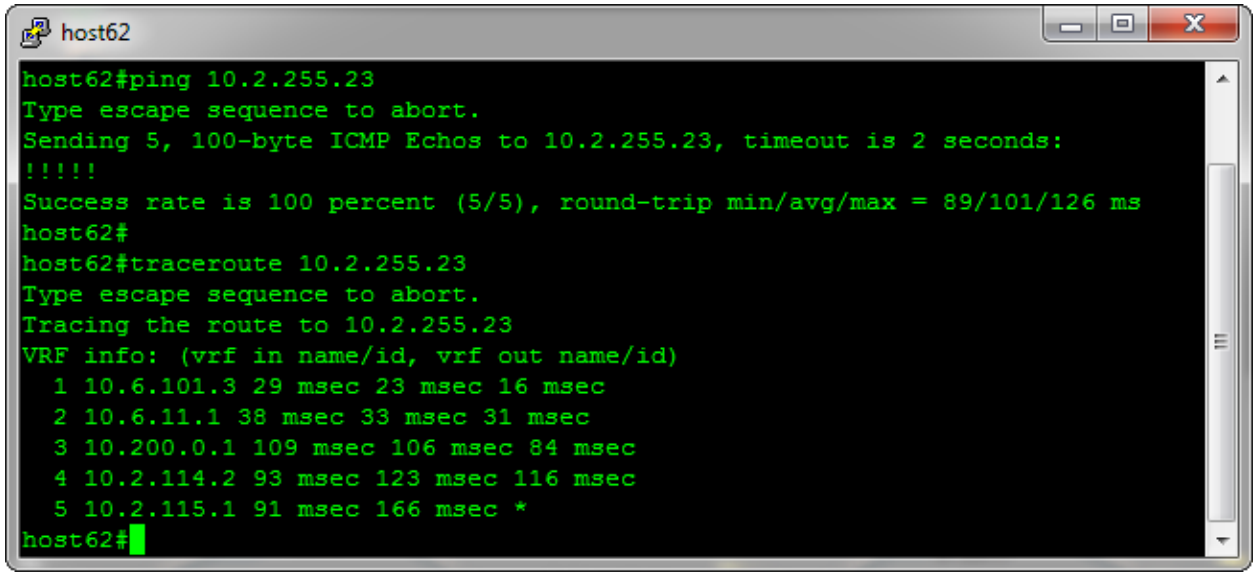




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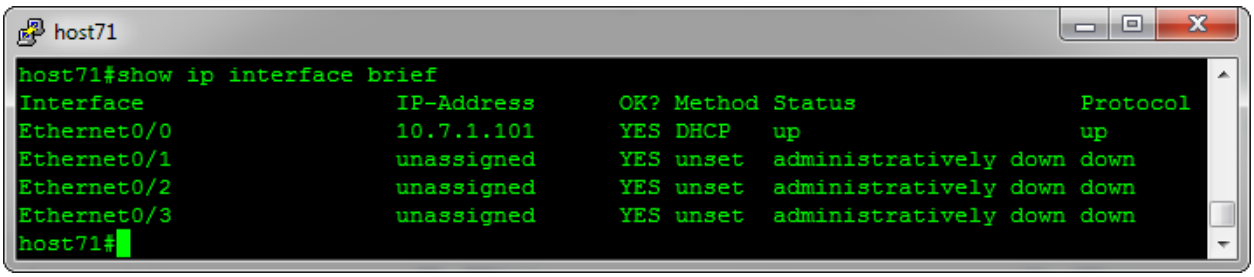
On host62

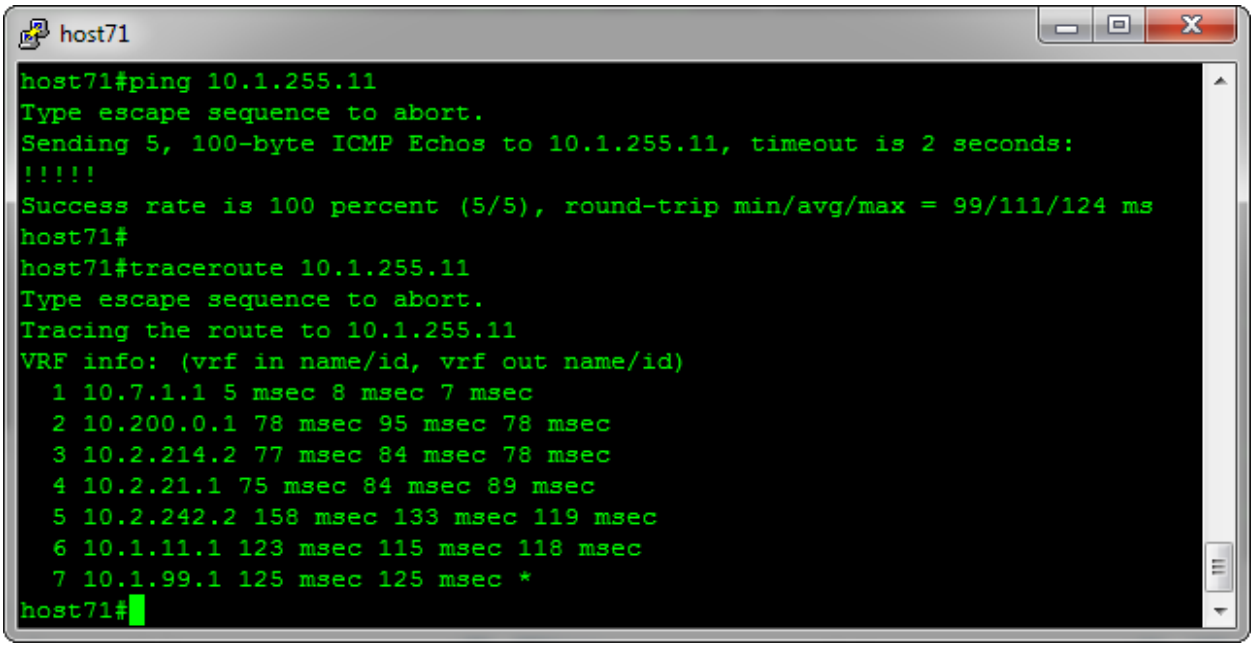




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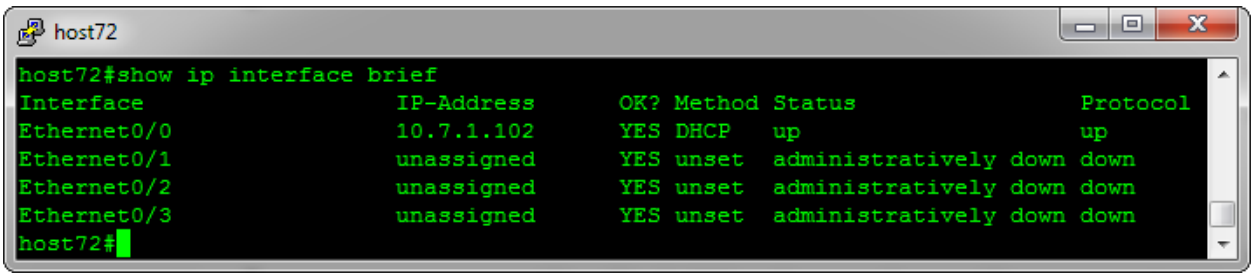
On host71

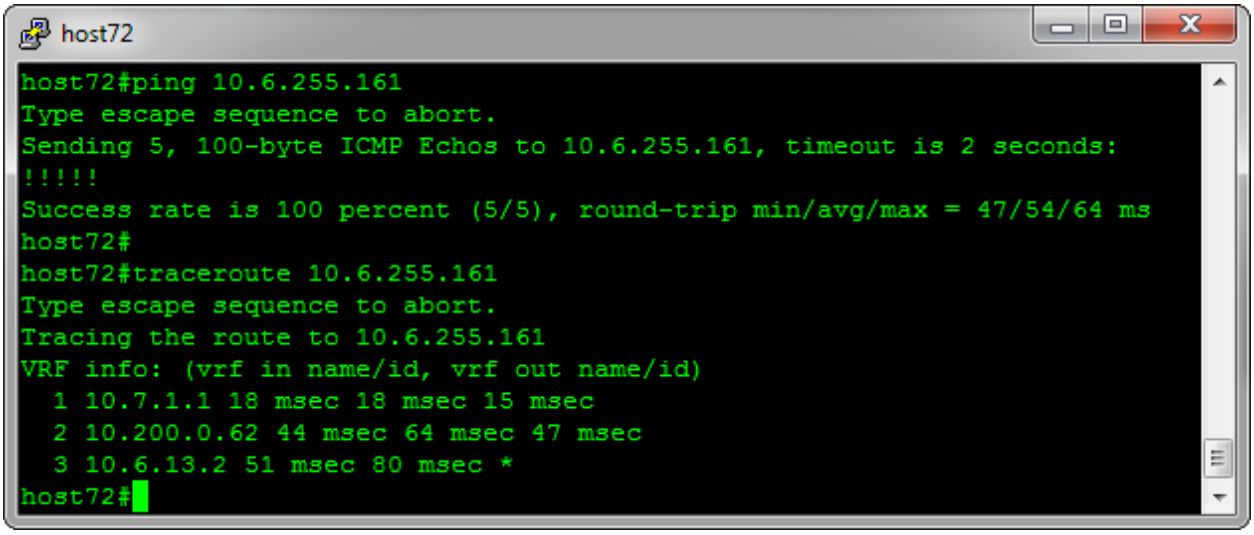




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On host72





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SECTION 1.14: Multicast in FADB2 (2 Points)

FABD2 is preparing to enable PIM Sparse Mode multicast routing in its network. As a part of validating the runbooks, FABD2 requires a sanity check to prevent inappropriate use of multicast related configuration commands on different router types.

- First Hop Routers – routers where multicast sources are connected.
- Last Hop Routers – routers where multicast receivers (subscribers) are connected.
- Intermediary Hop Routers – routers on the path between First Hop and Last Hop routers.

In the table below, for each configuration command, select all router types where the use of the command is appropriate. (Select all that apply)

	Router Type		
Command	**Last Hop Router**	**First Hop Router**	**Intermediary Hop Router**
ip igmp spt-threshold			
Ip igmp version			
Ip pim register-source			
Ip pim sparse-mode			
Ip pim rp-address			

Solution

	Router Type		
Command	**Last Hop Router**	**First Hop Router**	**Intermediary Hop Router**
ip igmp spt-threshold	✓		
Ip igmp version	✓		
Ip pim register-source		✓	
Ip pim sparse-mode	✓	✓	✓
Ip pim rp-address	✓	✓	✓

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SECTION 1.15: Extending connectivity to IaaS (3 Points)

Extend the IPv6 connectivity from HQ through the SP into the giosk VRF on the IaaS site according to these requirements.

1. Set up global IPv6 addressing on the link between r11 and r3.
   - On r11 assign 2001:2710:311::2/64 to g0/0
   - On r3 assign 2001:2710:311::1/64 to g1
2. Enable the existing IPv4 BGP session between r11 and r3 to also advertise IPv6 prefixes. Do not configure a standalone IPv6 BGP session between these two routers.
3. Perform bidirectional route redistribution between the IPv6 EIGRP and BGP processes on r11
4. Ensure that all current and future IPv6 prefixes advertised between r11 and r3 will be installed into the RIB of these routers with the next hop address set to the proper global unicast address on their interconnection. Any policy that accomplishes this requirement must be applied in the inbound direction
5. The giosk VRF on r4 that extends the IPv6 connectivity from r4 to r30 on the IaaS site is a separate VRF independent of fabd2 VRF. Any route leaking from fabd2 VRF into giosk VRF must be done on a per-site basis and only for this FABD2 sites that need connectivity with the IaaS site.
6. By configuring r3 and r4 only, ensure that the HQ FABD2 site will have mutual visibility with the IaaS site while preventing
   - any other FABD2 site from possibly learning about the routes on the IaaS site
   - the IaaS site from possibly learning about the routes on any other FABD2 site
7. Use the minimum number of commands necessary to accomplish this requirements. Do not remove any existing configuration. If necessary, you are allowed to use an additional route-target with the value of 10000:3681
8. Verify that host11 and host12 can ping 2001:db8:14::1 located at the IaaS site. It is permitted to modify one existing configuration command on one of the SP routers to meet this requirement

Solution

On r11

```
r11(config)#interface Ethernet0/0
r11(config-if)# ipv6 address 2001:2710:311::2/64
r11(config-if)#exit
r11(config)#

r11(config)#router bgp 65001
r11(config-router)#neighbor 2001:2710:311::1 remote-as 10000
r11(config-router)#address-family ipv4
```

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```
r11(config-router-af)#neighbor 2001:2710:311::1 activate
r11(config-router-af)#redistribute eigrp 65001
r11(config-router-af)#exit-address-family
r11(config-router)#exit
r11(config)#

r11(config)#router eigrp ccie
r11(config-router)# address-family ipv6 unicast autonomous-system 65001
r11(config-router-af)# topology base
r11(config-router-af-topology)#redistribute bgp 65001
r11(config-router-af-topology)#exit-af-topology
r11(config-router-af)#exit-address-family
r11(config-router)#exit
r11(config)#
```

On r3

```
r3(config)#interface Ethernet0/0
r3(config-if)# ipv6 address 2001:2710:311::1/64
r3(config-if)#exit
r3(config)#

r3(config)#router bgp 10000
r3(config-router)#address-family ipv4 vrf fabd2
r3(config-router-af)#redistribute connected
r3(config-router-af)#neighbor 2001:2710:311::2 remote-as 65001
r3(config-router-af)#neighbor 2001:2710:311::2 activate
r3(config-router-af)#exit-address-family
r3(config-router)#exit
r3(config)#
```

Solution omitted……..

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SECTION 1.16: Enabling Internet Access for FADB2 (1 Points)

QUESTION

Enable highly available internet access for the FABD2 company network according to these requirements:

1. On routers r12, r23 and r24, bring up IPv4 BGP peerings with the ISP. Make sure that a default route is revceived over these peerings.
2. On routers r12 and r23 inject default route into OSPF if it is present in the routing table from a different routing source than the OSPFv2 process 1. On each router, this requirement must be completed using minimum possible number of commands.
3. On router r24 inject a default route into OSPF if and only if it is learned from ISP over BGP. To accomplish this requirement, it is allowed to use a route-map that references both a prefix-list and a tag. This requirement must be completed using minimum possible number of commands.
4. Router r12 may be used as an internet exit for the FABD2 company network only if neither r23 nor r24 are advertising a default route in OSPF. This requirement must be accomplish exclusively in “router ospf” mode on router r12 without changing the default parameters on routers r23 and r24.
5. On routers r12, r23 and r24 configure PAT and translate the entire FABD2 internal network 10.0.0.0/8 to the router address on the link towards the ISP. Create a standard ACL named NAT for this purpose. Do not use NAT pools.
6. Ensure that the internet connectivity of the FABD2 company network makes use of high availability provided by r12, r23 and r24

Solution

On r12

```
r12(config)#router bgp 65001
r12(config-router)#neighbor 200.99.12.1 remote-as 19999
r12(config-router)#exit
r12(config)#

r12(config)#router ospf 1
r12(config-router)#default-information originate metric 50000
r12(config-router)#exit
r12(config)#

r12(config)#interface gigabitEthernet 0/0
r12(config-if)#ip nat outside
r12(config-if)#exit
```

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```
r12(config)#

r12(config)#interface gigabitEthernet 0/1
r12(config-if)#ip nat inside
r12(config-if)#exit
r12(config)#

r12(config)#interface gigabitEthernet 0/2
r12(config-if)#ip nat inside
r12(config-if)#exit
r12(config)#

r12(config)#interface gigabitEthernet 0/3
r12(config-if)#ip nat inside
r12(config-if)#exit
r12(config)#

r12(config)#ip access-list standard NAT
r12(config-std-nacl)#permit 10.0.0.0 0.255.255.255
r12(config-std-nacl)#exit
r12(config)#

r12(config)# ip nat inside source list NAT interface gigabitEthernet 0/0 overload
```

On r23

```
r23(config)#router bgp 65002
r23(config-router)#neighbor 200.99.23.1 remote-as 19999
r23(config-router)#exit
r23(config)#

r23(config)#router ospf 1
r23(config-router)#default-information originate
r23(config-router)#exit
r23(config)#

r23(config)#interface gigabitEthernet 1
r23(config-if)#ip nat outside
r23(config-if)#exit
r23(config)#

r23(config)#interface gigabitEthernet 2
r23(config-if)#ip nat inside
r23(config-if)#exit
r23(config)#
```

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```
r23(config)#interface gigabitEthernet 3
r23(config-if)#ip nat inside
r23(config-if)#exit
r23(config)#

r23(config)#ip access-list standard NAT
r23(config-std-nacl)#permit 10.0.0.0 0.255.255.255
r23(config-std-nacl)#exit
r23(config)#

r23(config)#ip nat inside source list NAT interface gigabitEthernet 0/0 overload
```

On r24

```
r24(config)#router bgp 65002
r24(config-router)#neighbor 200.99.24.1 remote-as 19999
r24(config-router)#distance 4 200.99.24.1 0.0.0.0
r24(config-router)#exit
r24(config)#

r24(config)#ip prefix-list INTERNET seq 1 permit 0.0.0.0/0

r24(config)#route-map INTRENET permit 1
r24(config-route-map)#match ip address prefix-list INTERNET
r24(config-route-map)#match tag 19999
r24(config-route-map)#exit
r24(config)#

r24(config)#route-map INTRENET permit 2
r24(config-route-map)#exit
r24(config)#

r24(config)#router ospf 1
r24(config-router)#default-information originate route-map INTERNET
r24(config-router)#exit
r24(config)#

r24(config)#interface gigabitEthernet 1
r24(config-if)#ip nat outside
r24(config-if)#exit
r24(config)#

r24(config)#interface gigabitEthernet 2
r24(config-if)#ip nat inside
r24(config-if)#exit
r24(config)#
```

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```
r24(config)#interface gigabitEthernet 3
r24(config-if)#ip nat inside
r24(config-if)#exit
r24(config)#

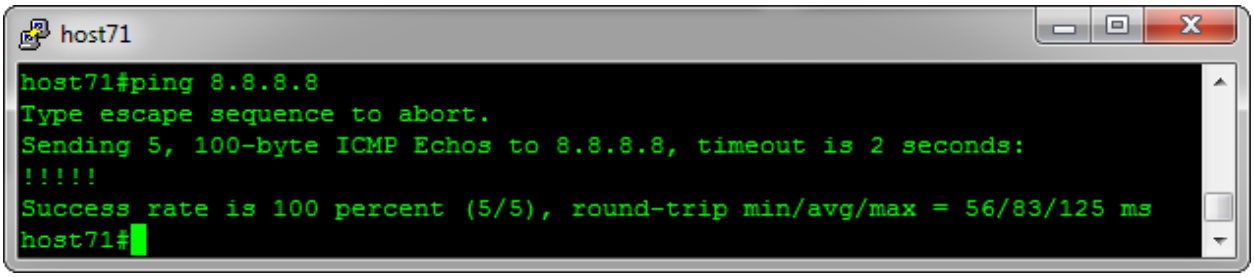
r24(config)#interface tunnel 0
r24(config-if)#ip nat inside
r24(config-if)#exit
r24(config)#

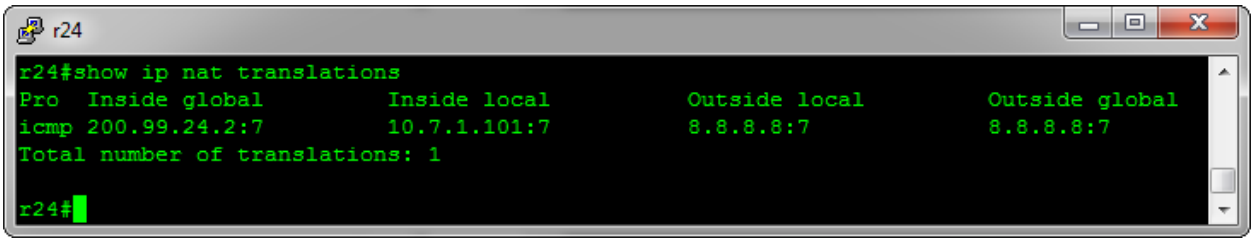
r24(config)#ip access-list standard NAT
r24(config-std-nacl)#permit 10.0.0.0 0.255.255.255
r24(config-std-nacl)#exit
r24(config)#

r24(config)#ip nat inside source list NAT interface gigabitEthernet 0/0 overload
```

Verification

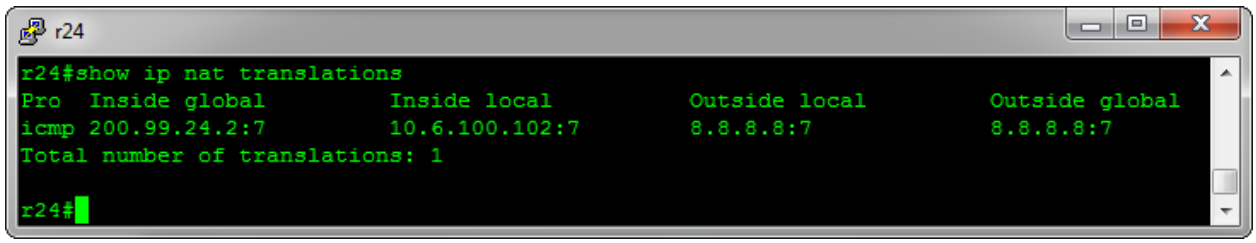
On host71 & r24



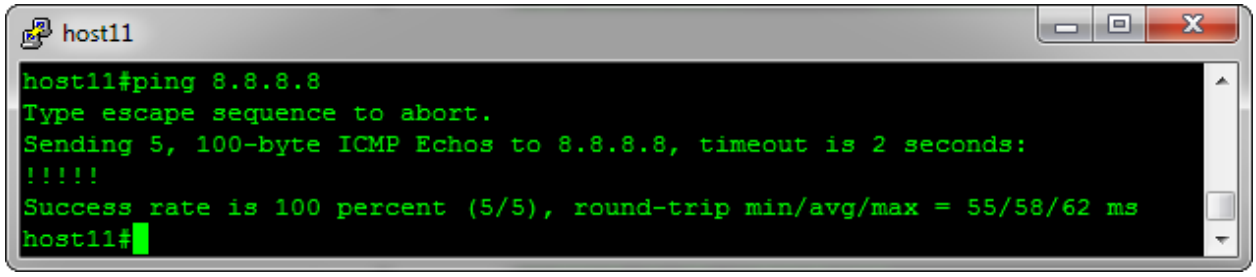


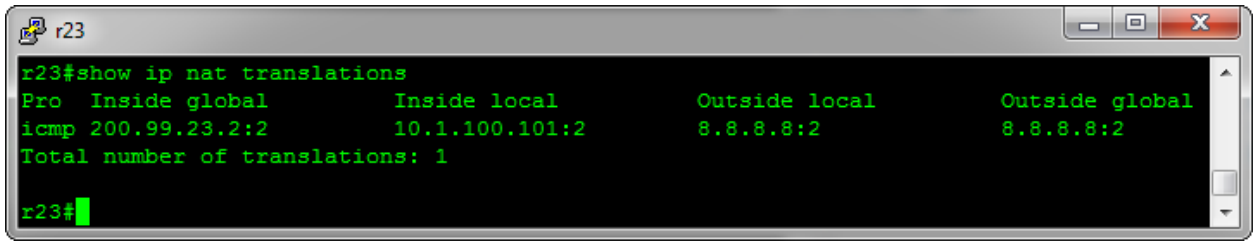
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On host61 & 24

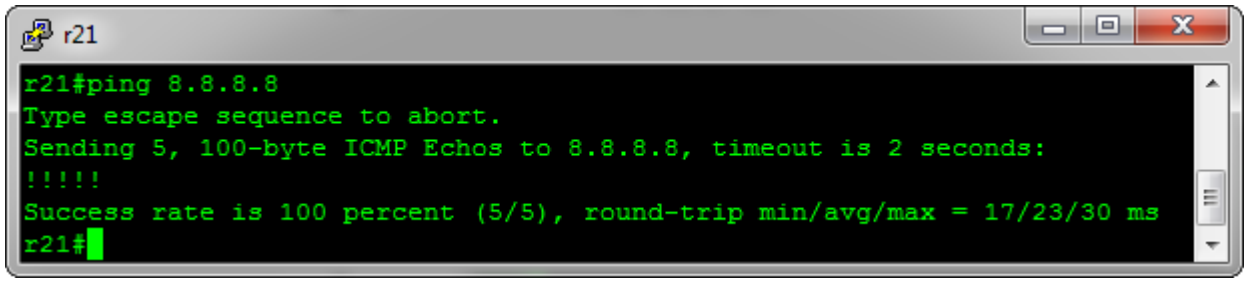


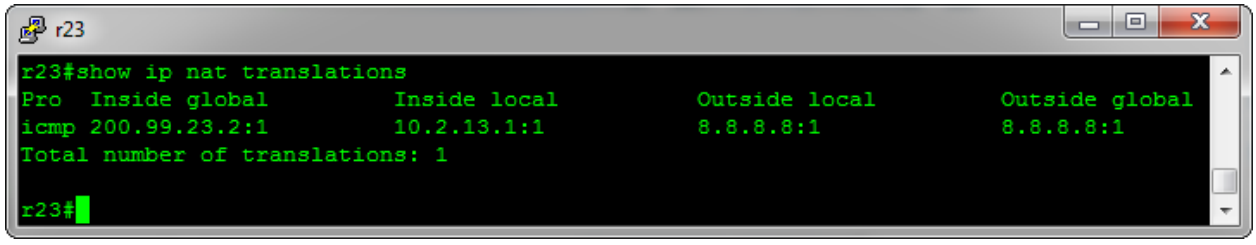
On host11 & r23





On r21 & r23





CCIE Lab Center

Section 2: Implementing Proof of Concept SDX Branches

SECTION 2.1: Correcting the IP Adresses of Managed Switches in DNA Center (3 Points)

After Cisco DNA Center first achieves IP connectivity with the managed switches in Branch #1 & #2, it will place them into Maintenance mode due to their serial number being different from the DNA Center remembers. In addition, their management IP addresses in DNA Center will be automatically changed by appending them with the "dummy.com" string. As a result, after an initial contact, DNA Center will lose connectivity with the switches unless their management IP addresses are corrected in the DNA Center settings.

Correct the IP addresses of managed switches in the DNA Center according to the following requirements:

1. Use the host, such as host11 to access the DNA Center GUI website at https://203.0.113.11 URL
2. Execute the Provision – Devices – Inventory – Global – Actions – Inventory – Resync Device action in DNA Center on all switches before proceeding further.
3. DNA Center API reference and sandbox is available at https://203.0.113.11/dna/apitester URL
4. The network-device/update-maintenance-device-ip-address API call description and sandbox are available in the inventory section of the API reference.
5. Use the network-device/update-maintenance-device-ip-address API call to correct the IP addresses of the switches in Branch #1 and #2 appended text.

Note: These IP addresses cannot be changed from DNA Center GUI directly because they will become automatically invalidated again. This a built-in DNA Center behavior.

Solution

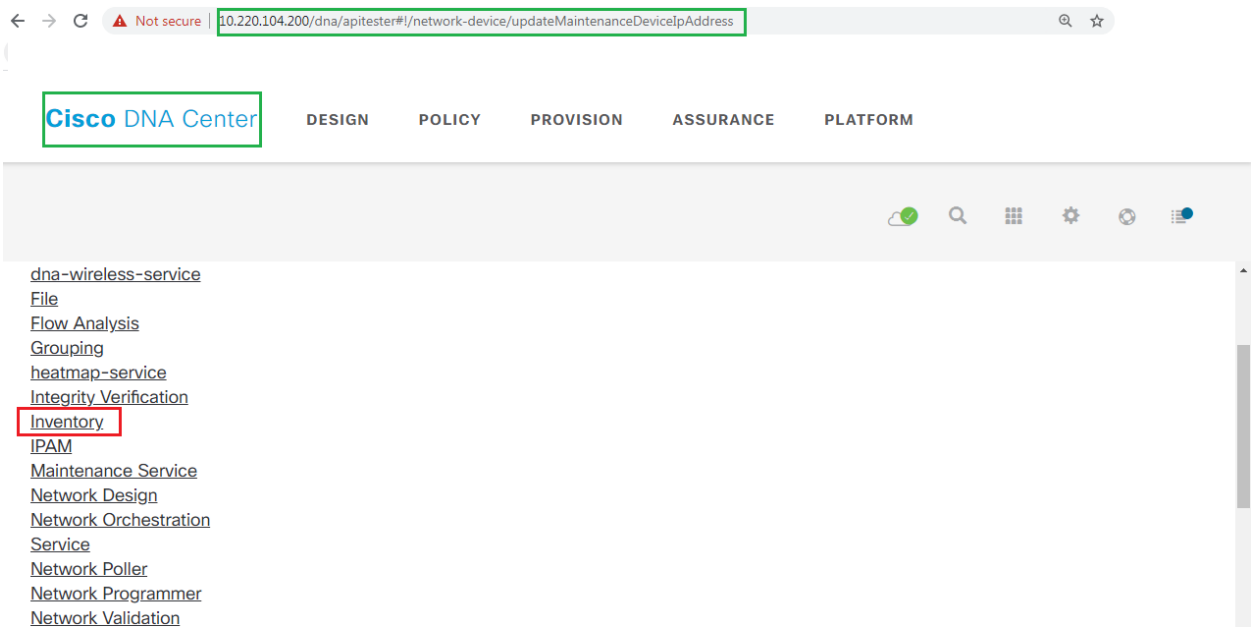
On DNAC (GUI access via host11)

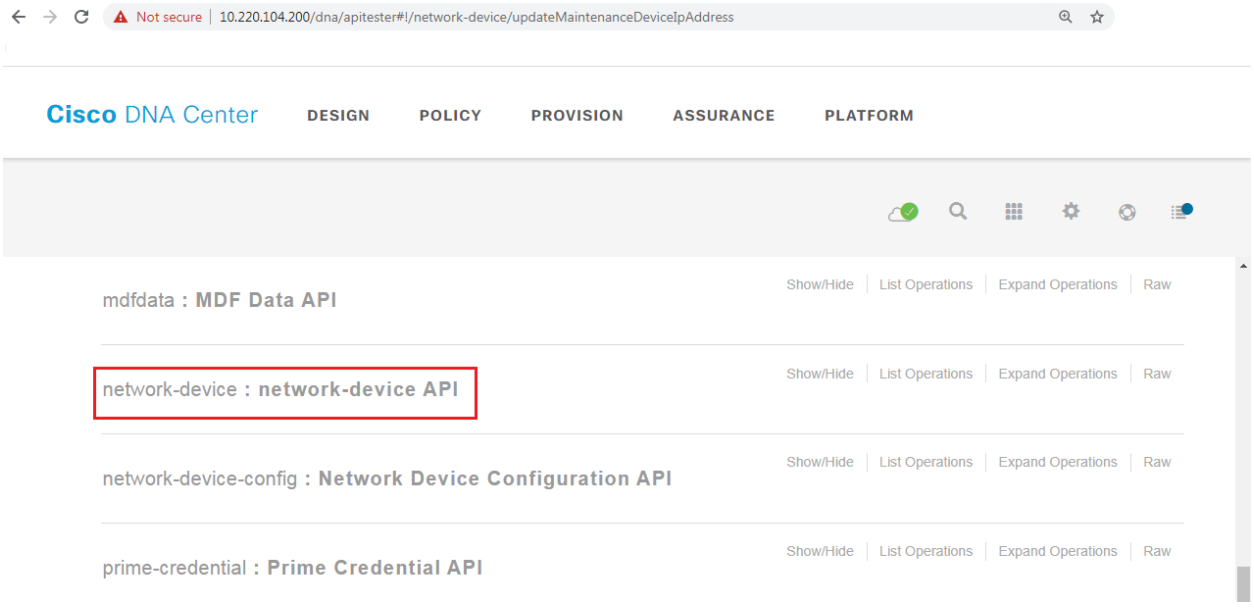
First Step: Changing Management ip address of sw400 via DNAC API Call

1. Navigate > Provision > Devices > Inventory > Global > Actions > Inventory > Resync Device
2. Use this link https://203.0.113.11/dna/apitester
3. Select Inventory
4. Select network-device : network-device API
5. Select network-device/update-maintenance-device-ip-address API call

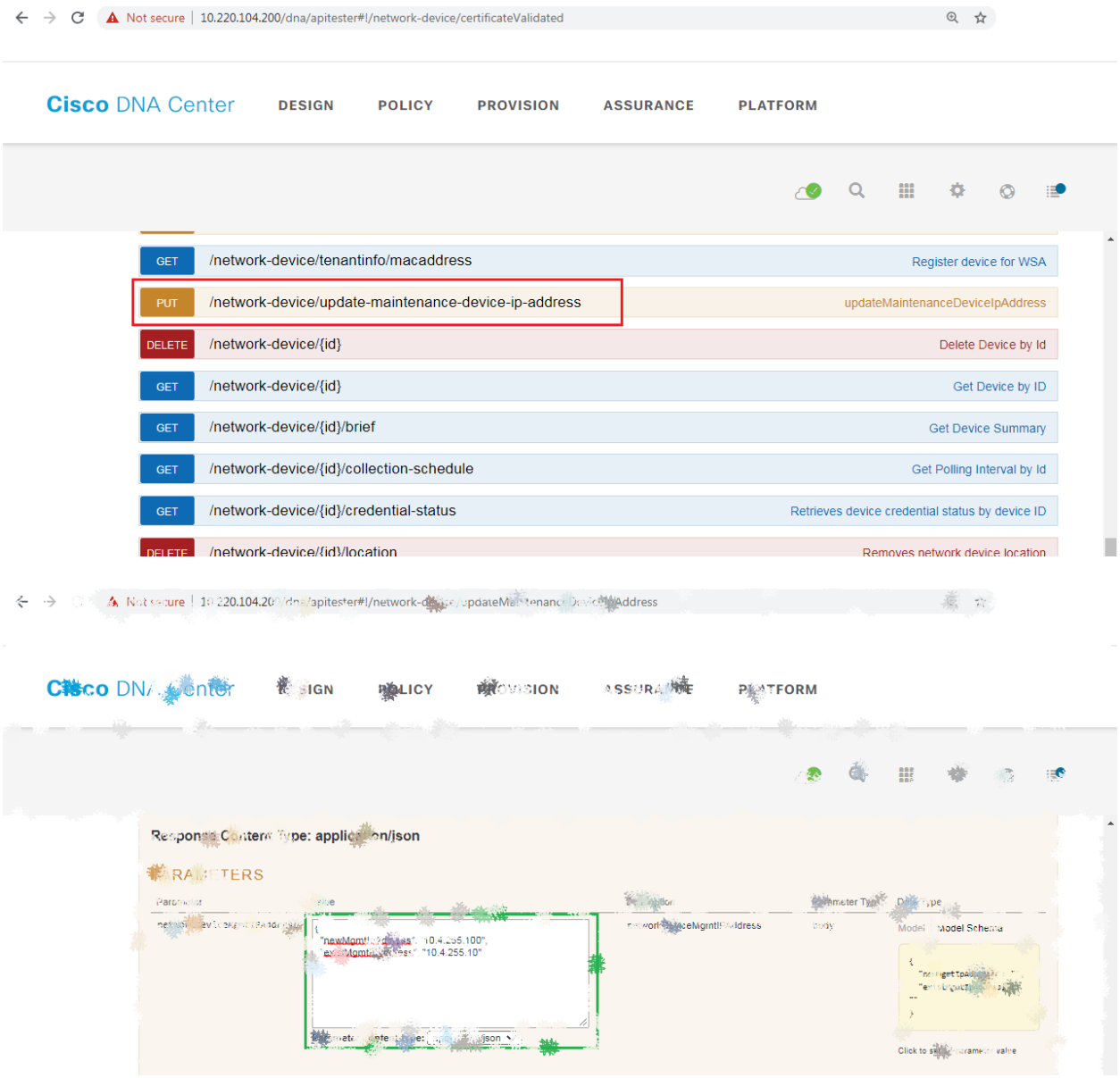
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6. In the box input the details as shown in screenshot
7. Click “Try it out!”
8. All Done



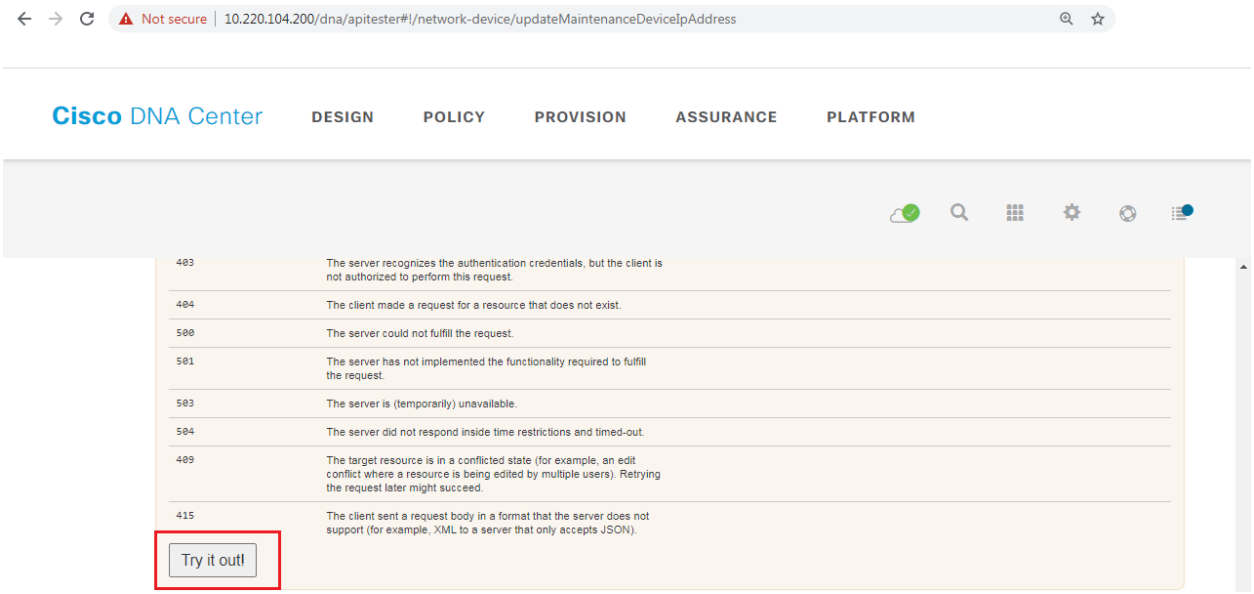


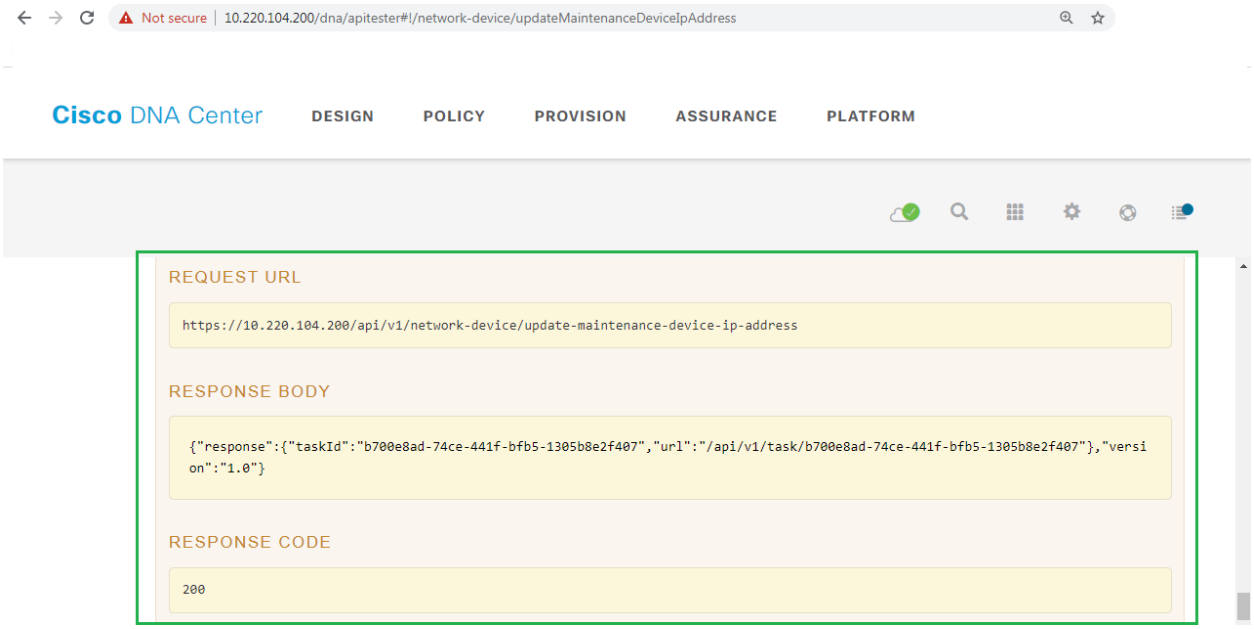
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The Above image is purposely blurred*

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SECTION 2.2: Completing VN Configuration in DNA Center

(3 Points)

Using the DNA Center GUI, perform configuration tasks according to these requirements.

1. Add new Virtual Network name IoT for the internet-of-things network on the Branches #1 and #2.
2. Create new addresses pools for the IoT VN named Branch1-ForIoT and Branch2-ForIoT on the global level, and Branch1-IoT Branch1-IoT on the branch level.
3. For Branch #1 IoT VN, allocate the subnet 10.4.198.0/24 and gateway IP address 10.4.198.1
4. For Branch #2 IoT VN, allocate the subnet 10.5.198.0/24 and gateway IP address 10.5.198.1
5. Associate the Branch1-IoT and Branch2-IoT pools with IoT VN on the respective branches.
6. Complete the configuration of the address pools for the Guest VN in the DNA Center so that Branch #1 and Branch #2 can accommodate guest connections, if a new address pool needs to be created and an address range allocated to it, follow the established addressing plan.
7. For all address pools, use the DHCP server 10.2.255.211 to allocate addresses to clients.

On sw211 complete the DHCP server configuration according to these requirements.

1. Create four new DHCP pools for the IoT and Employees VNs on the respective branches.
   - Pool named br1_iot for Branch #1 IoT VN
   - Pool named br1_emp for Branch #1 Employees VN
   - Pool named br2_iot for Branch #2 IoT VN
   - Pool named br2_emp for Branch #2 Employees VN
2. In each subnet, assign addresses from .101 to .254 inclusively, and the appropriate gateway to clients.

Solution

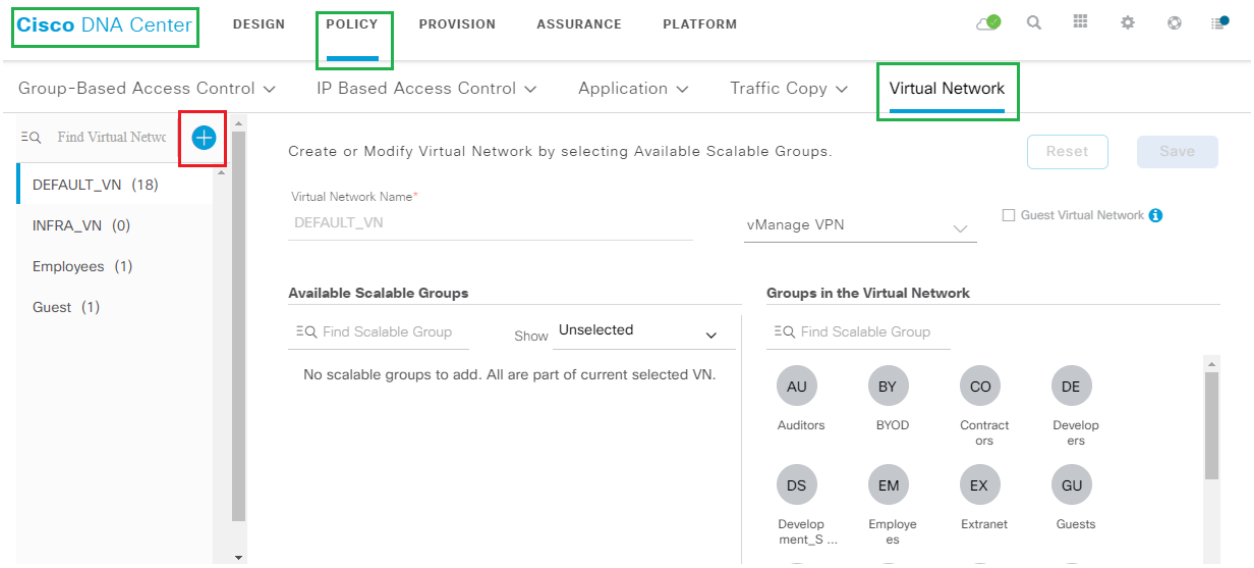
On DNAC (GUI access via host11)

First Step: Creating Virtual Network IoT

1. Navigate: DNA > Policy > Virtual Network
2. On Left Side Click “Add” Symbol
3. Input the Name as “IoT”

Solution omitted……

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Next Step: Creating IP Address Pools for IoT on Global Level

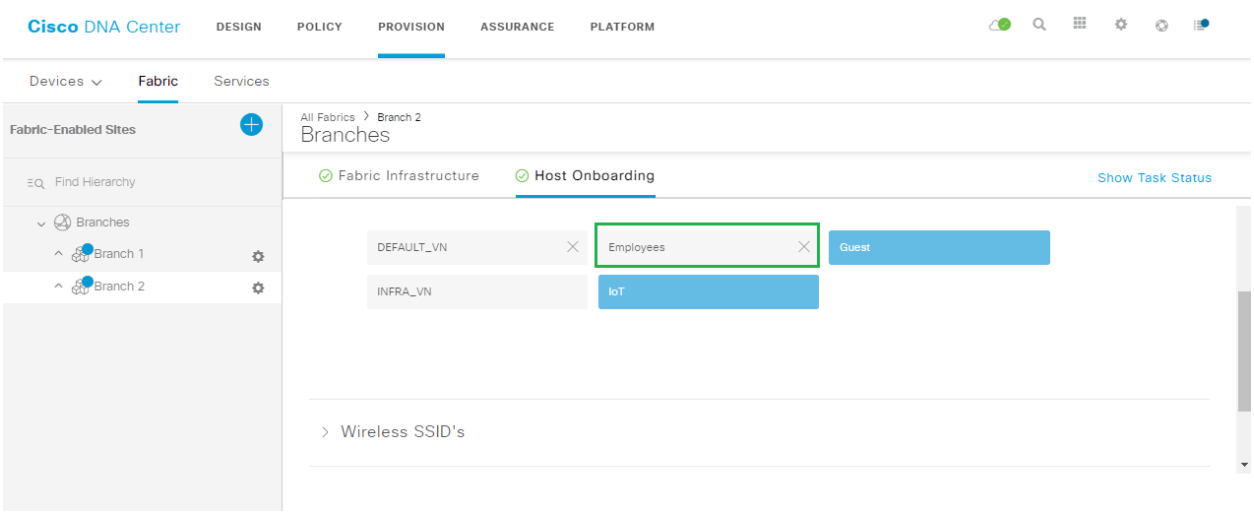
1. Navigate: DNA > Design > Network Setting
2. From the drop-down list on Network Setting: select IP Address Pools
3. Select Global on left side.
4. To add a new ip pool address click “Add”
5. Input the details as
    - IP Pool Name: Branch1-ForIoT

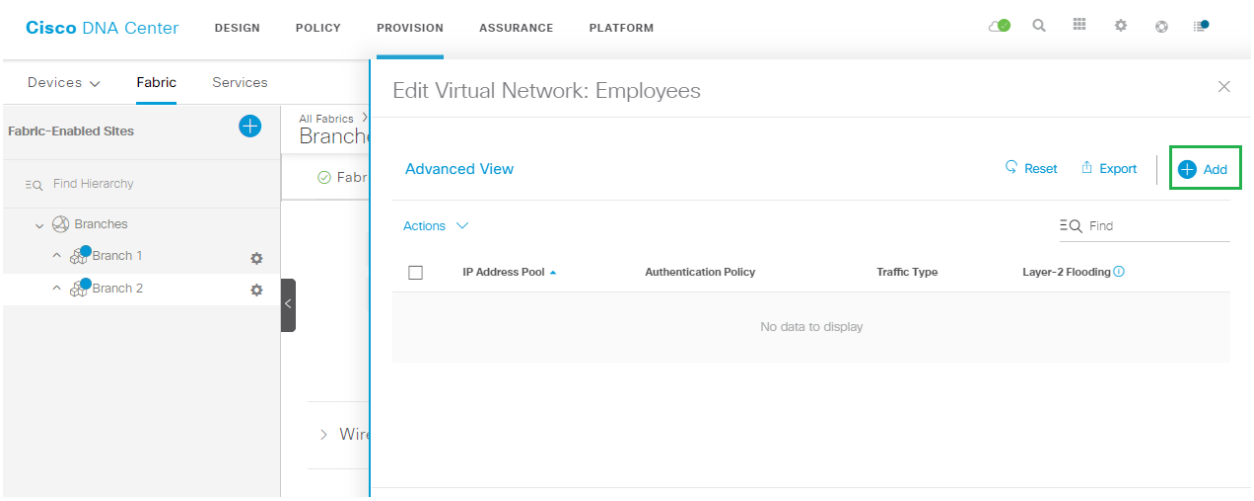
Solution omitted……

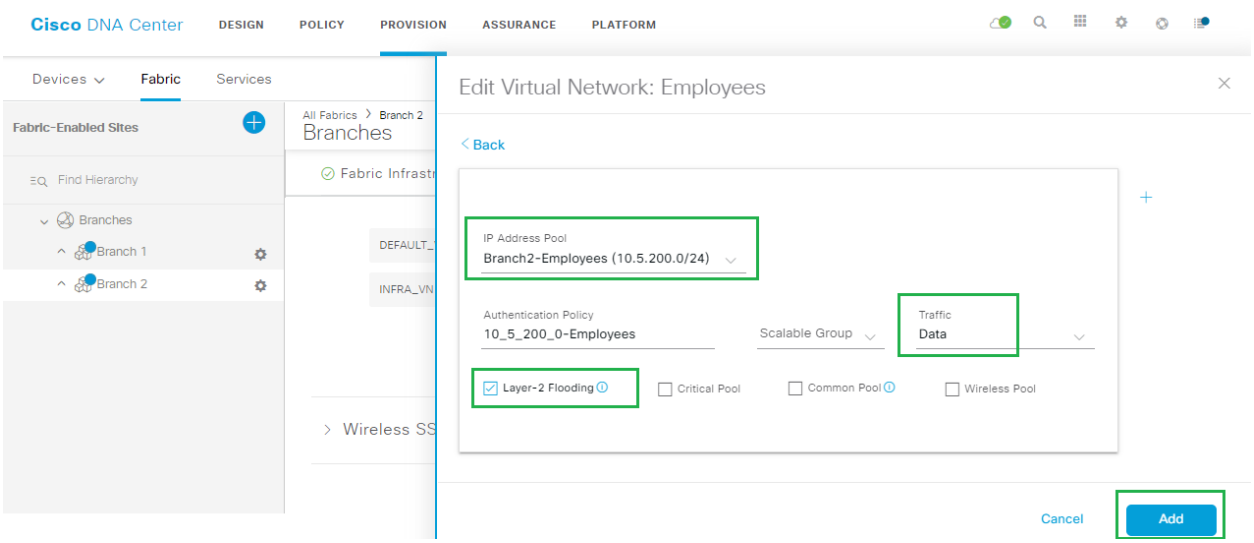
Next Step: Associating Branch2 with IoT Virtual Network & IP Pool Address

1. Navigate: DNA > Provision > Branches > Branch 2 > Host Onboarding
2. Click on Virtual Networks
3. Click “Add Virtual Network”
4. Select the box “Employees”
5. To associate ip pool address with Employees VN click on “Add”
6. Input the details as
    - IP Address Pool: Select from the drop-down: Branch2-Employees (10.5.200.0/24)
    - Traffic: Select from the drop-down: Data
    - Select the boxes
        - ✓ Layer-2 Flooding
7. Click “Add”
8. Click “Save”
9. Done

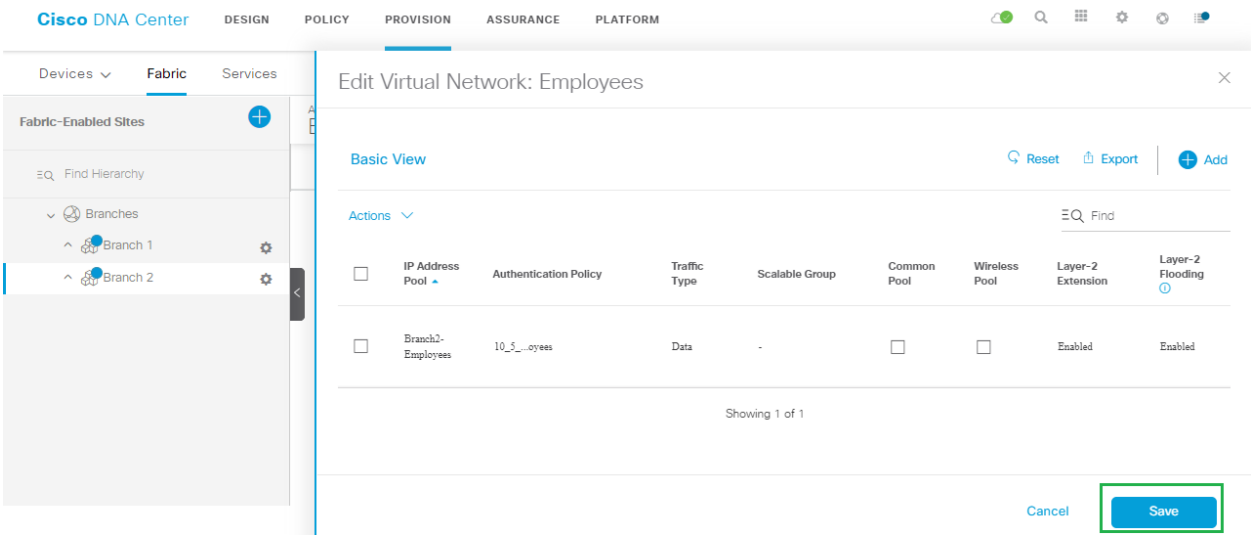
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On sw211

```
sw211(config)#ip dhcp pool br1_emp
sw211(dhcp-config)#network 10.4.200.0 255.255.255.0
sw211(dhcp-config)#default-router 10.4.200.1
sw211(dhcp-config)#exit
sw211(config)#

sw211(config)#ip dhcp pool br2_emp
sw211(dhcp-config)#network 10.5.200.0 255.255.255.0
sw211(dhcp-config)#default-router 10.5.200.1
sw211(dhcp-config)#exit
sw211(config)#

sw211(config)#ip dhcp pool br1_iot
sw211(dhcp-config)# network 10.4.198.0 255.255.255.0
sw211(dhcp-config)# default-router 10.4.198.1
sw211(dhcp-config)#exit
sw211(config)#

sw211(config)#ip dhcp pool br2_iot
sw211(dhcp-config)# network 10.5.198.0 255.255.255.0
sw211(dhcp-config)# default-router 10.5.198.1
sw211(dhcp-config)#exit
sw211(config)#

sw211(config)#ip dhcp excluded-address 10.4.198.1 10.4.198.100
sw211(config)#ip dhcp excluded-address 10.4.200.1 10.4.200.100
sw211(config)#ip dhcp excluded-address 10.5.198.1 10.5.198.100
sw211(config)#ip dhcp excluded-address 10.5.200.1 10.5.200.100
```

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SECTION 2.3: Mapping SDA VNs to SD-WAN VPNs (4 Points)

Using vManage GUI, perform configuration tasks according to these requirements.

1. Use any host, such as host11, to access the vManage GUI website at https://203.0.113.21 URL
2. Create three new SD-WAN VPNs to carry the SDA VN traffic
   - VPN ID 198 for IoT VN
   - VPN ID 199 for Guest VN
   - VPN ID 200 for Employees VN
3. On Branch #1 and Branch #2 vEdges, for each of these VPNs:
   - Create a new subinterface on the interface towards the SDA border switch. Align the VLAN ID and IP address on the subinterface with the configuration generated by DNA Center on the border switches for the appropriate VN.
   - Peer the vEdges and the SDA border switch using iBGP. Ensure reachability between all locations of the same VPN.

Solution

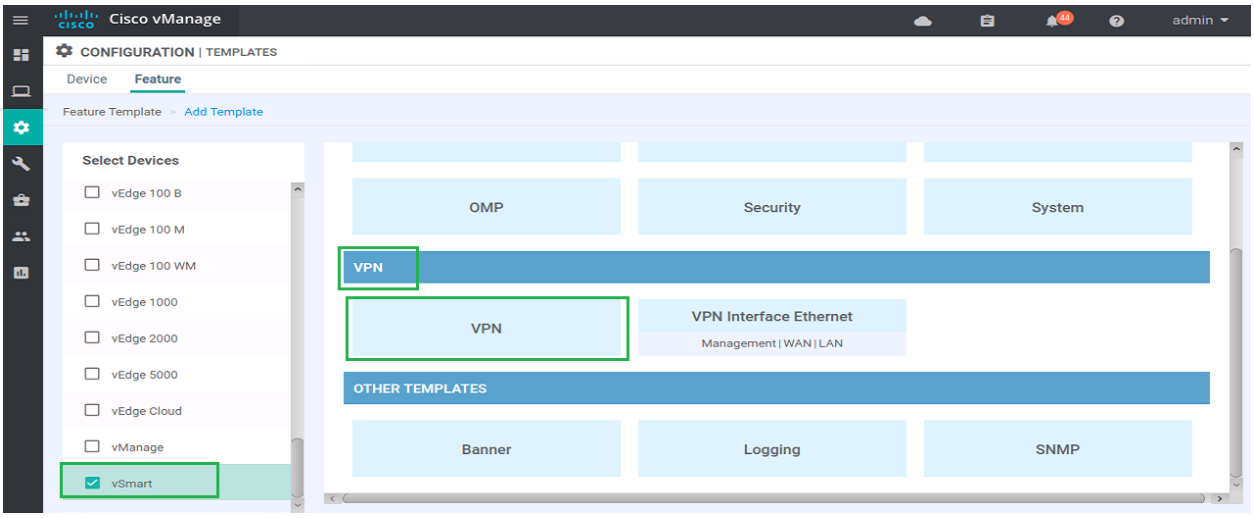
On vManage

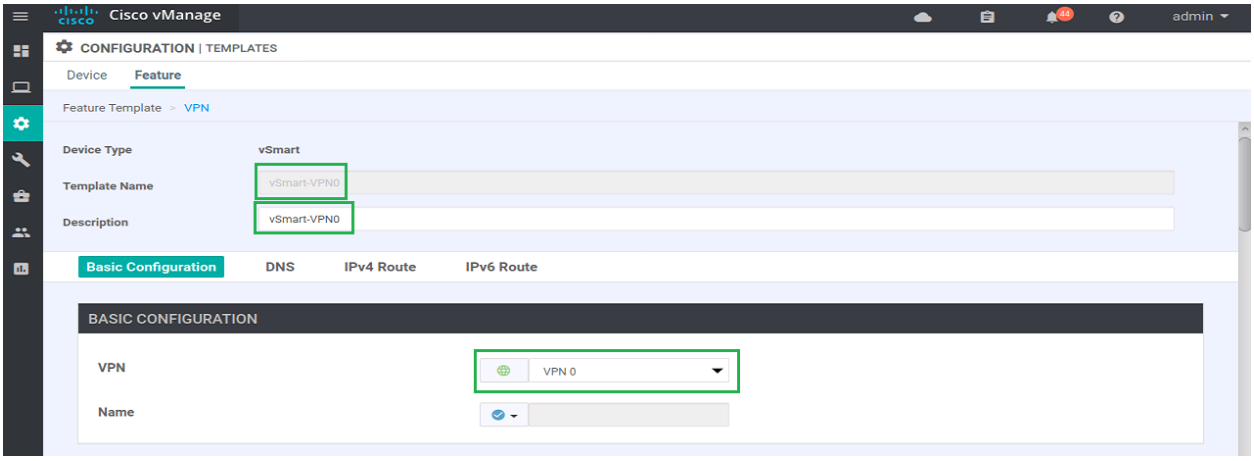
First Step: Creating vSmart Feature Template – VPN0 Template

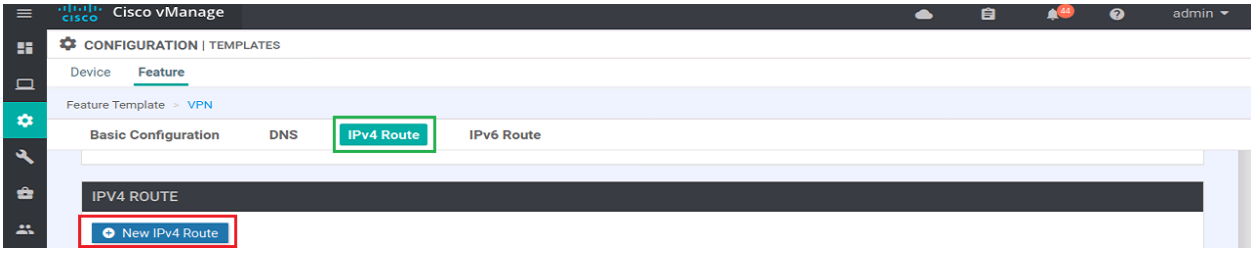
1. Navigate: Configuration > Template > Feature
2. Click “Add Template”
3. From the Drop-down list on left side: select vSmart
4. Right side under VPN: select “VPN”
5. Input the Template Name & Description: Name of your choice
6. Under Basic Configuration
   - VPN (Global): Select “VPN0”
7. Under IPv4 Route Configuration
   - Click New IPv4 Route
     - ✓ Prefix (Global): 0.0.0.0/0
     - ✓ Gateway (Default): “Next Hop”
8. Next Hop: Click on “Add Next Hop”
9. A dialog box will appear, again click on “Add Next Hop”
10. Input the Next Hop ip address: 10.2.251.1
11. Click on “Add”
12. Again Click on “Add”
13. Click “Save”

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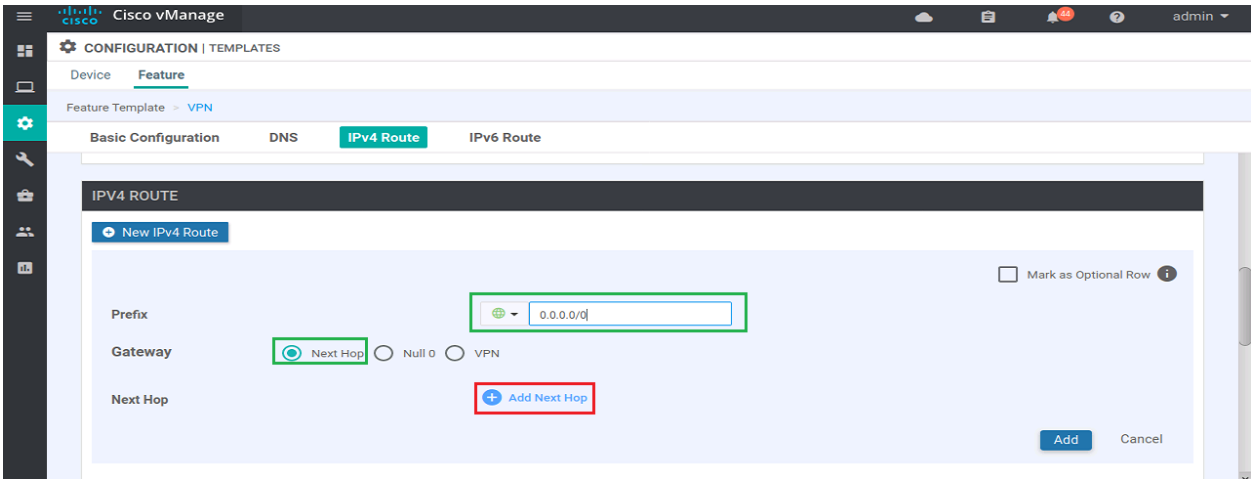


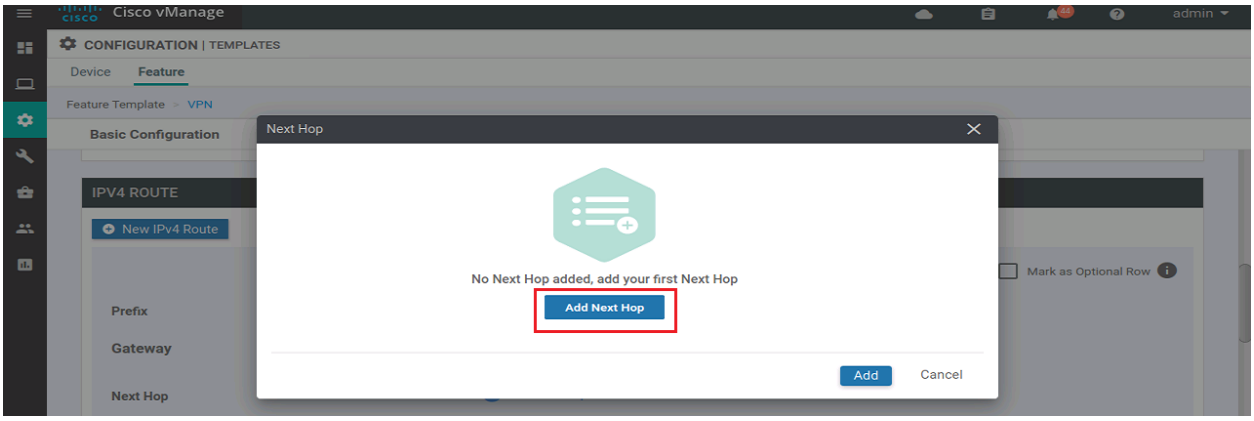


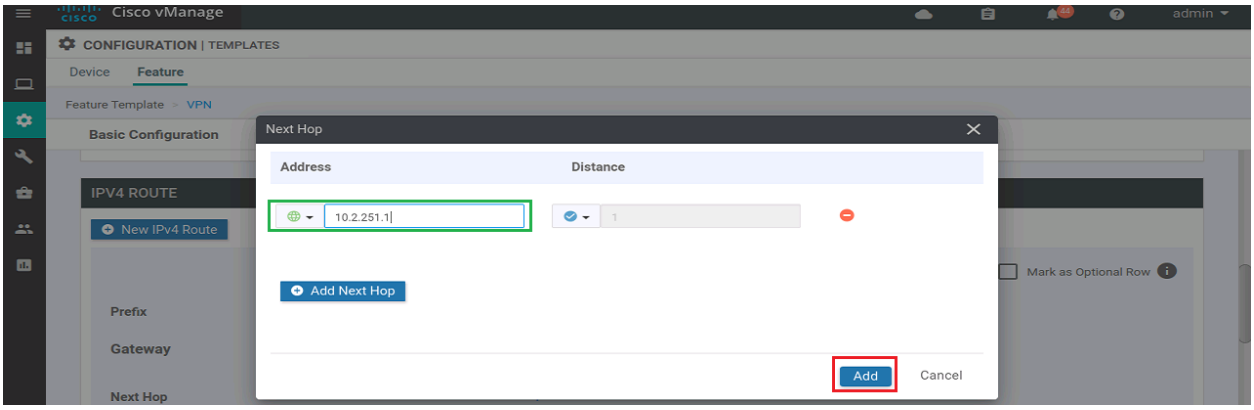




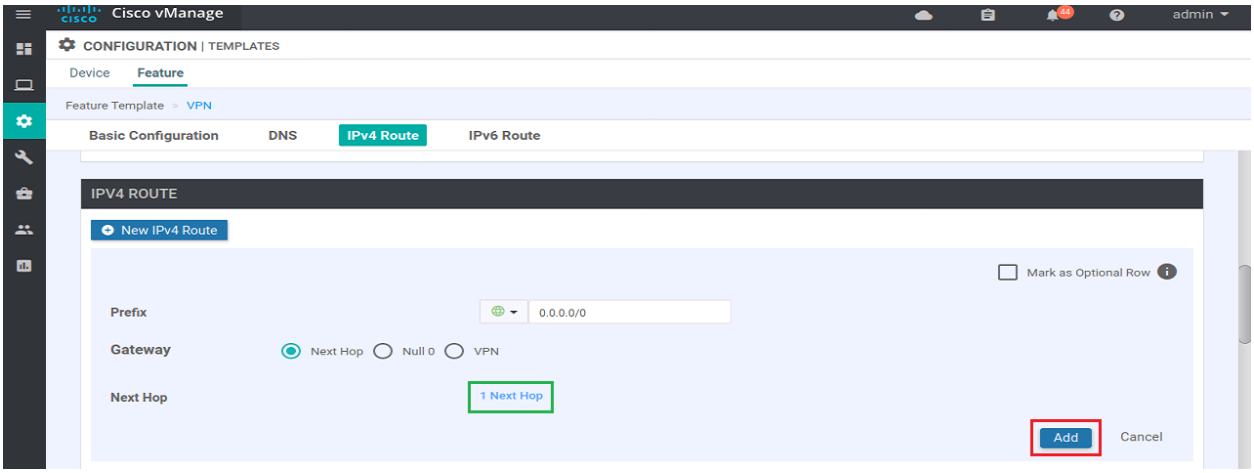
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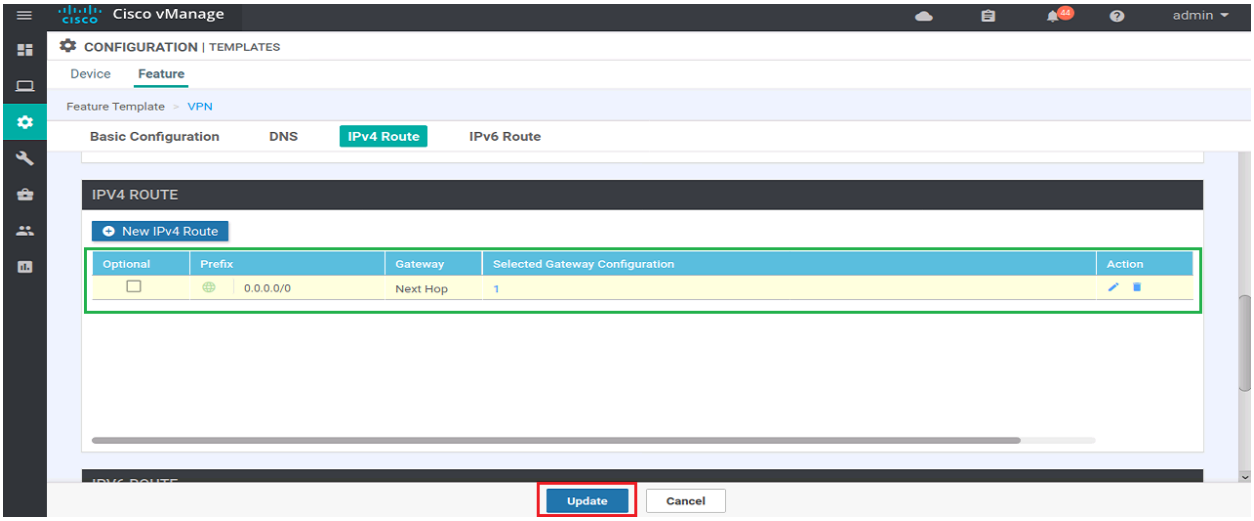






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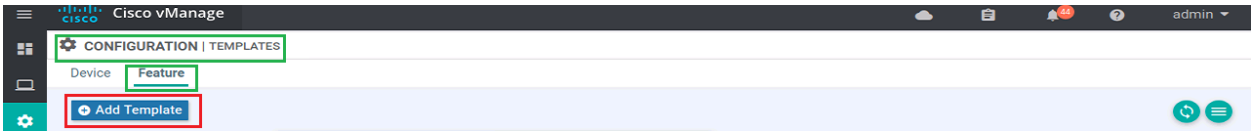


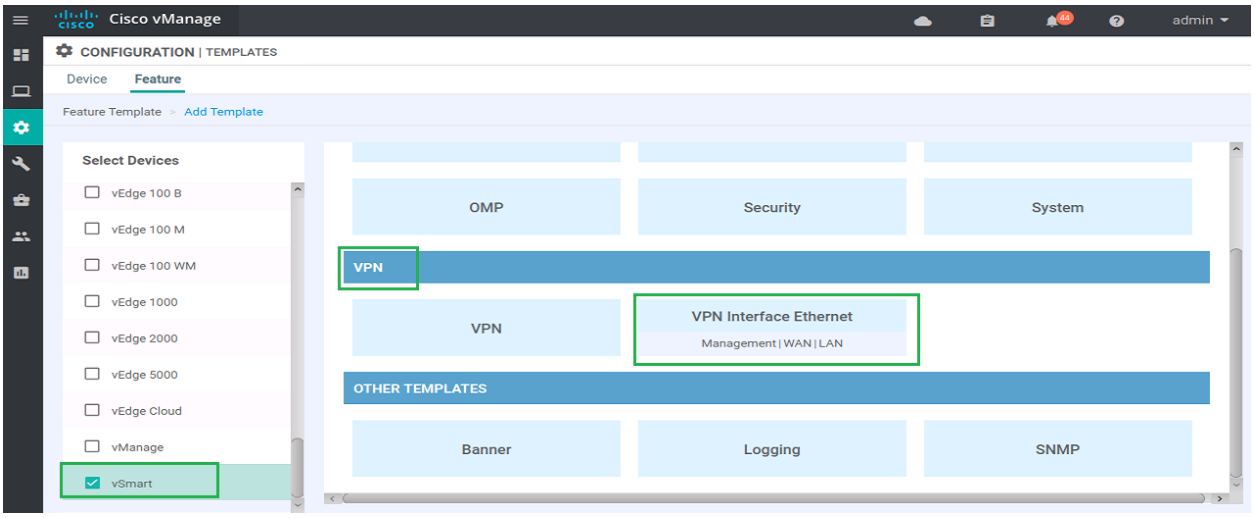


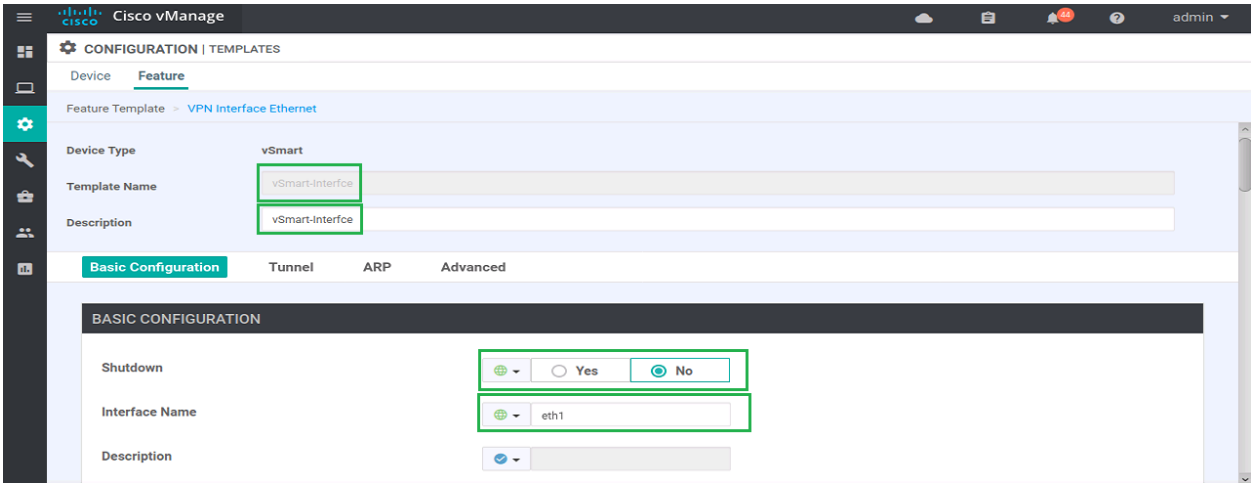
Next Step: Creating vSmart Feature Template – VPN0 Interface Template via GUI.

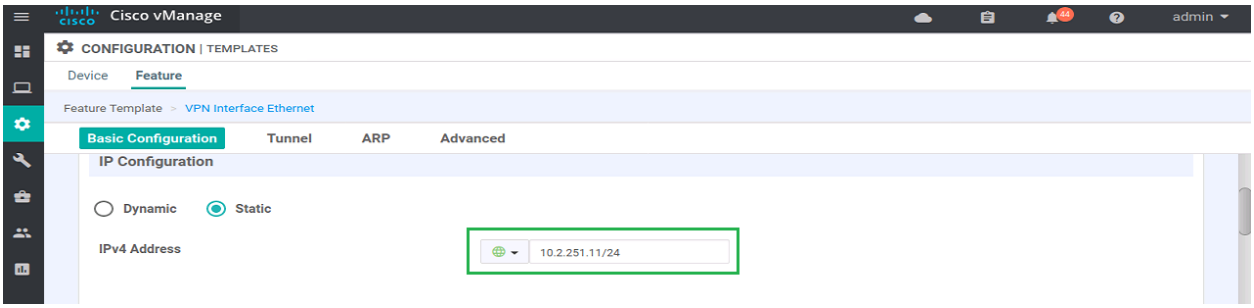
1. Navigate: Configuration > Template > Feature
2. Click “Add Template”
3. From the Drop-down list on left side: select vSmart
4. Right side under VPN: select “VPN Interface Ethernet”
5. Input the Template Name & Description: Name of your choice
6. Under Basic Configuration
   - Shutdown (Global): Select “No”
   - Interface Name (Global): eth1
   - IP address (Global): 10.2.251.11/24
7. Under Tunnel Configuration
   - Tunnel Interface (Global): Select “On”
8. Click “Save”

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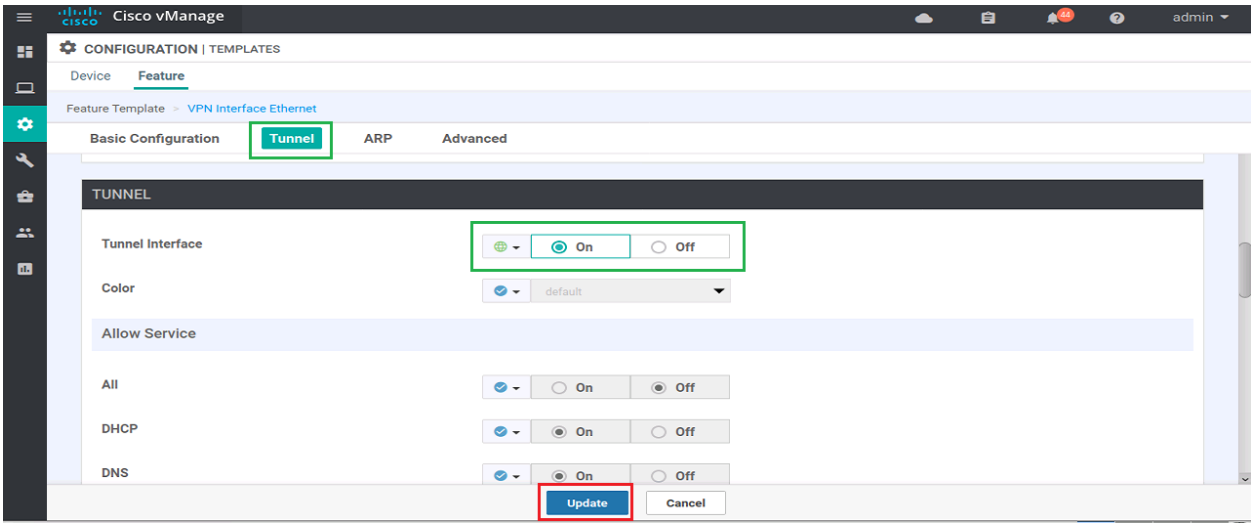






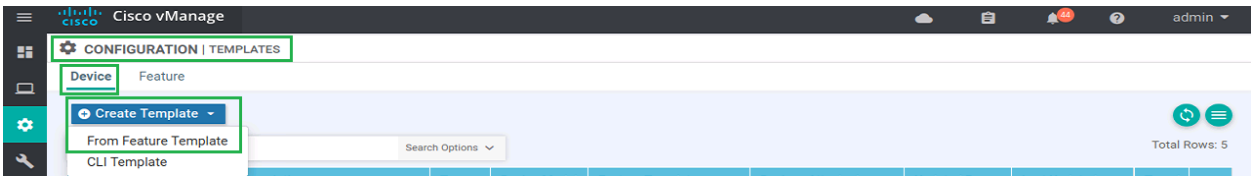


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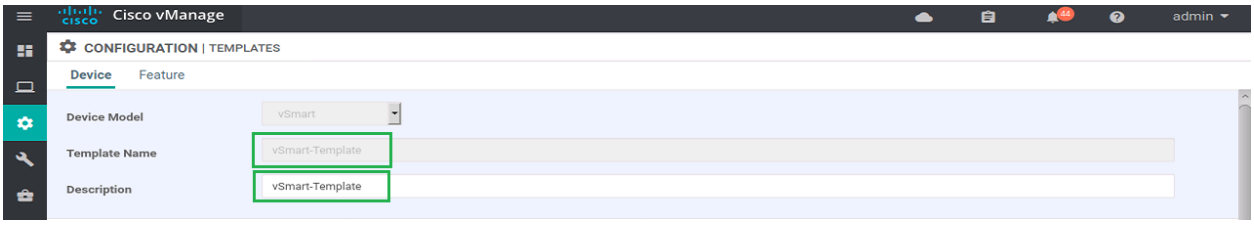


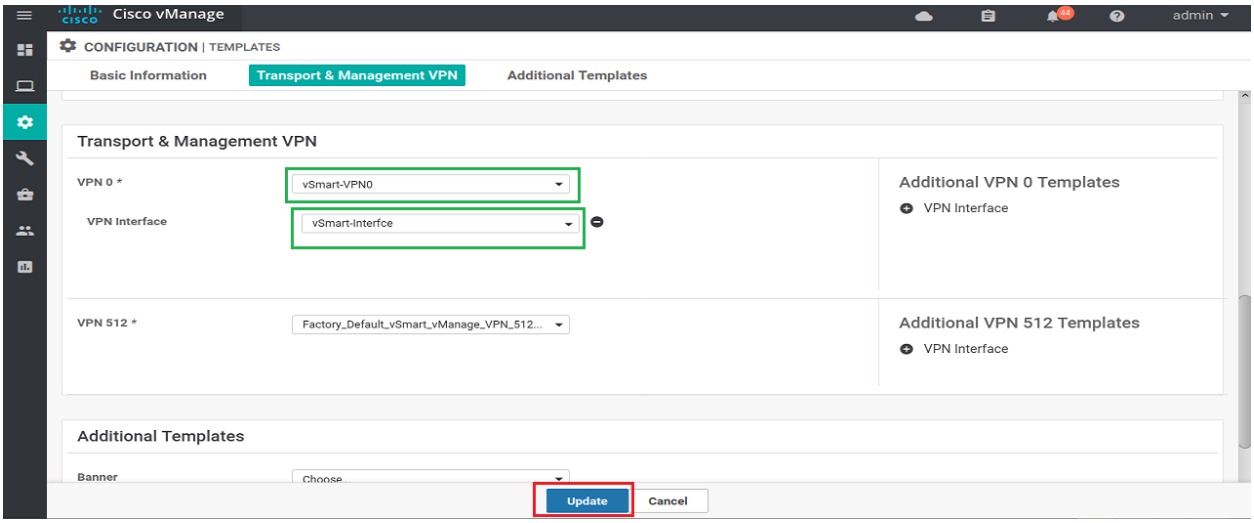
Next Step: Creating vSmart Device Template from Feature Templates via GUI.

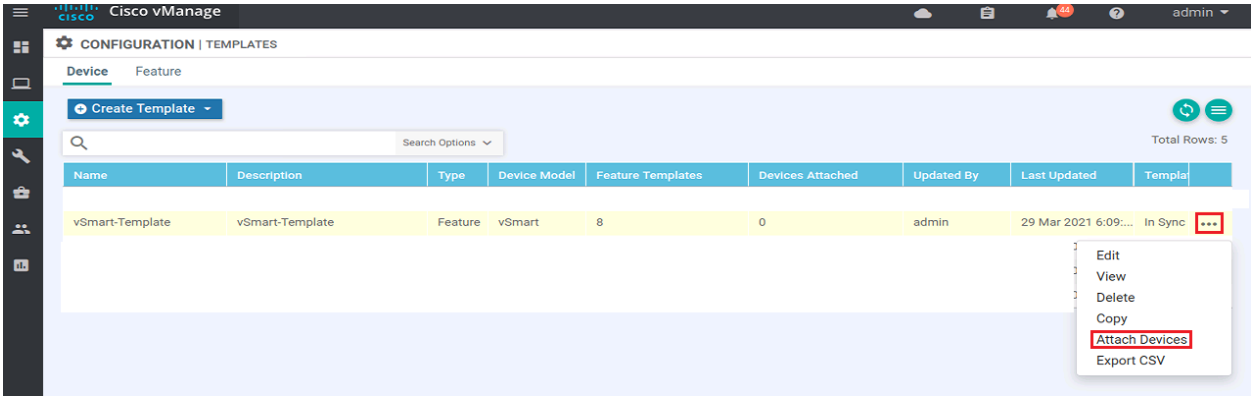
1. Navigate: Configuration > Template > Device
2. Click “Create Template”
3. Select “From Feature Template”
4. Details:
    - Device Model: vSmart
    - Template Name: Name of your choice
    - Description: Name of your choice
5. Click Transport & Management VPN
    - VPN0: Select “vSmart-VPN0”
    - Add the vpn interface by clicking “VPN Interface” on right side
    - VPN Interface: Select “vSmart-Interface”
6. Click “Create”
7. A Device Template is now created
8. Click on the three dots on right side and click “Attach Device”
9. A Windows will appear: Select the “vSmart” and click on the “Right Arrow”
10. Click “Attach”
11. Click on the three dots on right side and click “Edit Device Template”
12. Input the proper values and click “Update”
13. Click “Next”
14. Click “Configure Devices”
15. All Done



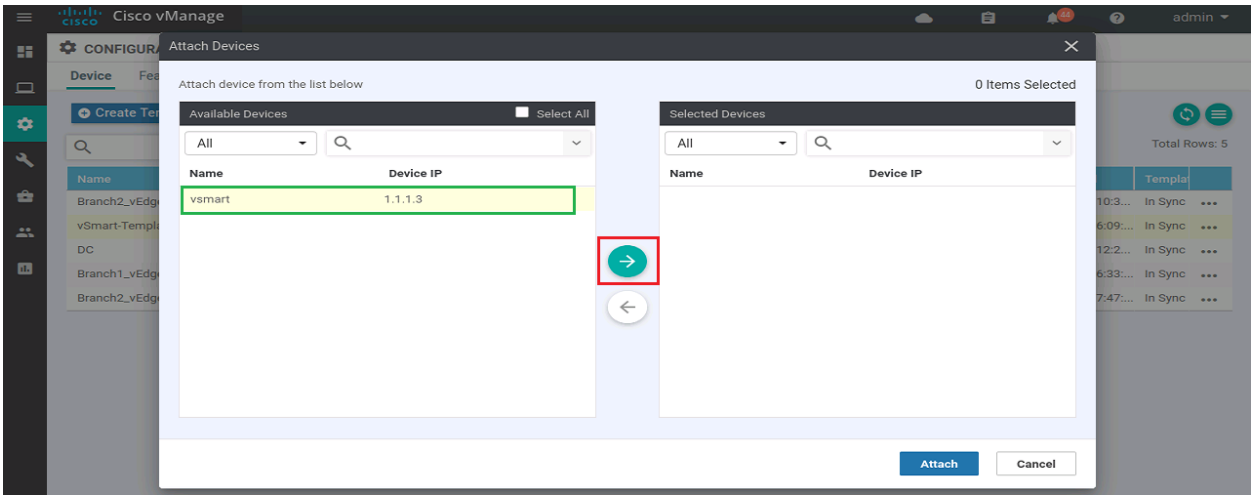
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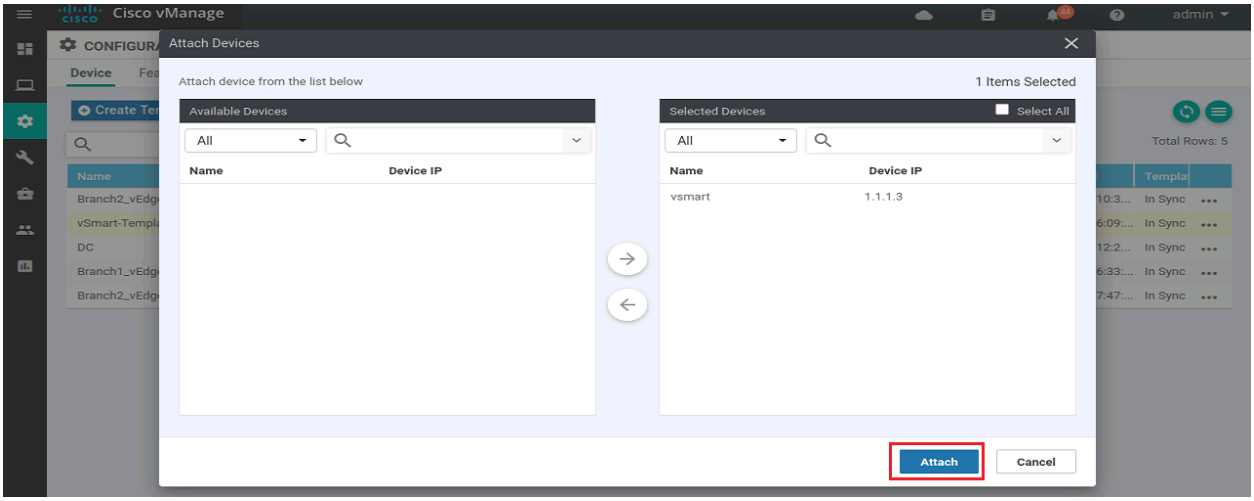


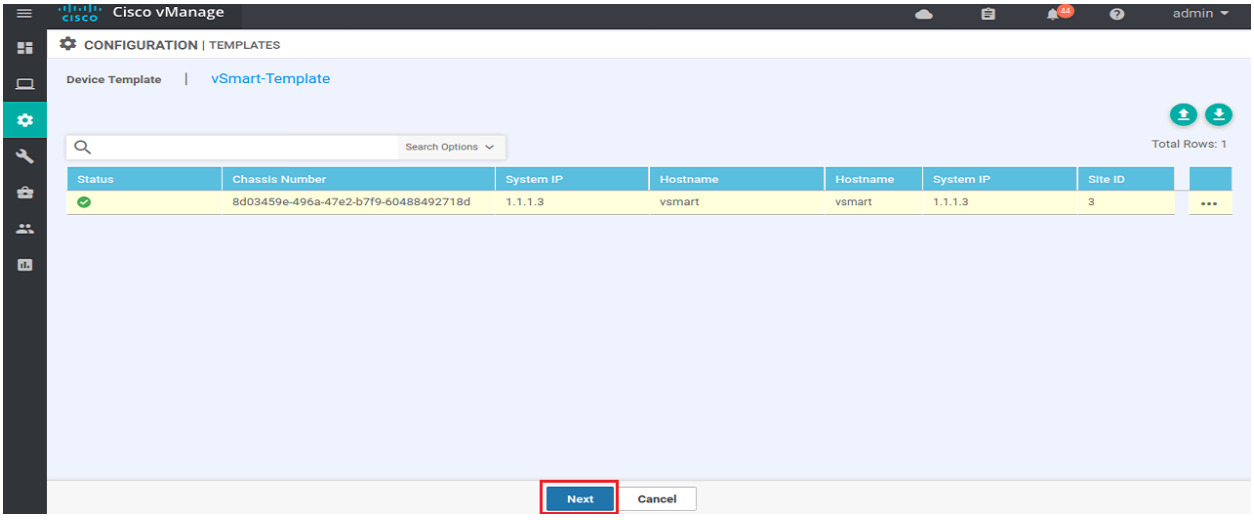




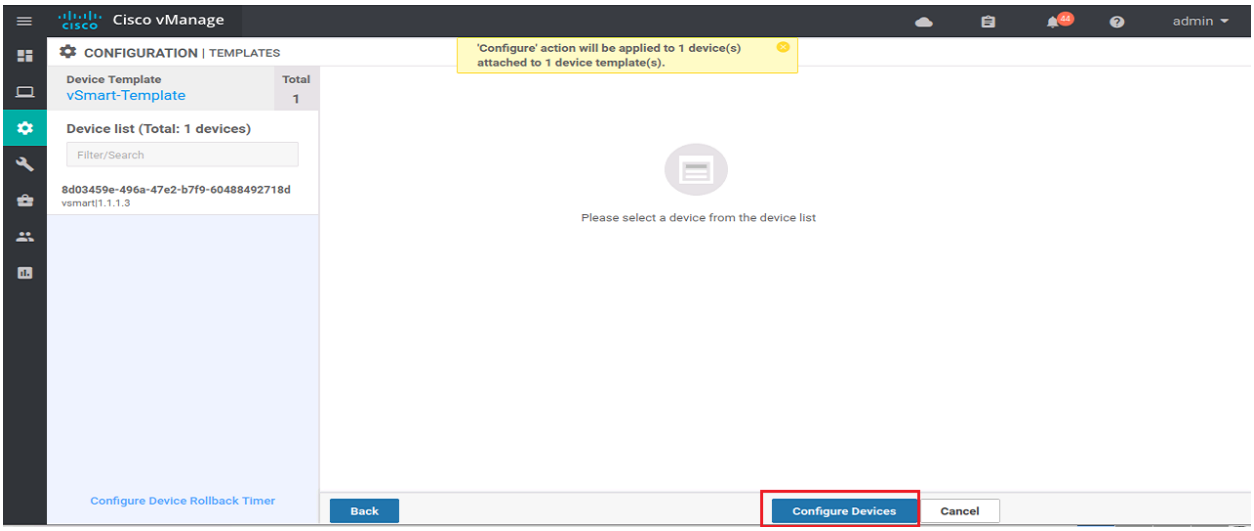
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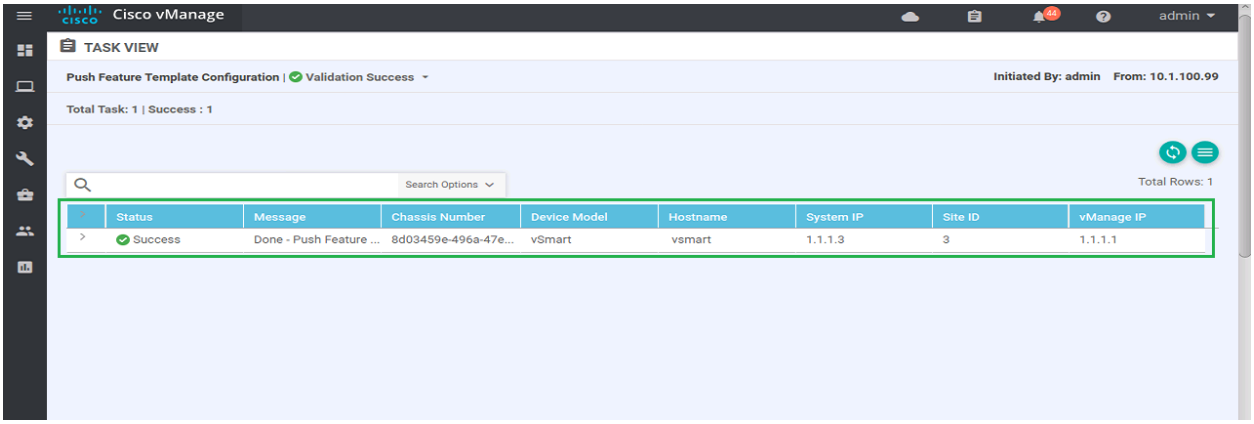






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Next Step: Creating Feature Template - VPN's Template for IoT & Guest & Employees

For VPN198

1. Navigate: Configuration > Template > Feature
2. Click "Add Template"
3. From the Drop-down list on left side: select vEdge Cloud
4. Right side under VPN: select "VPN"
5. Input the Template Name & Description: You can use any name of your choice

Solution omitted……

Next Step: Creating Device Template From Feature Template – Branch2 vedge52

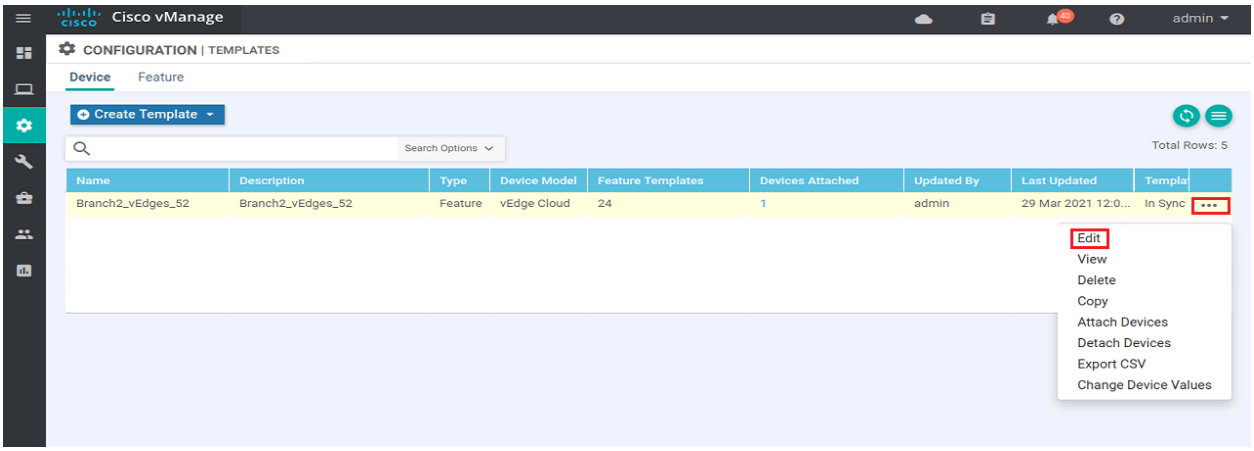
1. Navigate: Configuration > Template > Device
2. Select Branch2_vedges_52
3. Click on the three dots on the right side corner to edit the template
4. Under Basic Information

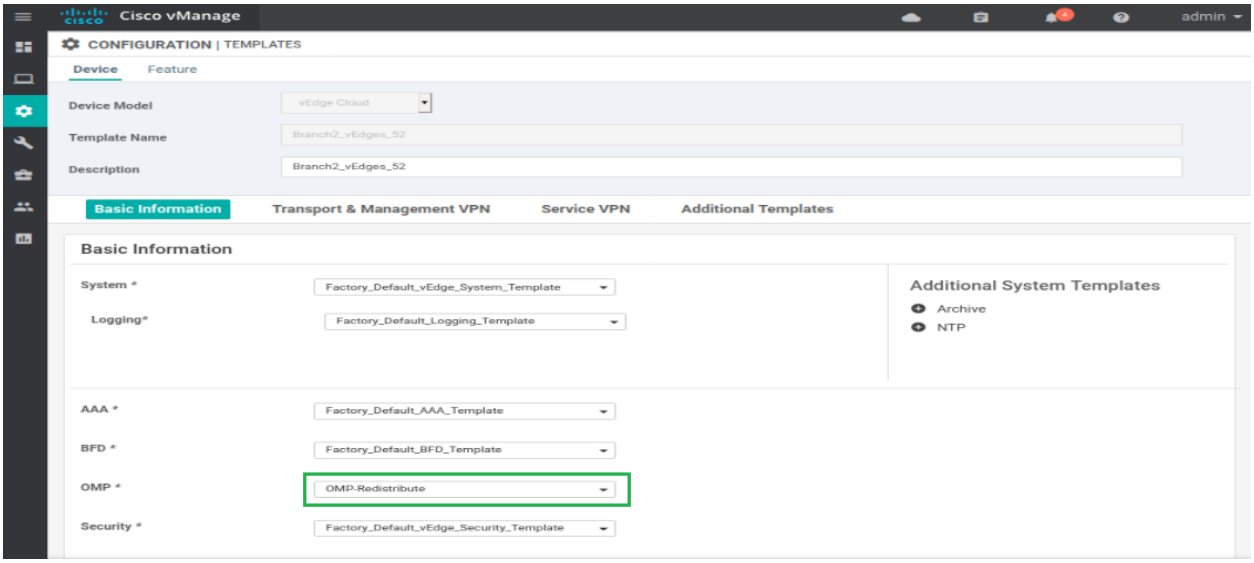
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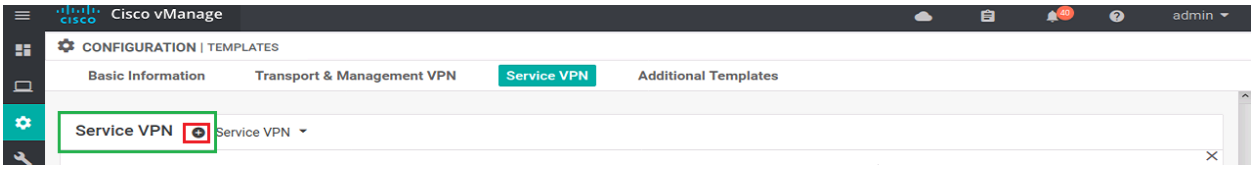
- Select: OMP Feature Template

5. Under Service VPN: Click on “Add Symbol”
6. To Add BGP and VPN Interface: Click on the right side on BGP and VPN Interface
   - VPN: Employees-VN ---->Select the template name which you have given
   - BGP: Employees-IBP-VPN-200 --->Select the template name which you have given
   - VPN Interface: Branch1_ge0/3.3002 ---> Select the template name which you have given

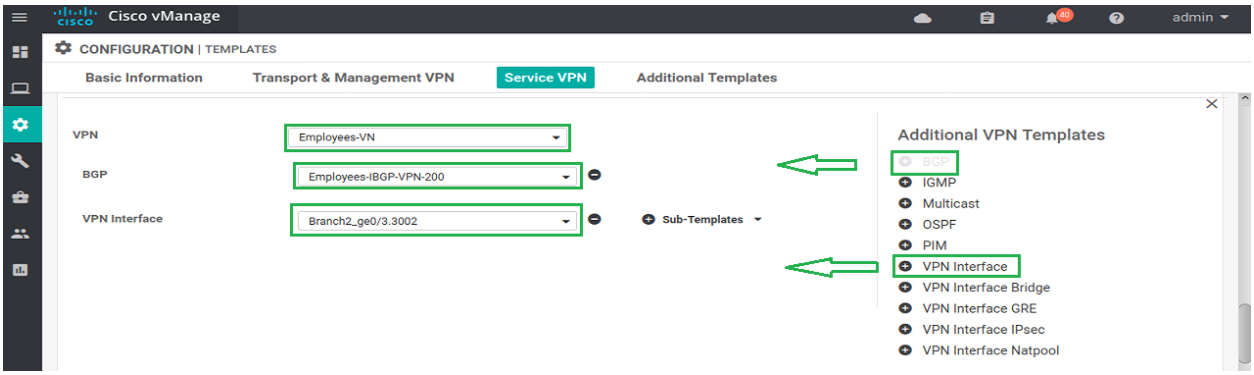
Solution omitted……

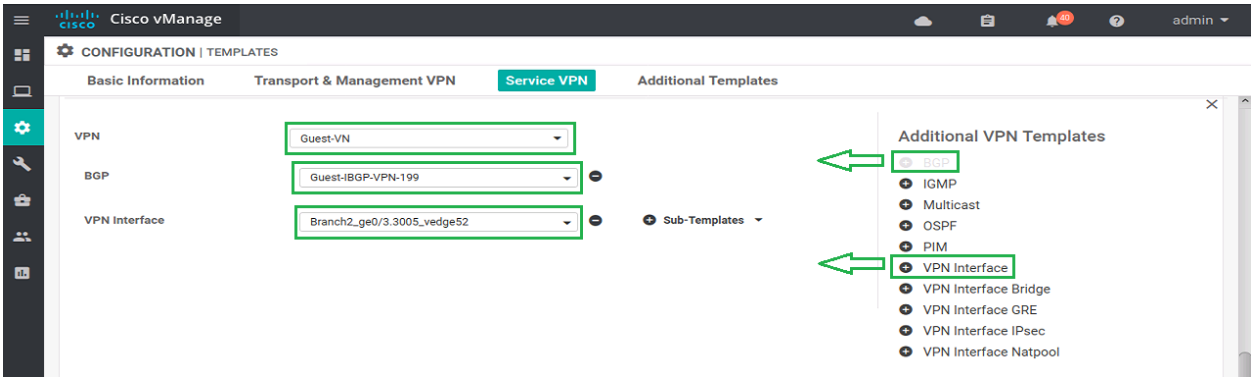


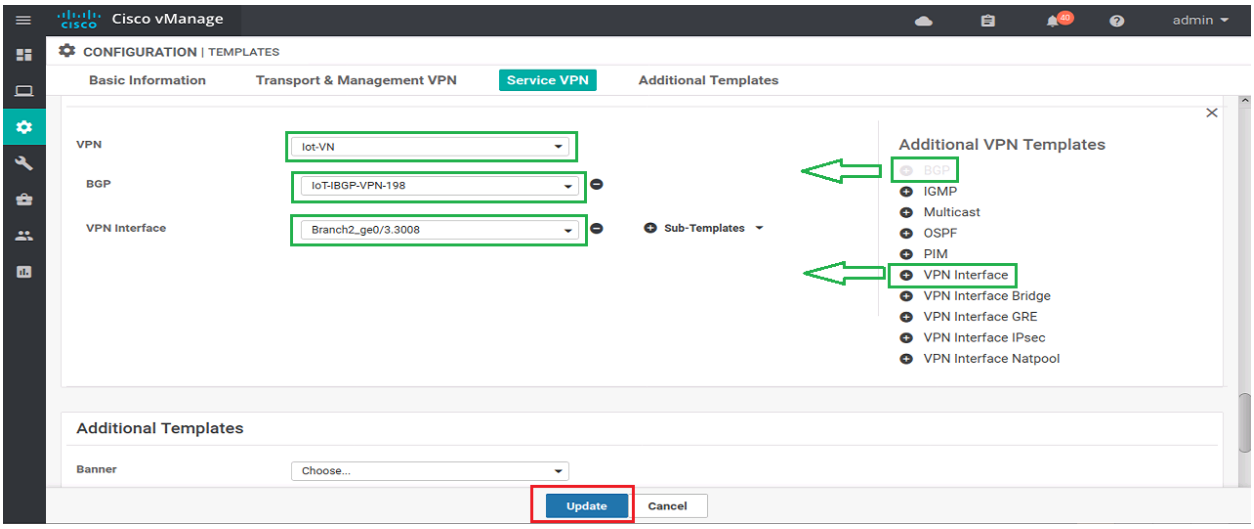




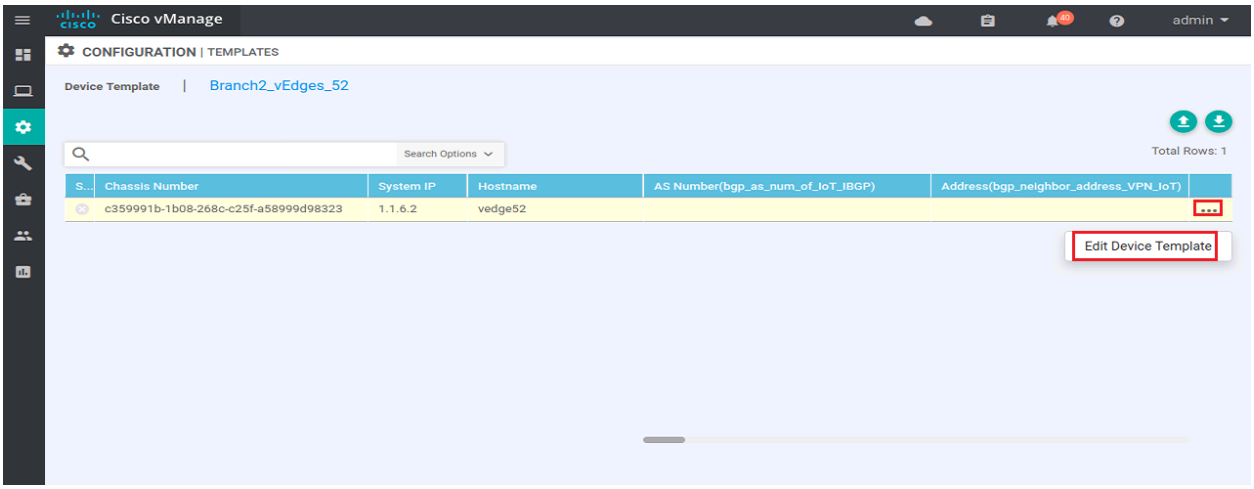
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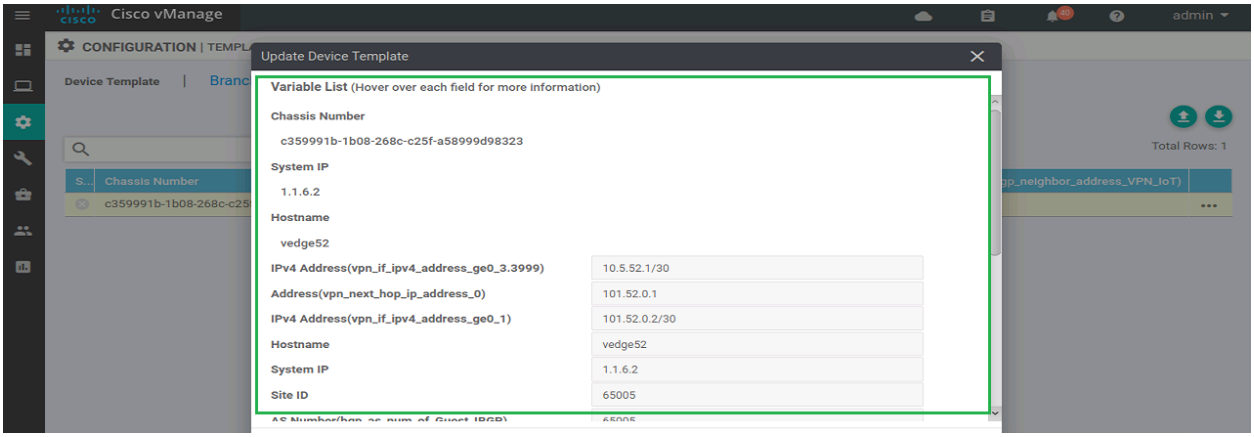


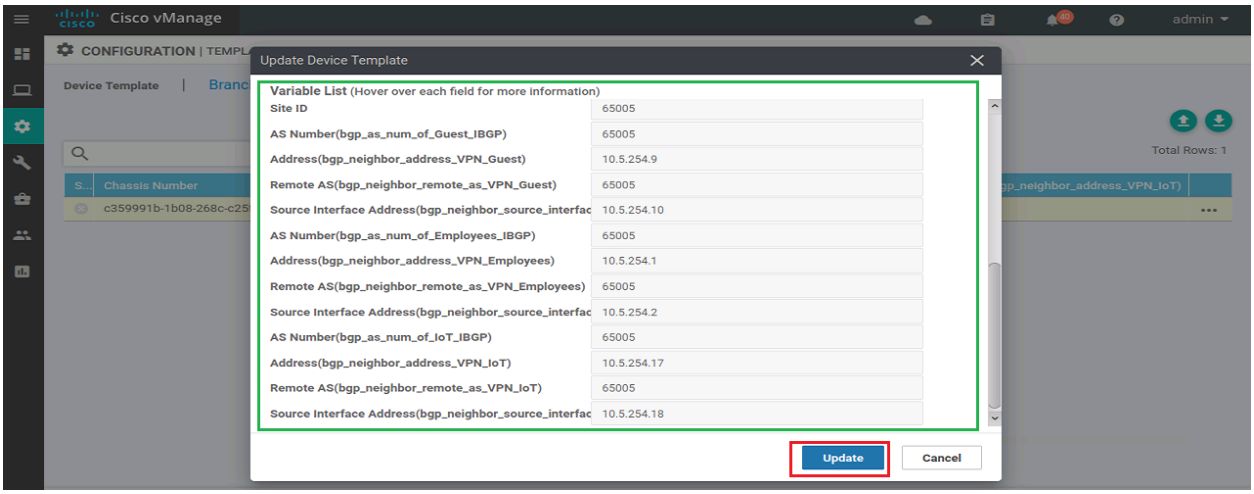




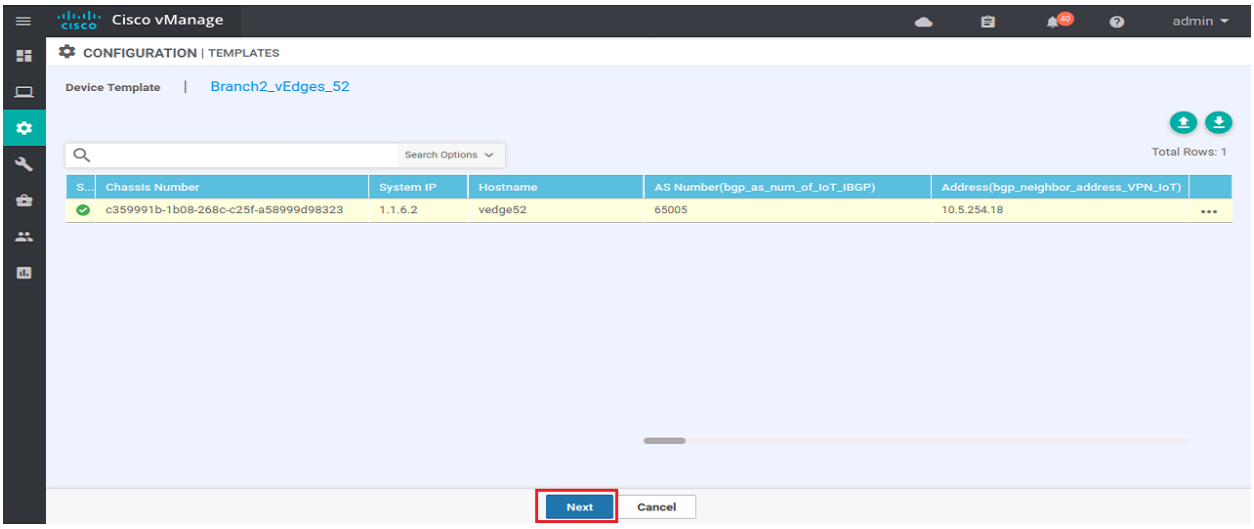
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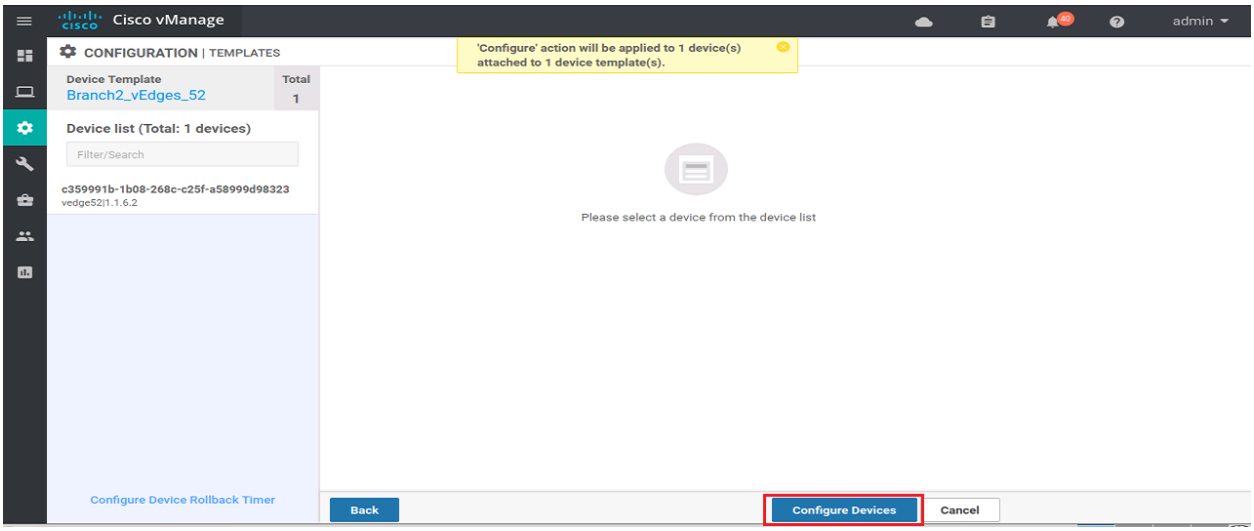


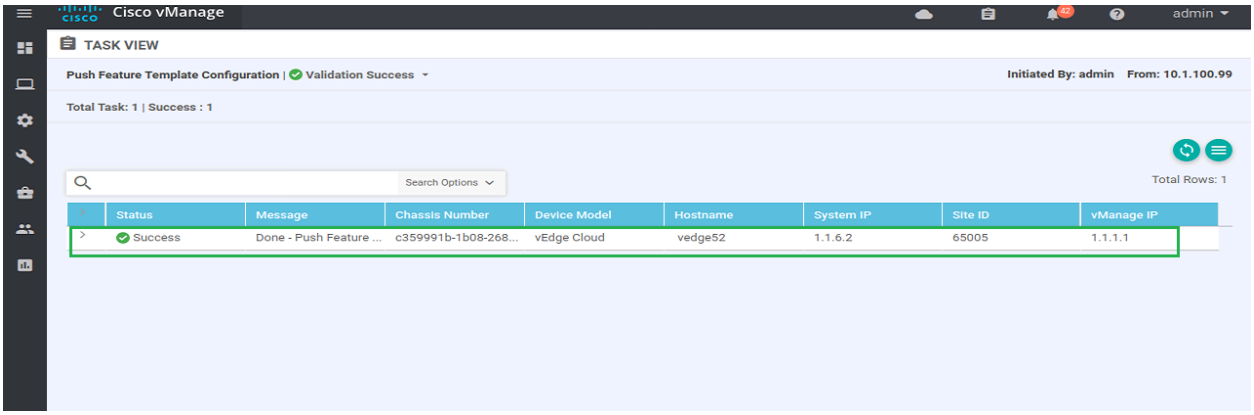




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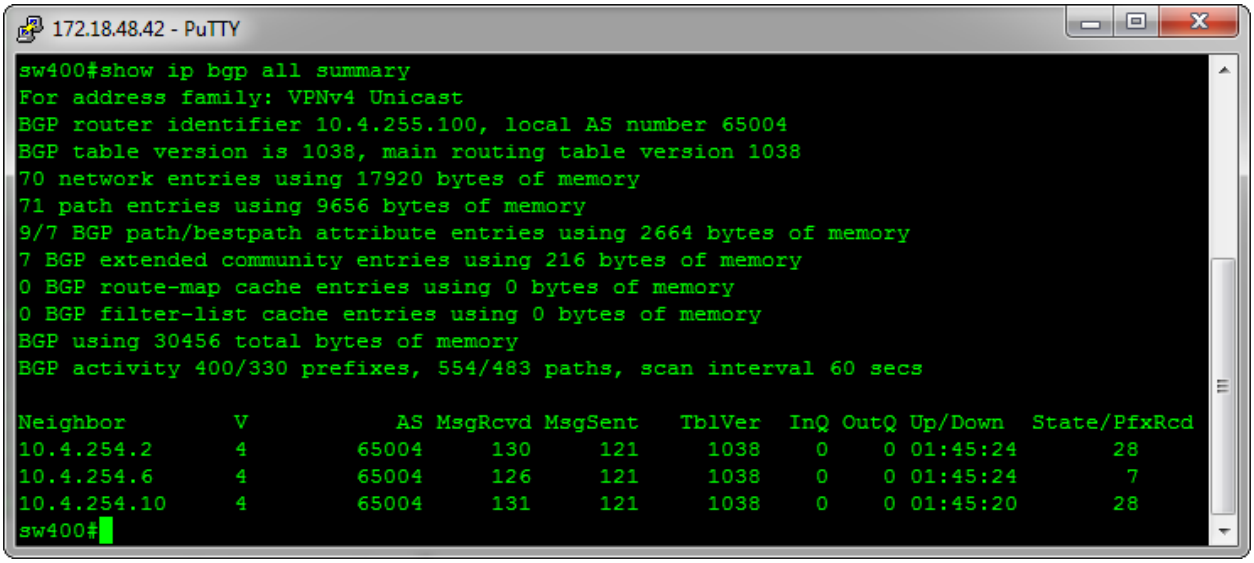




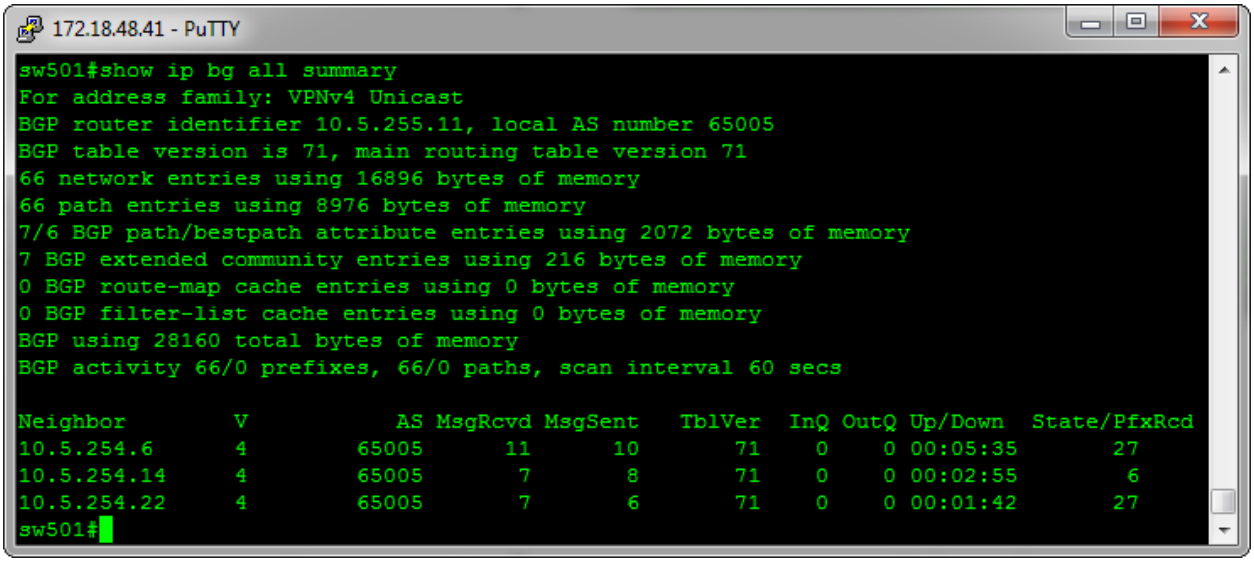
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Verification

On sw400

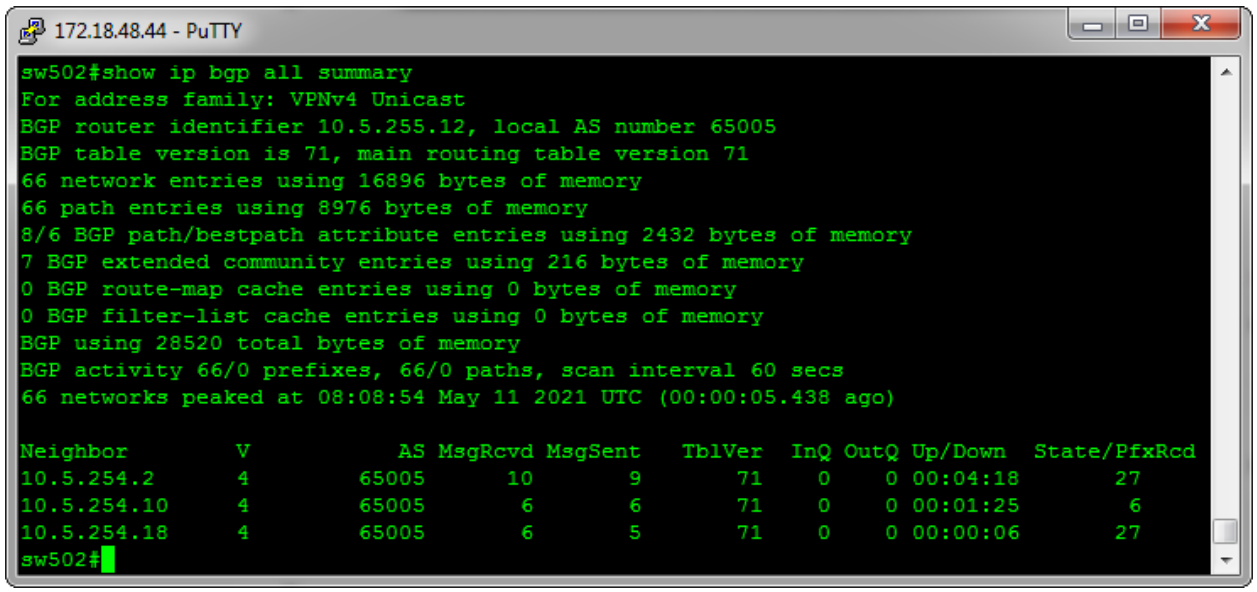


On sw501



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On sw502



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SECTION 2.4: Configuring SD-WAN VPN Route Leaking (3 Points)

To allow the traditional parts of FABD2 network to communicate with the Employees and IoT VPNs/VNs, configure route leaking in SD-WAN according to these requirements.

1. Prefixes in the IoT VPN 198 must be imported into the existing SDA underlay VPN 999 and tagged with the tag value of 198.
2. Prefixes in the Employees VPN 200 must be imported into the existing SDA underlay VPN 999 and tagged with the tag value of 200.
3. Prefixes in the SDA Underlay VPN 999 advertised from the DC that are within the 10.4.0.0/15 range must be rejected. Other prefixes in the SDA Underlay VPN 999 advertised from the DC must be accepted and also imported into IoT VPN 198 and Employees VPN 200
4. Redistribution from OMP into OSPF on Branches #1 and #2 in VPN 999 must exclude vRoutes tagged with value of 198 and 200
5. Place host41 into Employees VN, Place host51 into IoT VN. Make sure both hosts receive their IP settings from DHCP.
6. Ensure that IoT and Employees VPNs on Branches #1 and #2 have reachability to Branches #3 and #4. It is allowed to modify the VPN 999 settings to accomplish this requirement.

Solution

First Step: Creating Lists & Topology For Route-Leaking & Denying Data Traffic as per task requirements

1. Creating Lists:- Navigate: Configuration > Policies
   - On top-right corner > Custom Options > Centralized Policy > Select Lists
   - Select “Site” and than click on “New Site List”

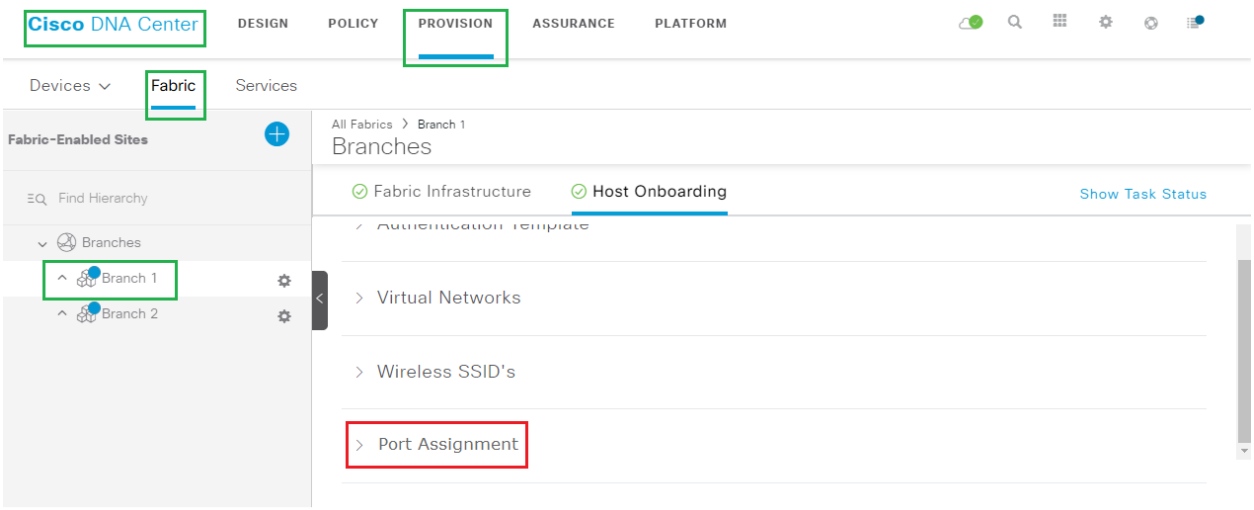
Solution omitted……

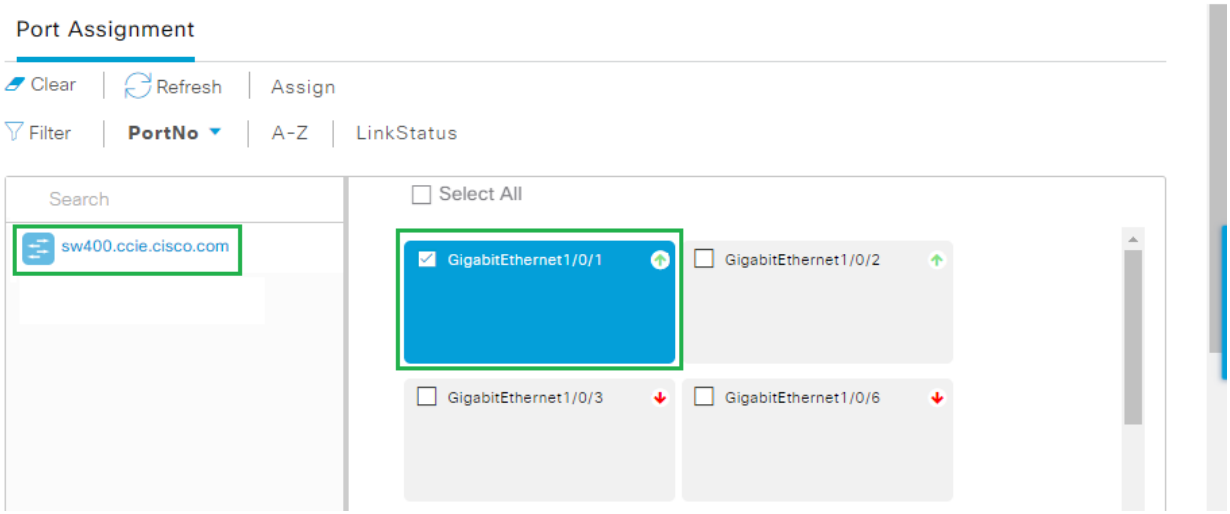
Next Step: Associating host41 in Employees VN & host51 in IoT VN

1. Navigate: Provision > Fabric
2. Select Branch1 > Select Host Onboarding > Select Port Assignment
3. Select sw400 > Click GigabitEthernet 1/0/1
4. Click “Save/Update”
5. Repeat the above steps

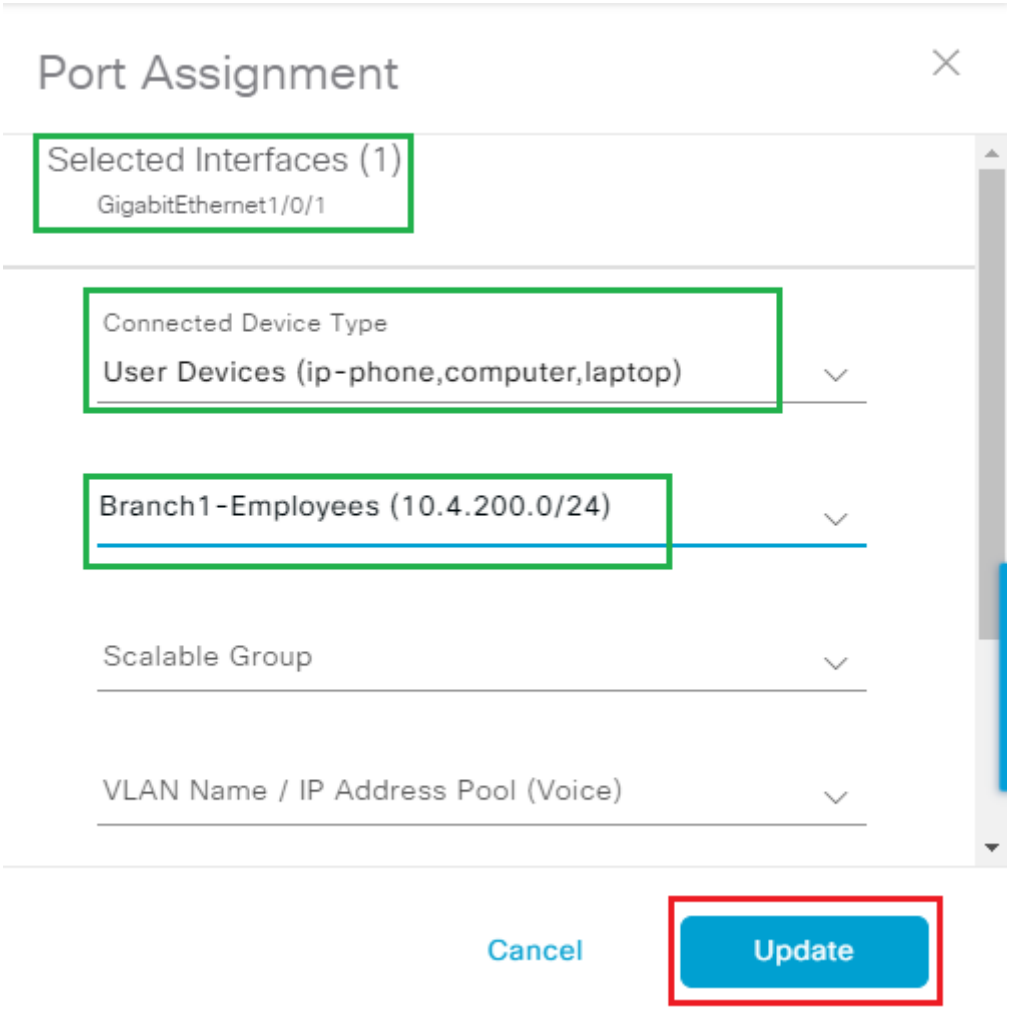
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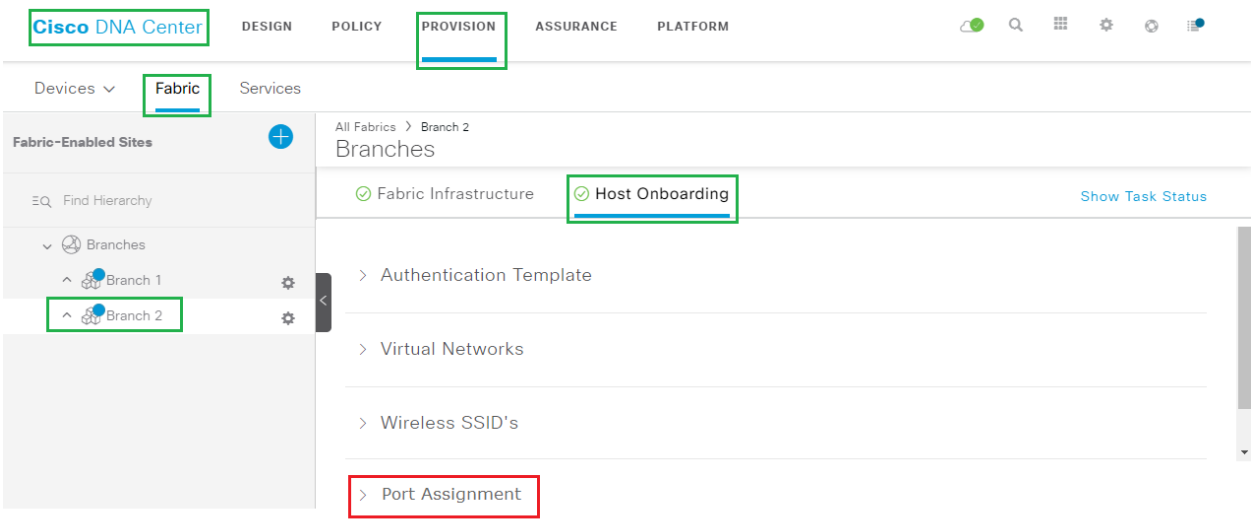
6. Navigate: Provision > Fabric
7. Select Branch2 > Select Host Onboarding > Select Port Assignment
8. Select sw510 > Click GigabitEthernet 1/0/1
9. Click “Save/Update”



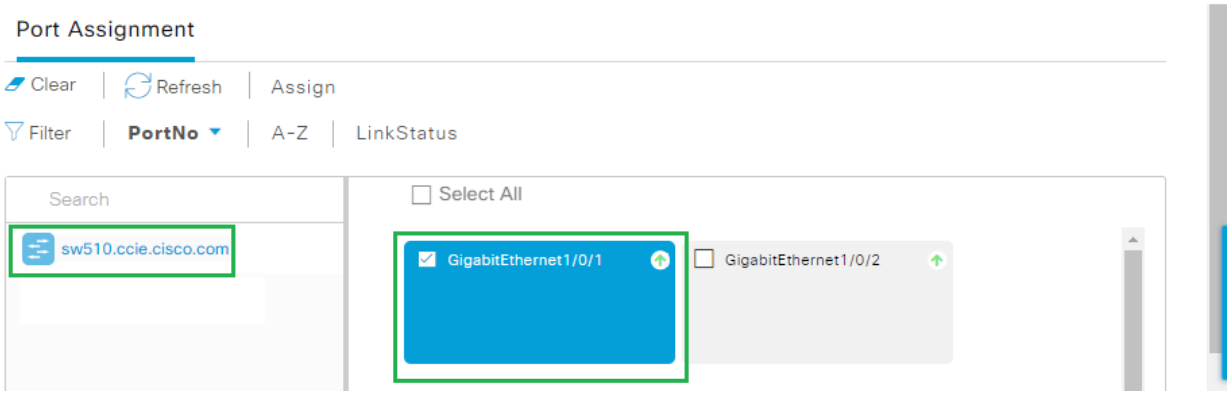


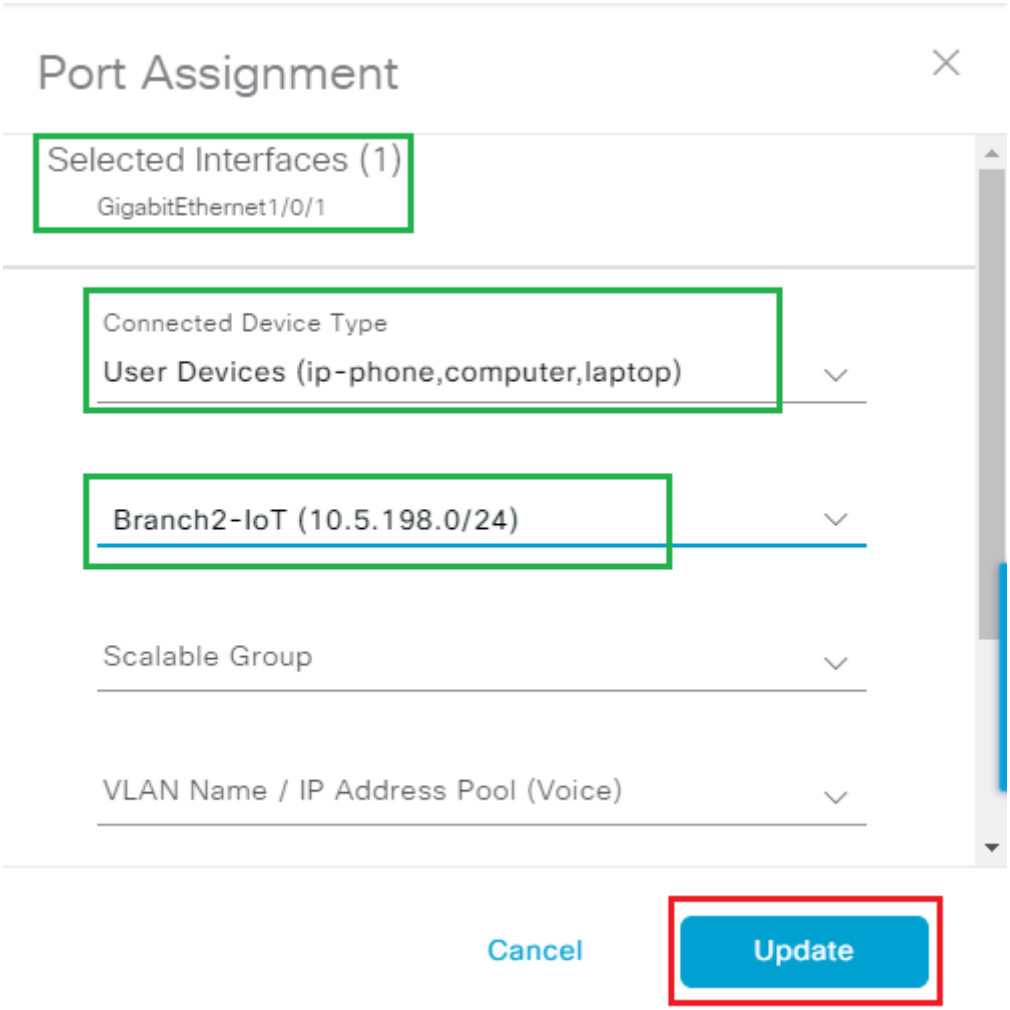
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SECTION 2.5: Handling Guest Traffic (4 Points)

The Guest VN/VPN on Branches #1 and #2 must remain isolated from the rest of the company network. It is only allowed to reach the internet through r23 and r24 in the DC. Enable internet connectivity for the Guest VPN according to these requirements.

1. On vedge21 and vedge22, place the ge0/2 interfaces into the Guest VPN 199
2. On r23 and r24, create a new VRF named Guest using the RD of 65002:199, and place the g4 interface into this VRF.
3. Assign addresses to these interfaces:
   - r23 g4: 10.2.123.1/24
   - r24 g4: 10.2.224.1/24
   - vedge21 ge0/2: 10.2.123.2/24
   - vedge22 ge0/2: 10.2.224.2/24
4. Peer r23 and vedge21 in the Guest VRF/VPN using iBGP
5. Peer r24 and vedge22 in the Guest VRF/VPN using iBGP
6. Ensure that r23 and r24 learn the routes in the Guest VRF/VPN over iBGP.
7. On r23 and r24 configure a static default route in the Guest VRF and point it to the ISP's IP Address 200.99.23.1 or 200.99.24.1 as appropriate. Advertise this default route in iBGP to vedge21 and vedge22
8. On r23 and r24 configure PAT to allow the Guest VPN to access internet by translating in to the router address on the link towards the ISP. Resue the NAT ACL already created on the router. Do not use NAT pools

Configure r23 as a DHCP server for Guest VPN according to these requirements.

1. Create Loopback1 interface on r23 assocaited with the Guest VRF and having the IP address 10.2.255.211/32
2. Advertise this prefix in BGP toward vedge21
3. Create DHCP pool named br1_guest for Branch #1 Guest subnet
4. Create DHCP pool named br2_guest for Branch #2 Guest subnet
5. Explicitly associate both DHCP pools with the VRF Guest
6. In each subnet assign addresses from .101 up to .254 inclusively and the appropriate gateway to clients.
7. Associate host42 and host52 with the Guest VN in DNAC, and make sure that both hosts receive the appropriate address.
8. Make sure that host42 and host52 can ping 8.8.8.8 in the ISP cloud

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Solution

On vManage

First Step: Creating Feature Template – VPN Interface for DC vEdge21 & vEdge22: For Guest VPN

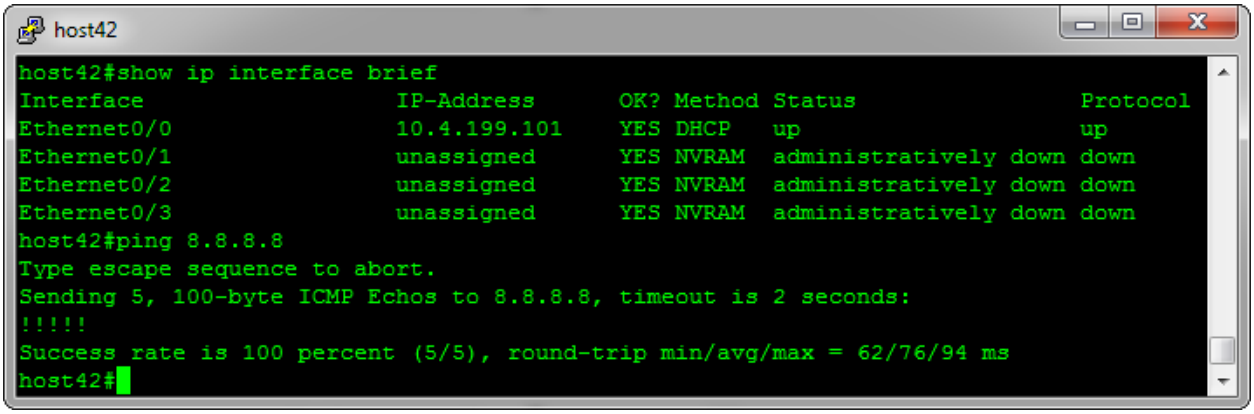
1. Navigate: Configuration > Template > Feature

Solution omitted……

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Verification

On host42



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SECTION 2.6: Support for Silent Hosts in Branch #2 (3 Points)

The Item consist of multiple questions. You may need to scroll down to be able to see all questions.

In Future Branch #2 will be equipped with IP-based IoT endpoints operating in speak-when-spoken-to mode, also called silent hosts. Which of the following SDA features enables a working connectivity with thses IoT endpoints?

a. Layer 2 Extension
b. Native Multicast
c. Layer 2 Flooding
d. Endpoint Mobility

In the statement below, select one of the option from the drop-down list to complete the sentence and form a correct statement.

For SDA to Support Silent hosts, in the underlay as a prerequisite.

a. IP Multicast routing with PIM-SM must be enabled
b. No additional capability aside from unicast IP connectivity is required
c. IS-IS must be used as a routing protocol
d. DHCP Snooping must be enabled

Solution

Part 1
Answer: c

Part 2
Answer:

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Section 3: Making use of Programmability

SECTION 3.1: Enabling CLI Access to R30 (2 Points)

There is no direct console access provided to the router r30. Moreover, r30 does not accept any remote connections because its VTY lines are configured with transport input none. Using RESTCONF enable remote access to r30 for all remote access protocols, according to the following requirements:

1. You can use host31 to access router r30 using IP address 10.3.11.1
2. You can use any method of accessing the RESTCONF API on r30 from host31, including curl, Python, or Postman
3. You must change the input transport protocol on all configurable VTY lines.
4. The input transport protocol value setting must be changed from none to all

Important parameters:

1. Username/Password for HTTP authentication
   - admin/admin
2. URL
   - https://10.3.11.1:443/restconf/data/Cisco-IOS-XE-native:native/line/vty
3. HTTP method to retrieve the configuration
   - GET
4. HTTP method to modify the configuration
   - PATCH
5. HTTP headers
   - Content-Type: application/yang-data+json
   - Accept: application/yang-data+json
6. Recommended curl switches
   - -I, -k, -X, -H, -u, -d

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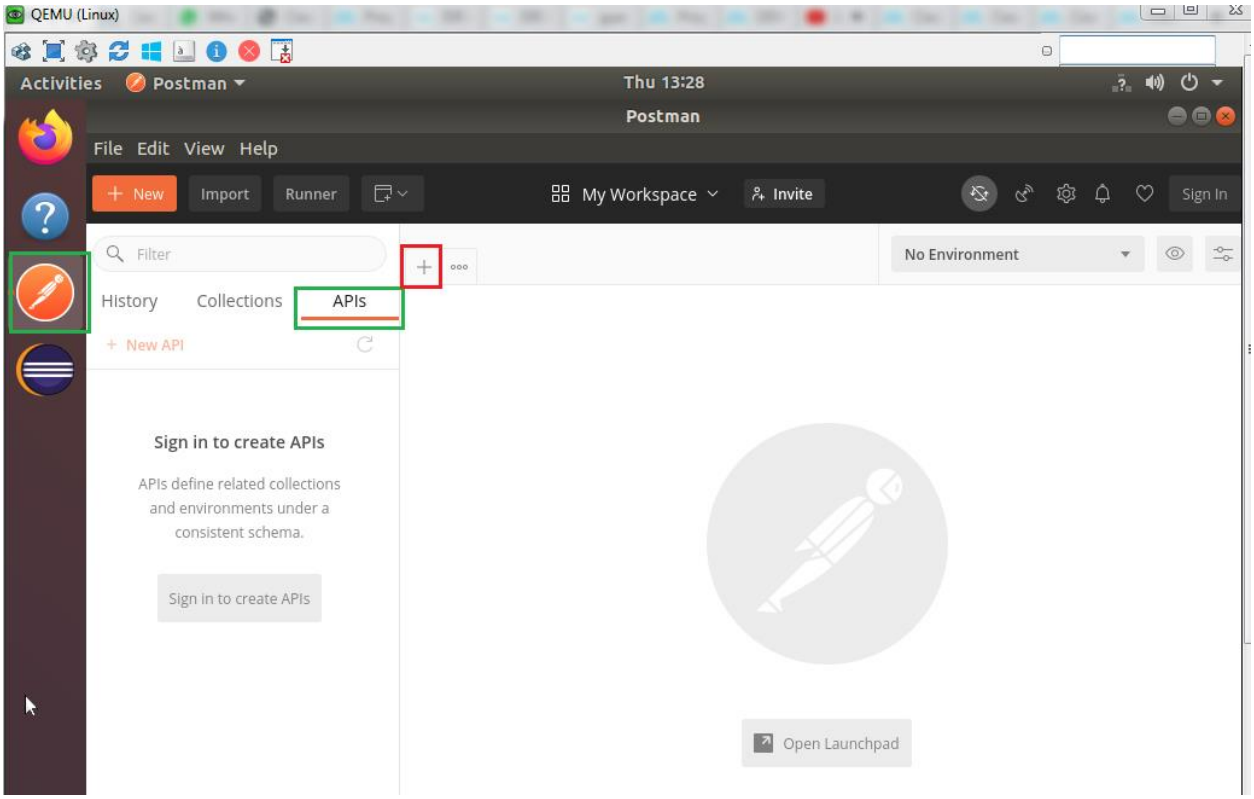
Solution

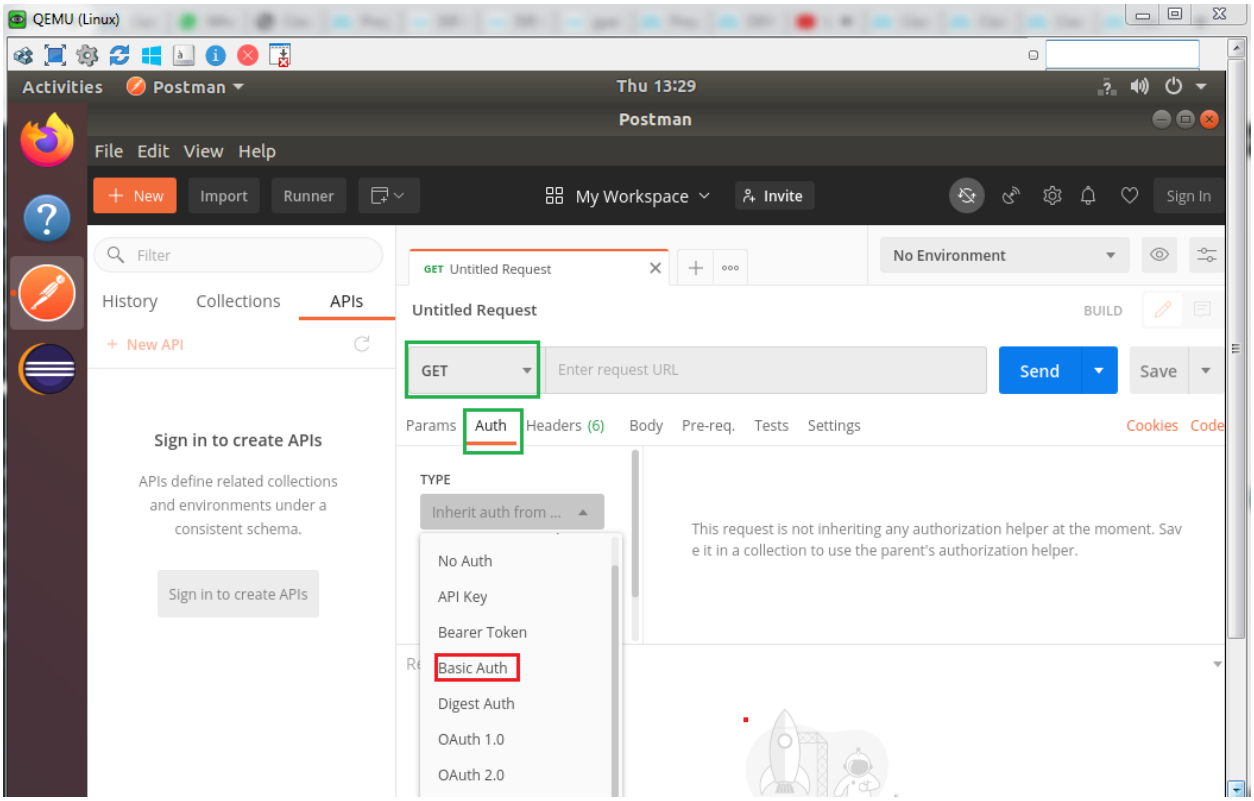
On host31

Steps:

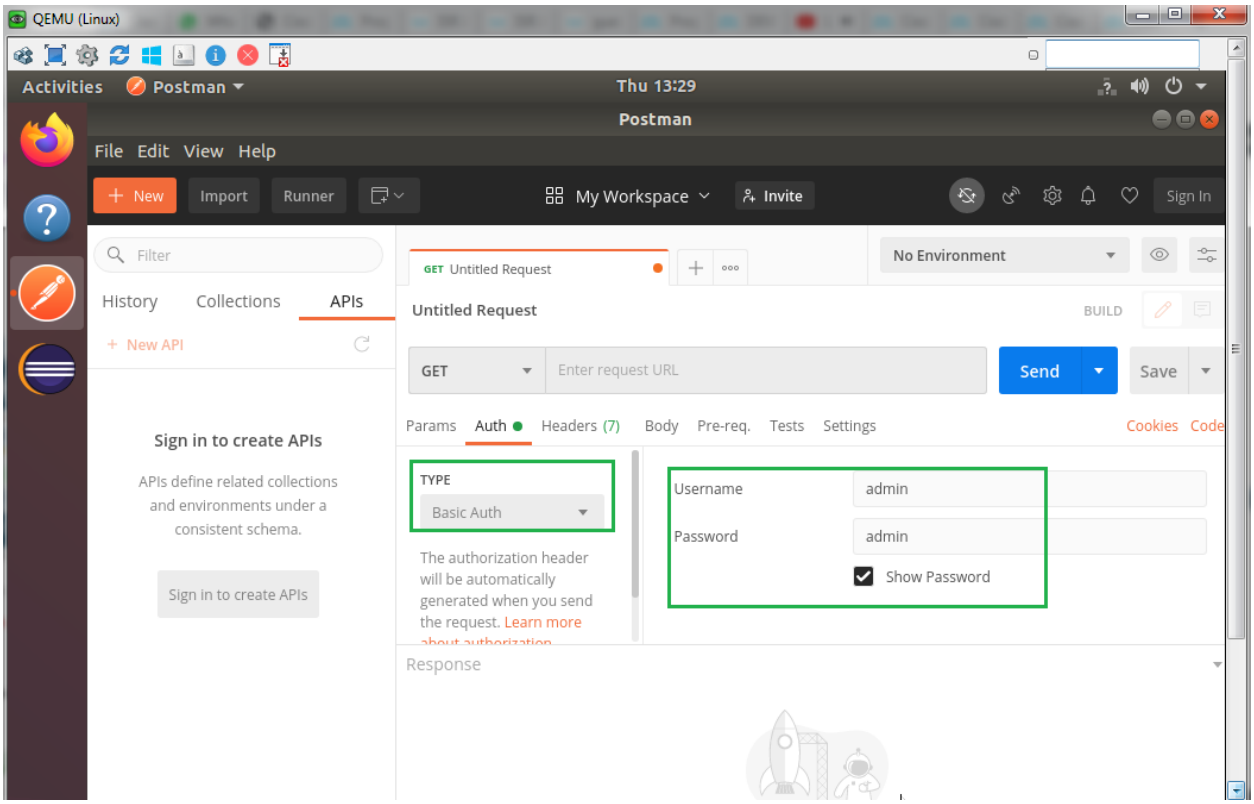
1. Open the host31 console (VNC)
2. Open Postman
3. Select API
4. Click on "+ " Symbol
5. Select the Method as "GET"
6. Select the Auth as "Basic Auth"
7. Input the Username/Password: admin/admin
8. Input the executable link: https://10.3.11.1:443/restconf/data/Cisco-IOS-XE-native:native/line/vty
9. Input the header details as shown in screenshots
10. Under Setttings: Switch "Off" SSL
11. Click "Send"
12. Click on the button as shown in the screenshot to copy the generated response and click on "+ " Symbol
13. Select the Method as "Patch"
14. Select the Auth as "Basic Auth"
15. Input the Username/Password: admin/admin
16. Input the executable link: https://10.3.11.1:443/restconf/data/Cisco-IOS-XE-native:native/line/vty
17. Input the header details as shown in screenshots
18. Under Setttings: Switch "Off" SSL
19. Paste the copied text in the body (raw --json)
20. Replace the "none" to "all" under line vty (pasted texts)
21. Click "Send"
22. All Done

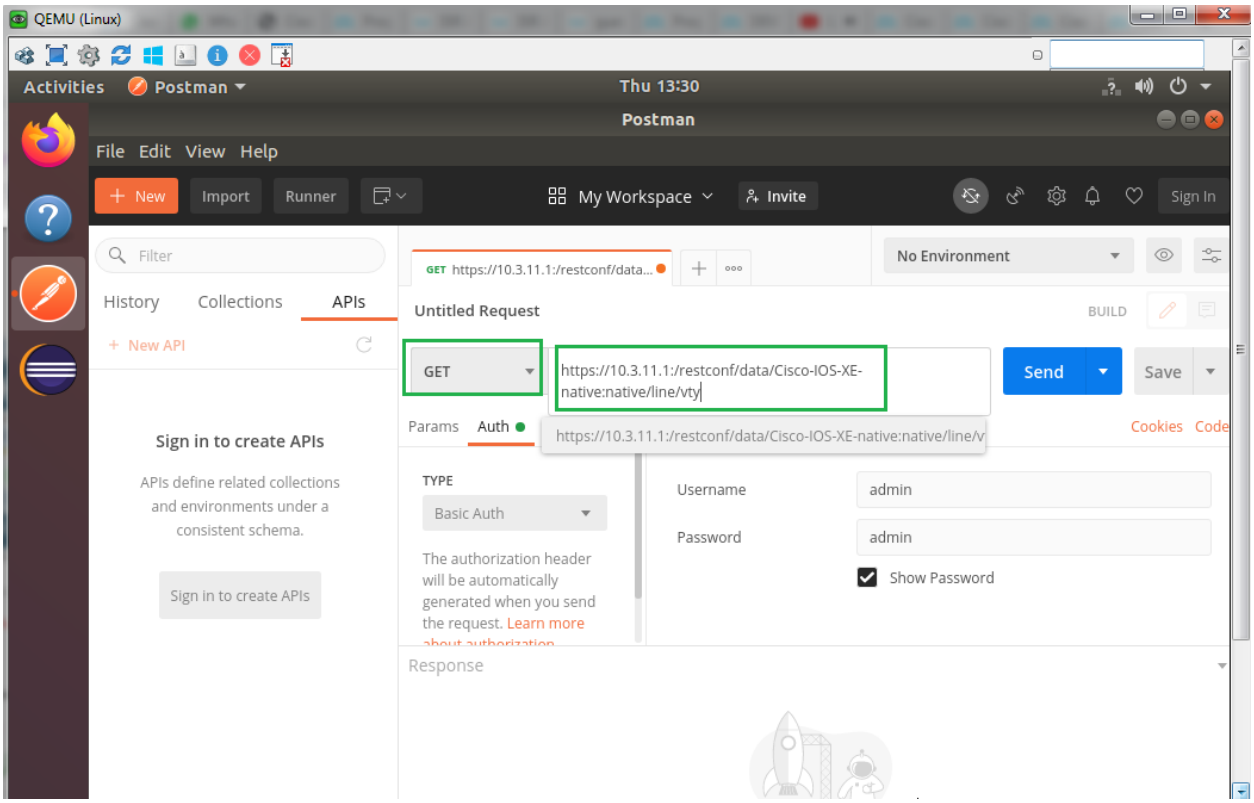
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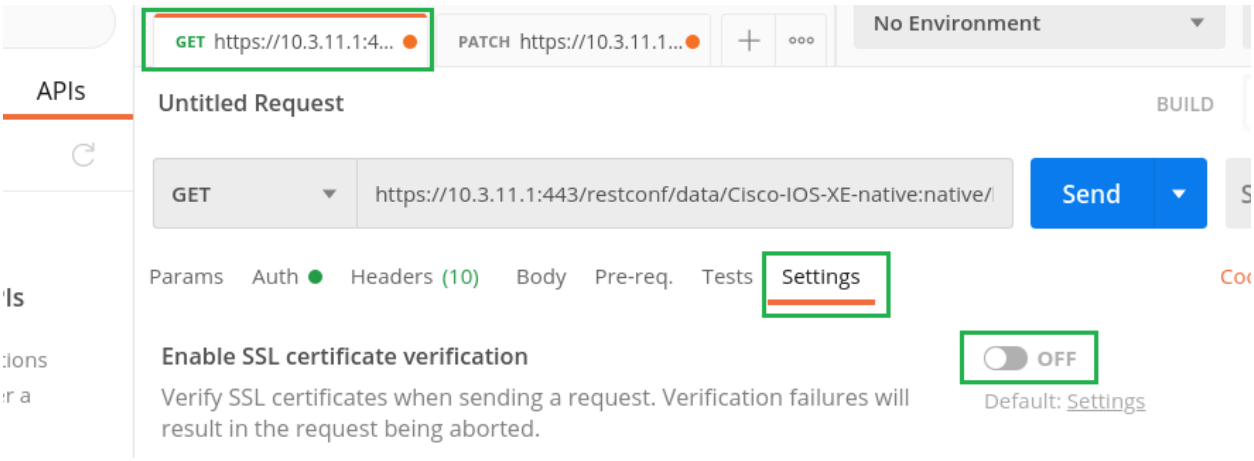


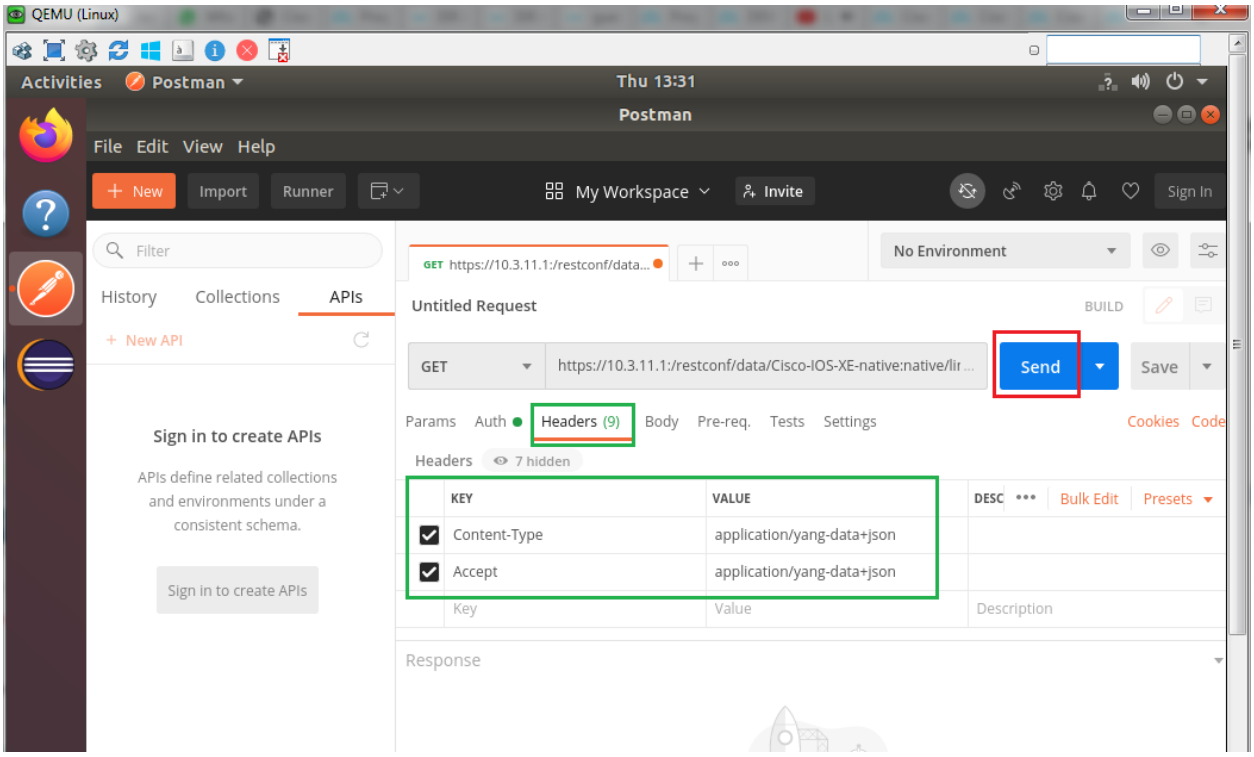
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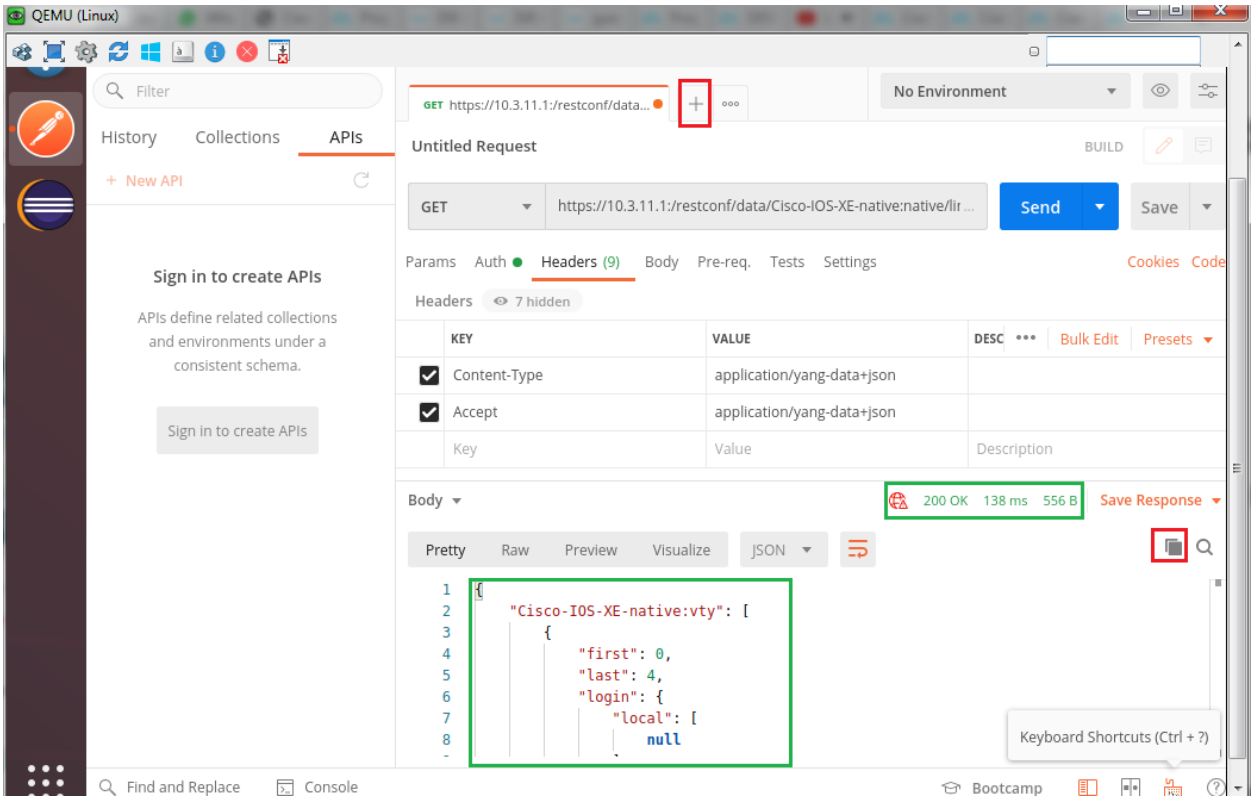


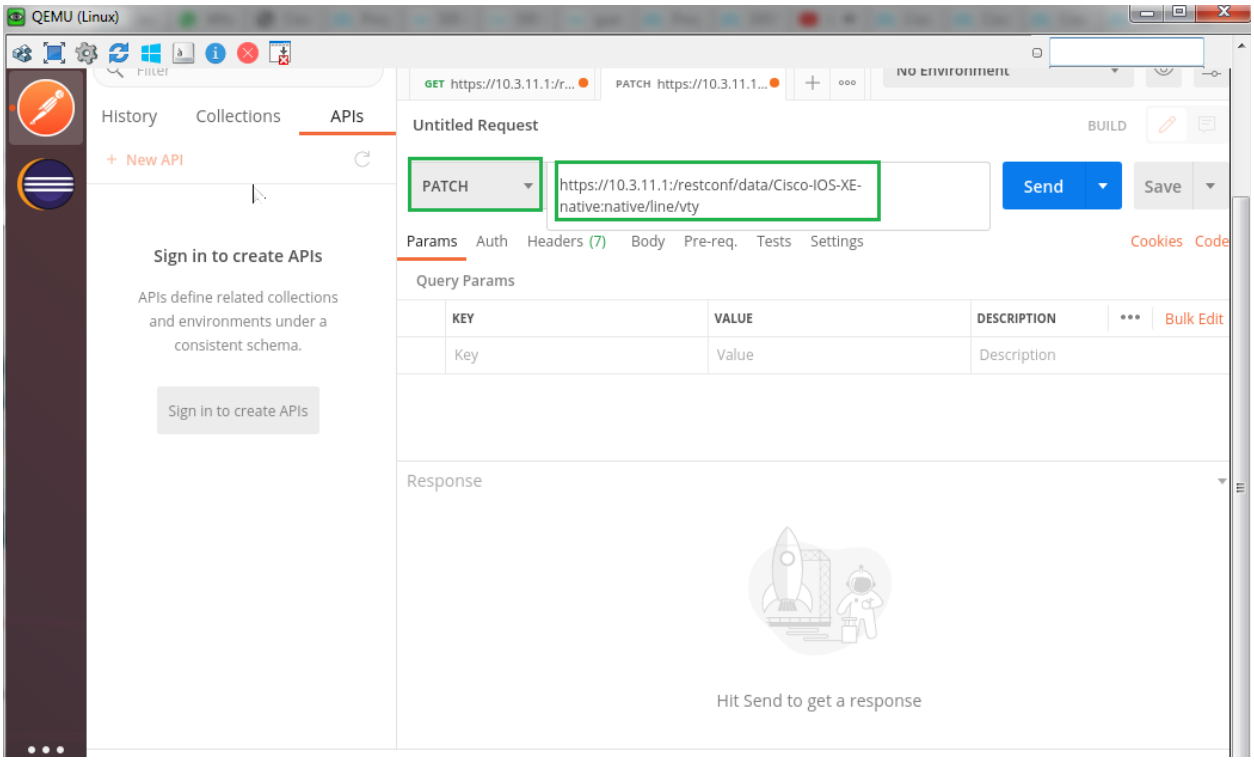
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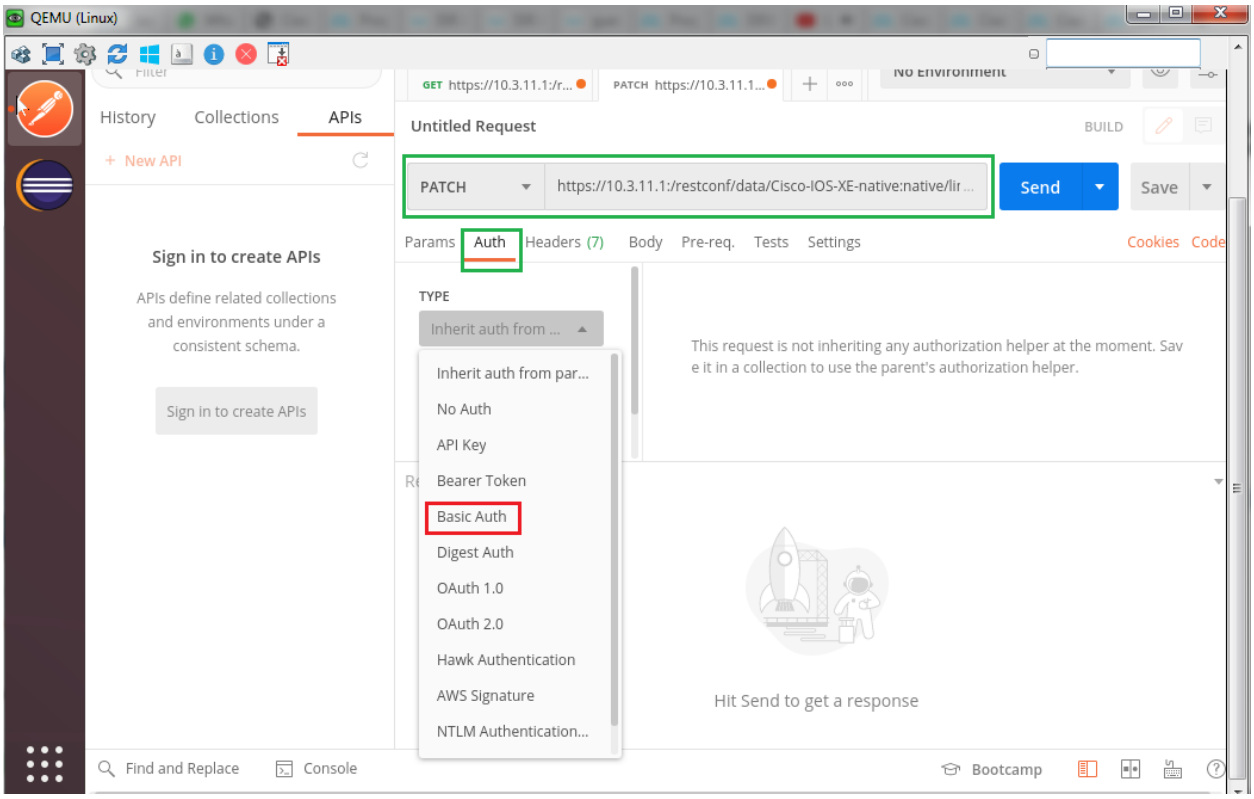


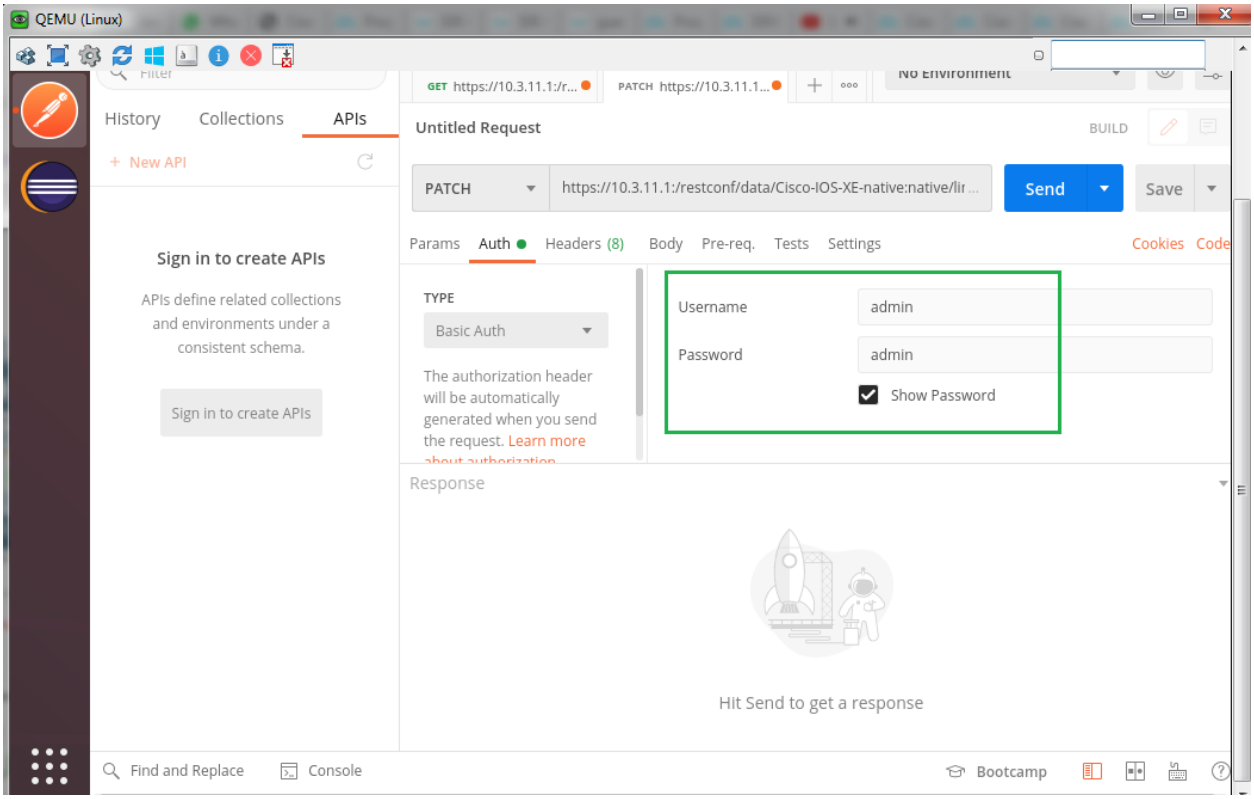
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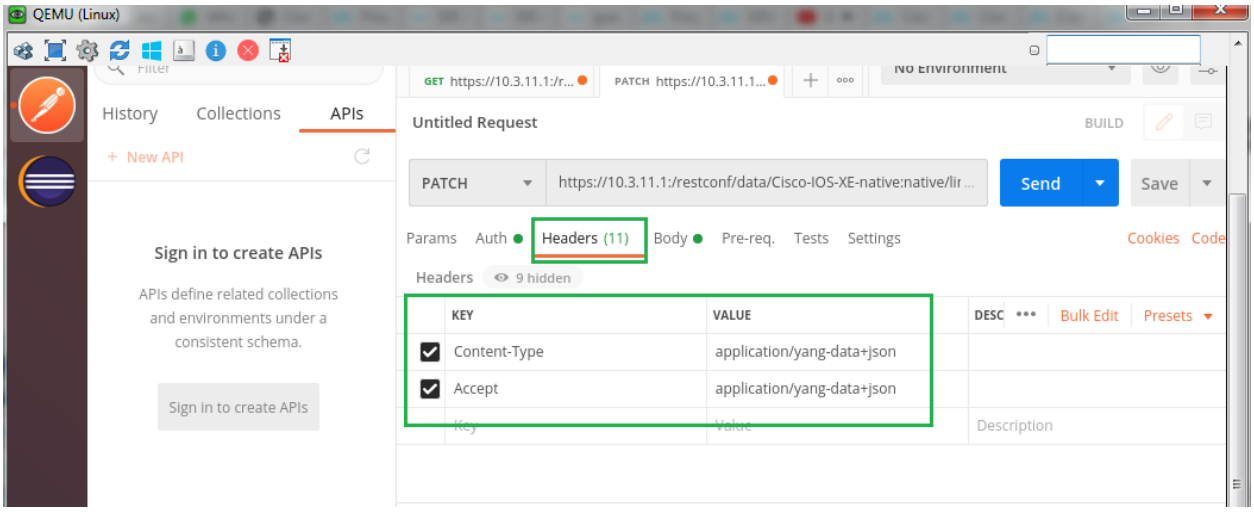


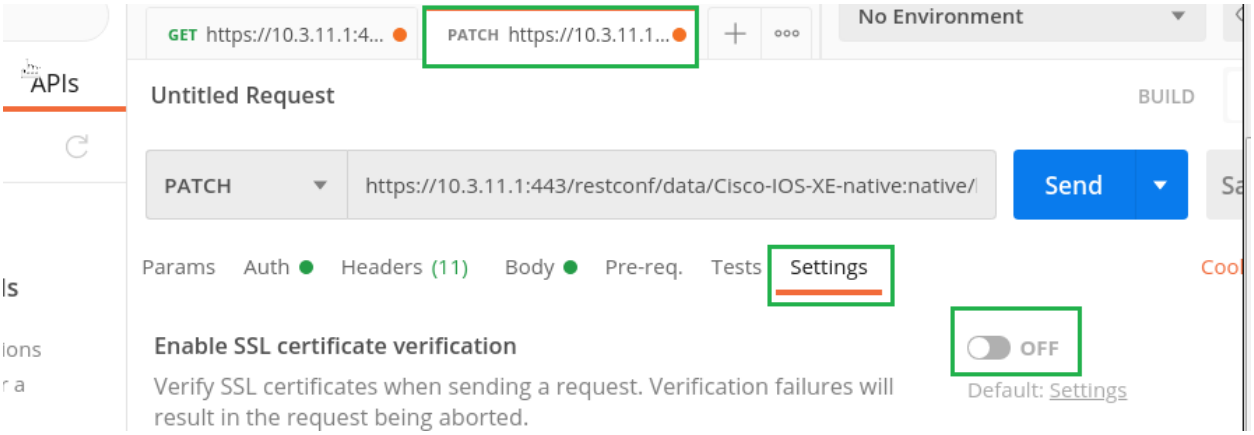
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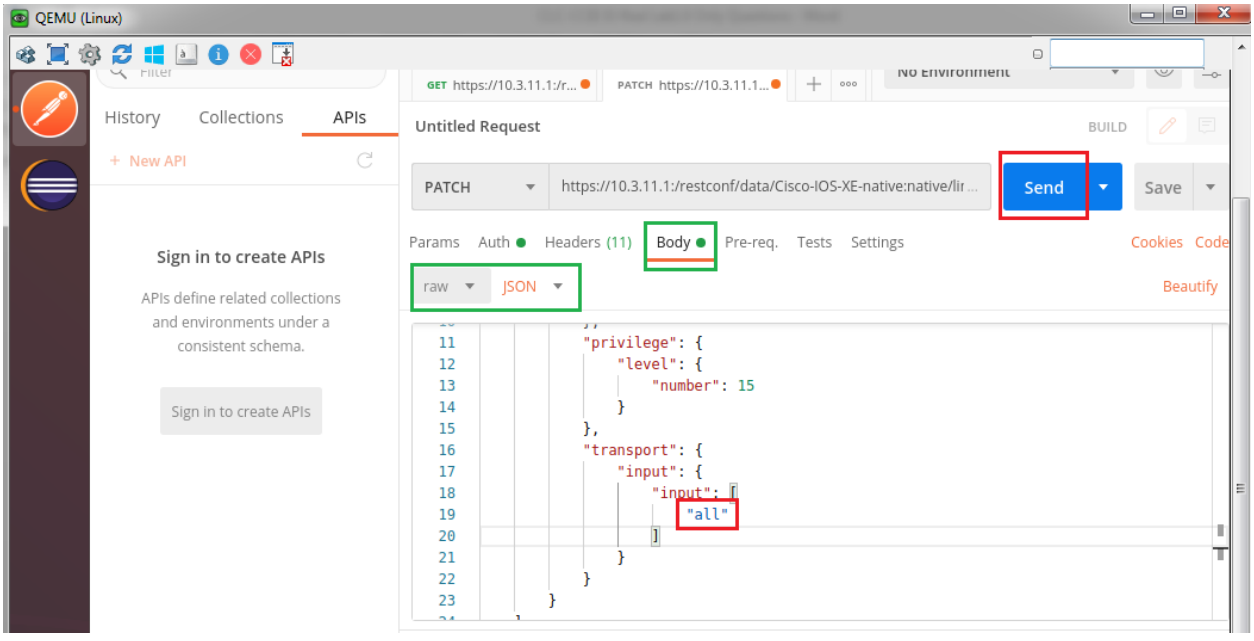




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SECTION 3.2: Using Guest Shell and Python on r30 (3 Points)

On r30, enable guestshell and create a Python script named "ribdump.py" in the guestshell according to the following requirements:

1. If an additional IP network is necessary to start guestshell, you are allowed to use addresses from the range 192.168.255.0/24. This range must not be advertised in any routing protocol.
2. The Python script must be saved under the name ribdump.py in the home directory of the guestshell user.
3. The purpose of the script is to display the complete contents of all routing tables in no-default VRFs created on the router.
4. The script must execute the **show ip route vrf…..** or **show ipv6 vrf…..** command for every non-default VRF created on the router depending on what address families are enabled in that VRF
5. The script must determine the list of created VRFs and enabled address families dynamically every time it is run using, for example, **show vrf brief | include ipv**
6. The script must not attempt to display the VRF routing table for an address family that is not in the VRF
7. It must be possible to run script using the **guestshell run python ribdump.py** command from privileged EXEC mode

Solution

On host31

Steps:

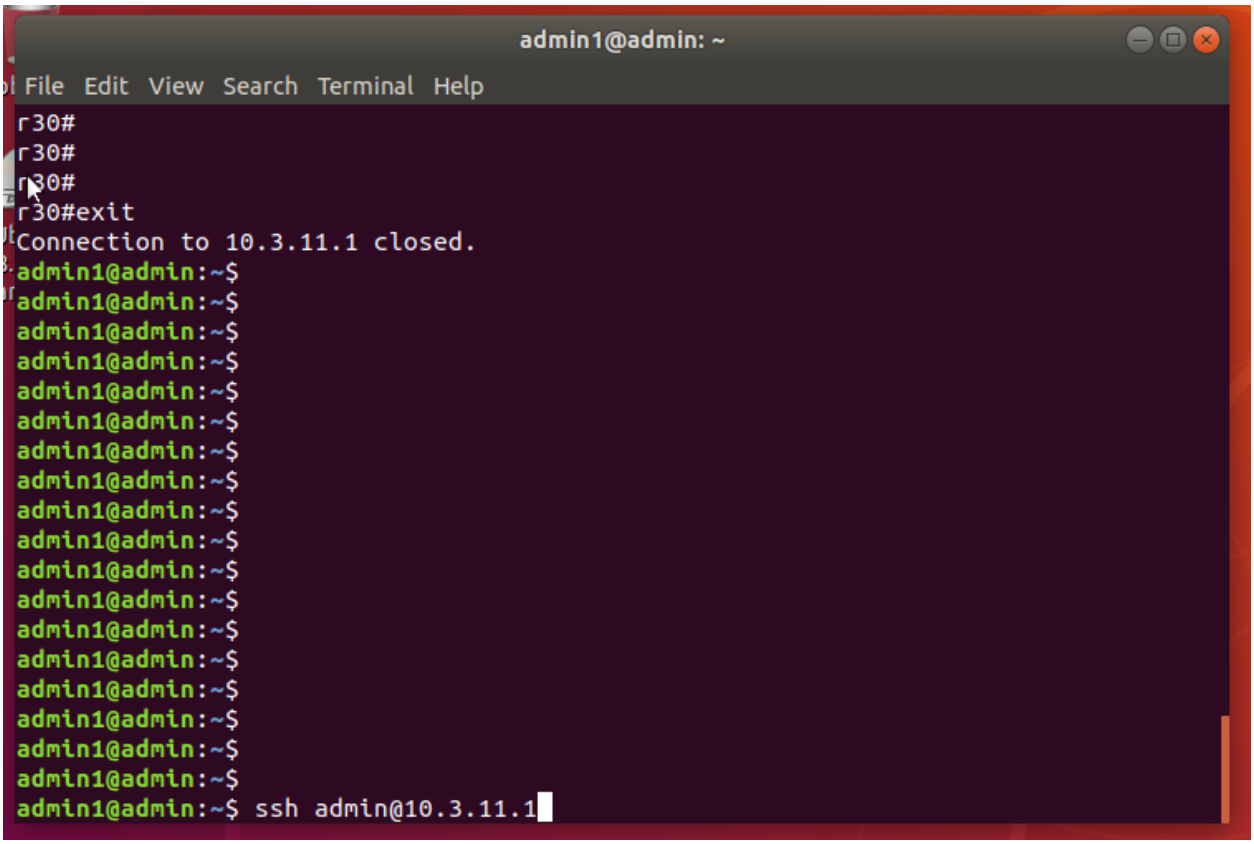
1. Open the host31 console (VNC)
2. Open Terminal
3. ssh admin@10.3.11.1
4. Enter the password: admin
5. Check the router configuration with "show run"
6. **Important steps to know**
   - If virtualportgroup is not created than create it as below
     - Interface vitualportgroup 0
   - If virtualportgroup is pre-configured or you have manually configured it regardless, next step
     - Ip address 192.168.255.1 255.255.255.0
     - Ip nat inside (optional)
7. Enable guestshell with command
   - guestshell enable
8. If you see any error on step 4, than use below commads (optional)
   - Configure terminal
   - lox

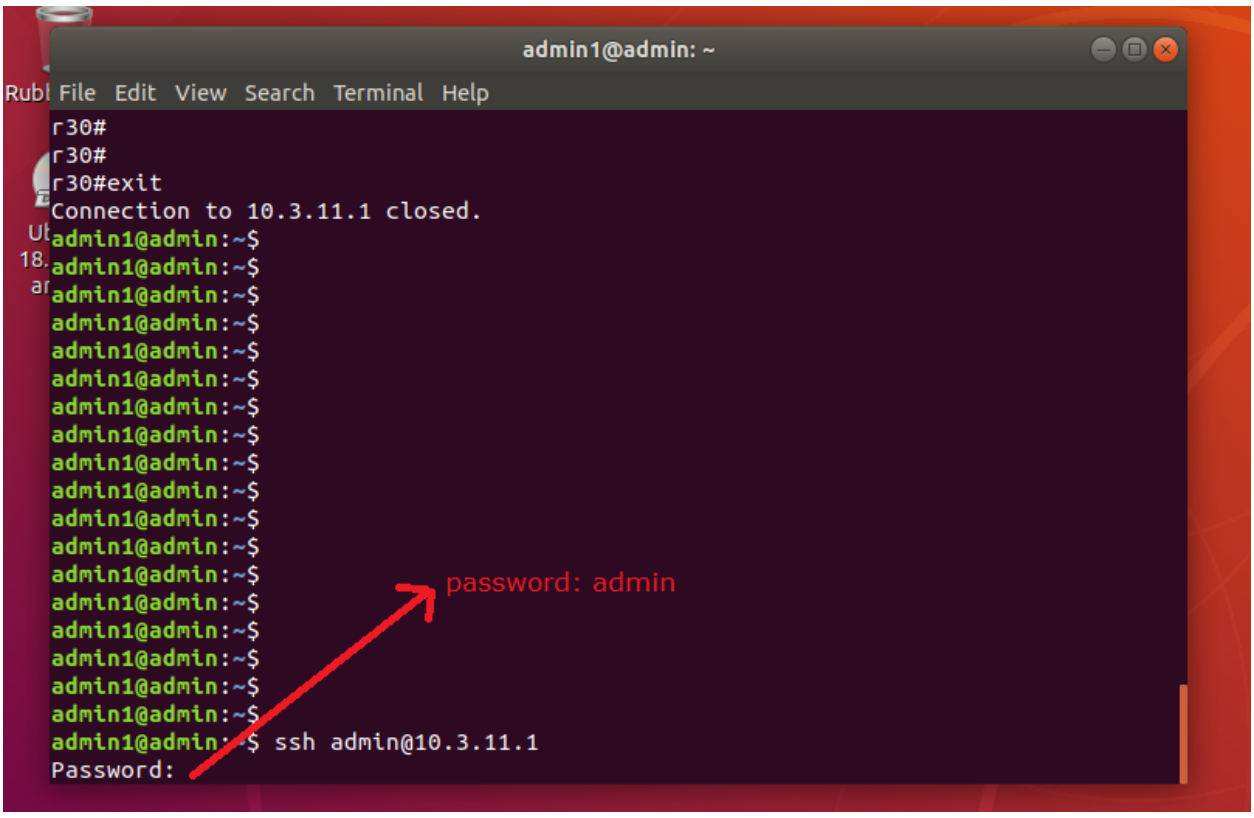
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- app-hosting appid guestshell
- app-vnic management guest-interface 0
- End
- Now try again to enable guestshell

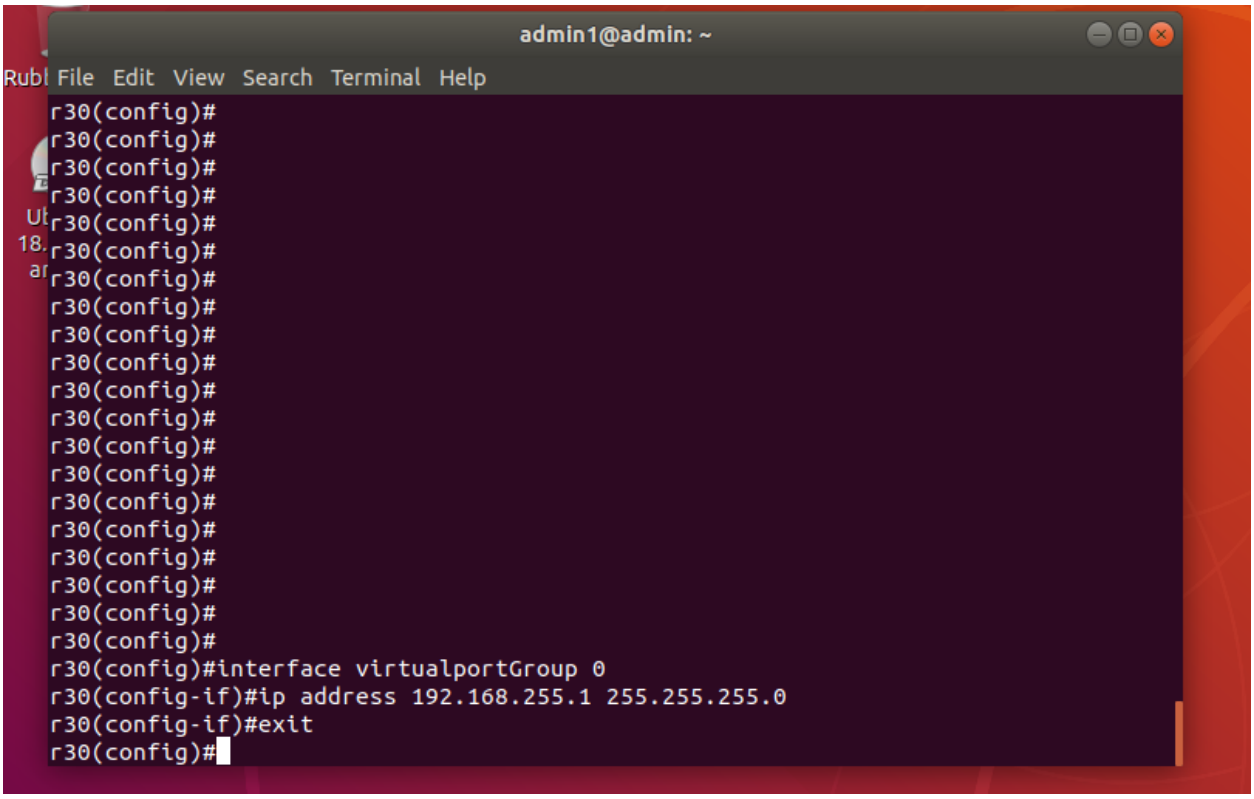
9. To enter guestshell mode use command
    - guestshell
10. To make python script use command
    - sudo vi ribdump.py
11. Now the python file is created and you have to type/make the python script
12. Presss “i” to enter insert mode
13. Open a text editor and type the script (optional)
14. Create the script in the text editor (optional)
15. Copy and paste the script from the text editor in the terminal (optional)
16. You can directly write the script in the terminal and avoid using text editor. Its completely your choice
17. To save and exit the vi editor use below command
    - Press esc
    - Than type :wq
    - Press enter
18. Now run the script form exec privilege mode
    - R30#guestshell run python ribdump.py
19. If you see the proper output as shown in screenshots than the task is completed
20. All Done

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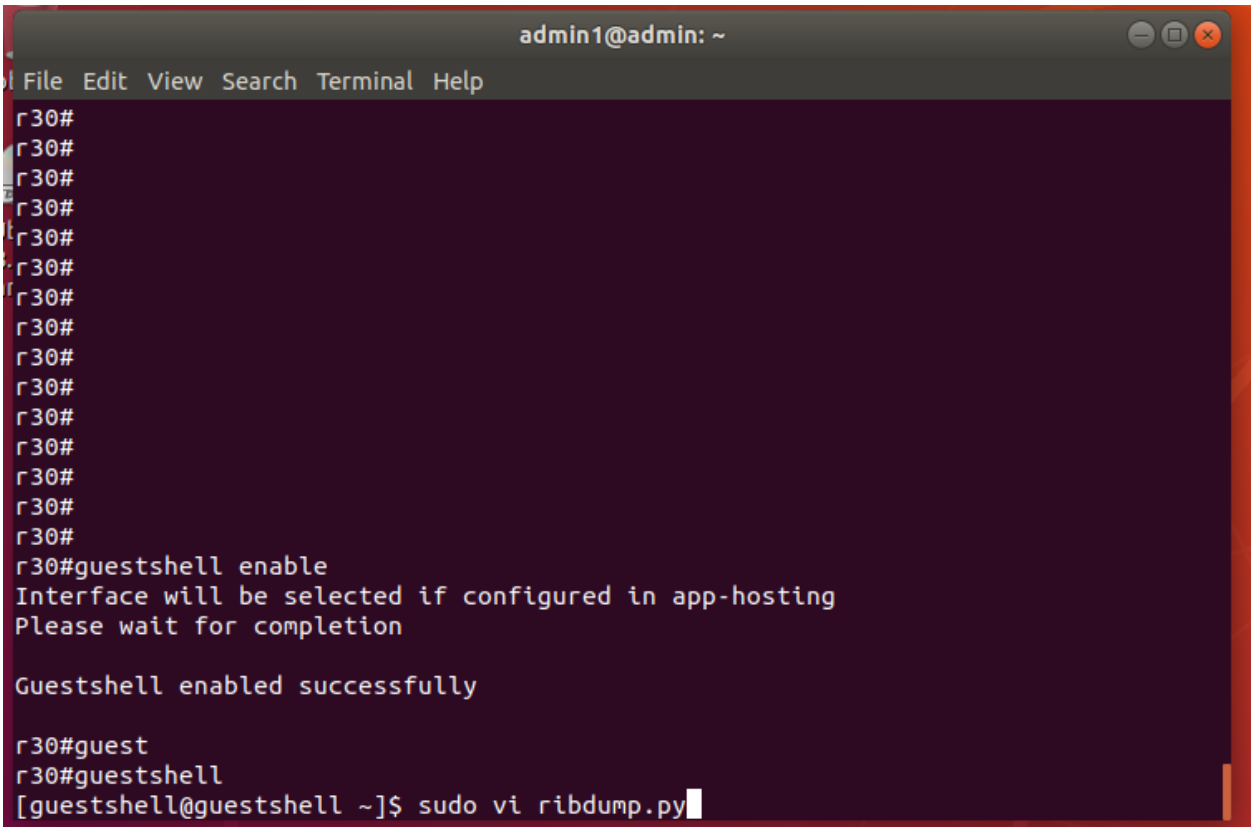


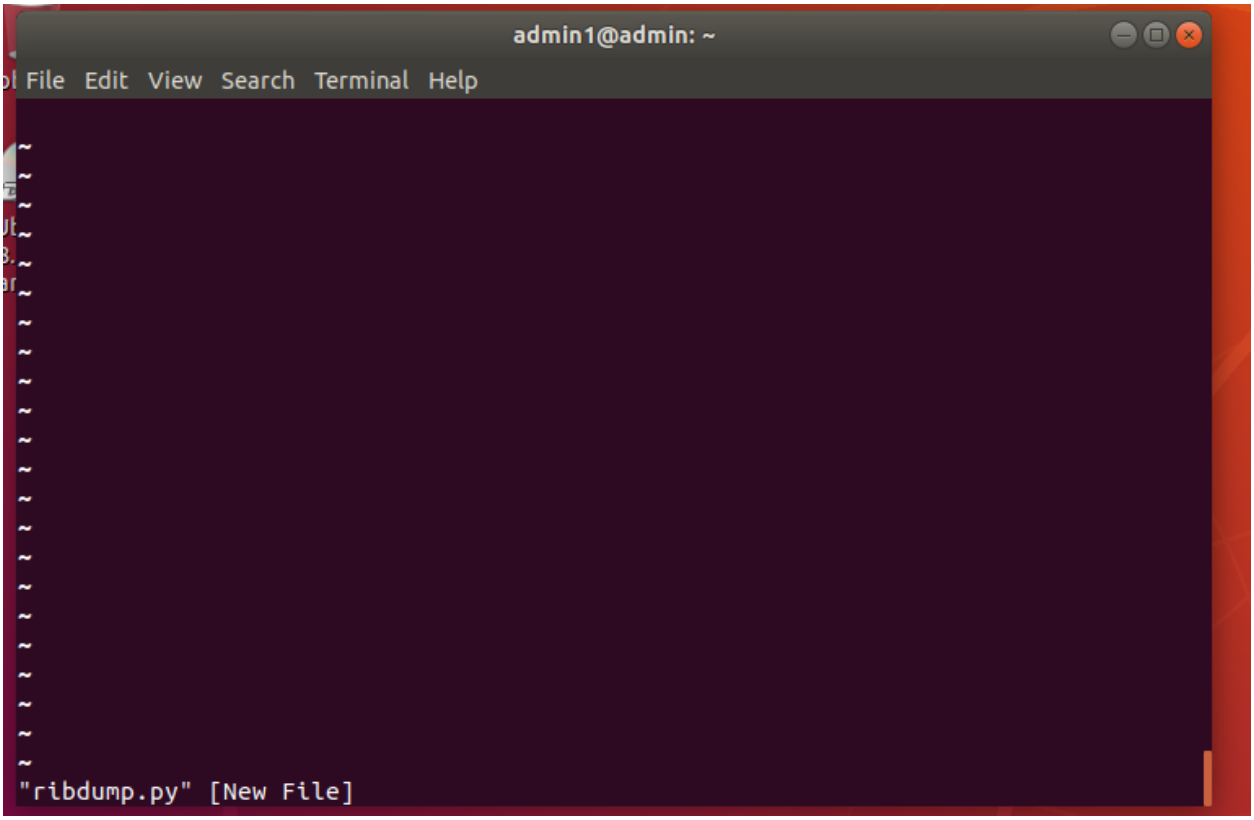


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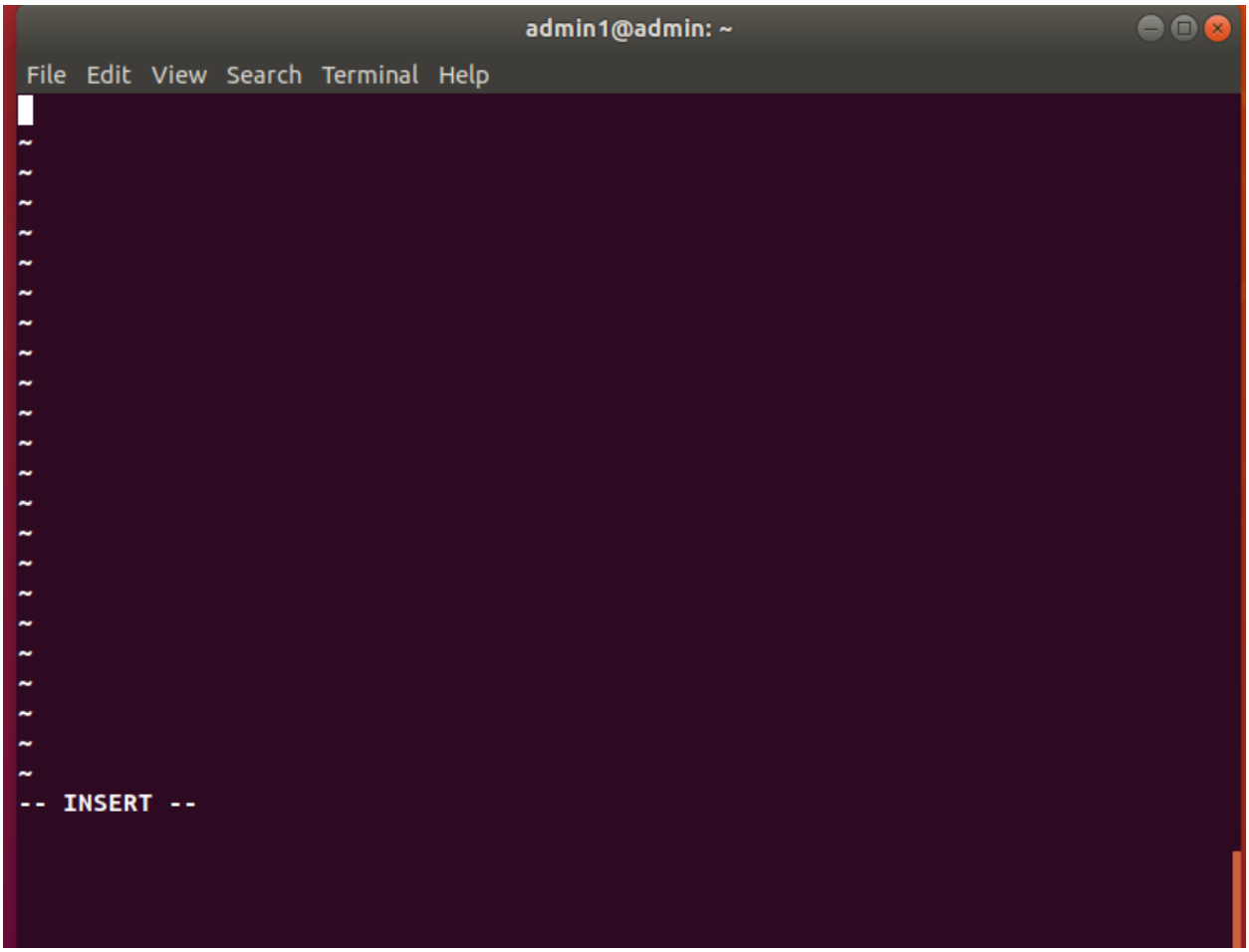

admin1@admin: ~
File Edit View Search Terminal Help
r30(config)#
r30(config)#
r30(config)#

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```
import cli
import os
from collections import defaultdict

response = cli.execute("show vrf brief | include ipv")
print response

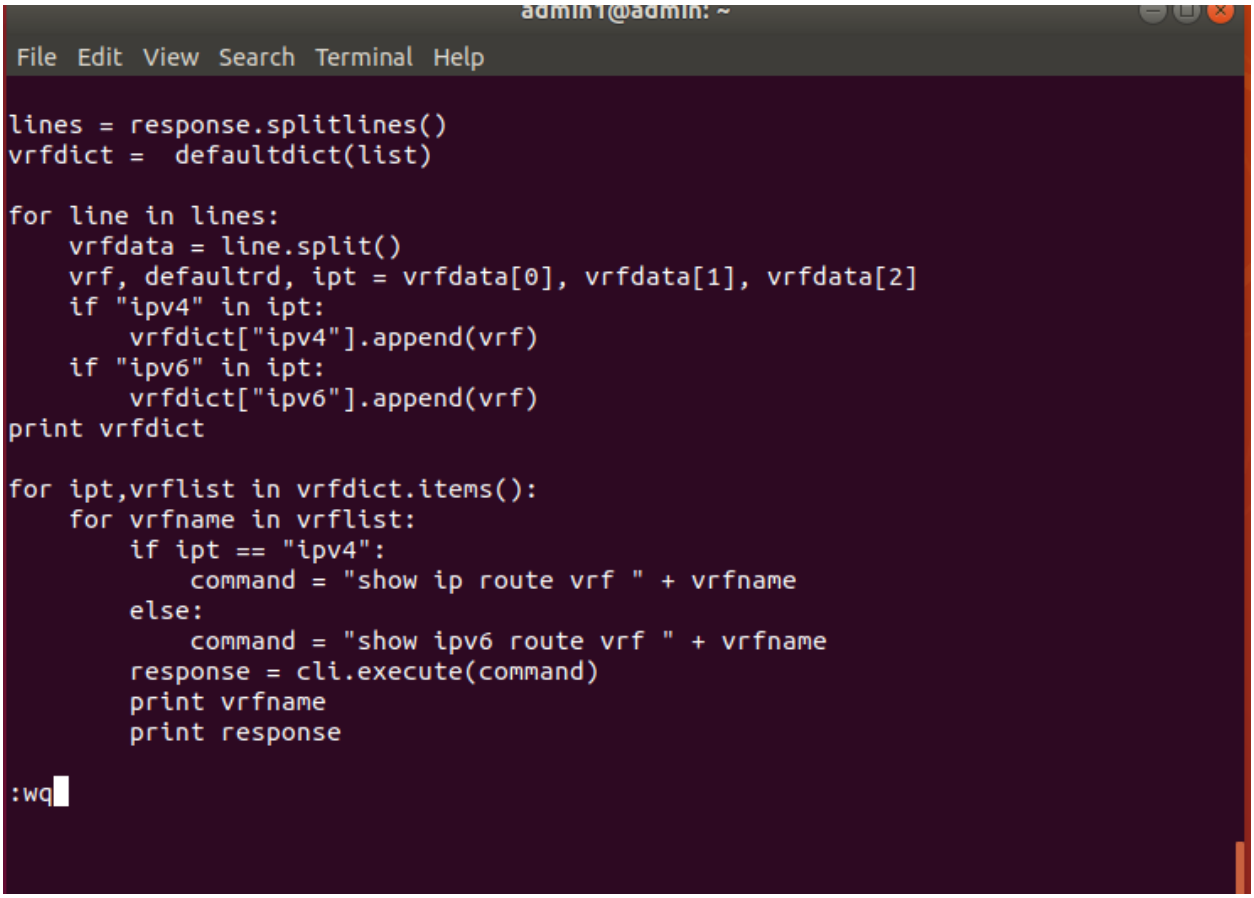
lines = response.splitlines()
vrfdict =  defaultdict(list)

for line in lines:
   vrfdata = line.split()
   vrf, defaultrd, ipt = vrfdata[0], vrfdata[1], vrfdata[2]
   if "ipv4" in ipt:
      vrfdict["ipv4"].append(vrf)
   if "ipv6" in ipt:
      vrfdict["ipv6"].append(vrf)
print vrfdict

for ipt,vrflist in vrfdict.items():
```

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```
for vrfname in vrflist:
    if ipt == "ipv4":
        command = "show ip route vrf " + vrfname
    else:
        command = "show ipv6 route vrf " + vrfname
    response = cli.execute(command)
    print vrfname
    print response
```



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admin1@admin: ~

File Edit View Search Terminal Help

```
lines = response.splitlines()
vrfdict =  defaultdict(list)

for line in lines:
    vrfdata = line.split()
    vrf, defaultrd, ipt = vrfdata[0], vrfdata[1], vrfdata[2]
    if "ipv4" in ipt:
        vrfdict["ipv4"].append(vrf)
    if "ipv6" in ipt:
        vrfdict["ipv6"].append(vrf)
print vrfdict

for ipt,vrflist in vrfdict.items():
    for vrfname in vrflist:
        if ipt == "ipv4":
            command = "show ip route vrf " + vrfname
        else:
            command = "show ipv6 route vrf " + vrfname
        response = cli.execute(command)
        print vrfname
        print response

"ribdump.py" [New] 29L, 725C written
[guestshell@guestshell ~]$
```

File Edit View Search Terminal Help

```
r30#guestshell run python ribdump.py
  WAN                           65003:30             ipv4        Gi1.100
  giosk                         65530:1              ipv4,ipv6   Gi1.414
defaultdict(<type 'list'>, {'ipv4': ['WAN', 'giosk'], 'ipv6': ['giosk']})
WAN
Routing Table: WAN
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
Gateway of last resort is not set
      100.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C        100.4.30.0/30 is directly connected, GigabitEthernet1.100
L        100.4.30.2/32 is directly connected, GigabitEthernet1.100
C        100.5.30.0/30 is directly connected, GigabitEthernet1.200
L        100.5.30.2/32 is directly connected, GigabitEthernet1.200
giosk
Routing Table: giosk
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
```

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SECTION 3.3: Automated Configuration Backup Script (2 Points)

This item consist of multiple questions. You may need to scroll down to be able to see all questions.

1. You are tasked with writing a Python script to back-up the configuration of a number of IOS-XE devices through RESTCONF, and store the configuration in the text files. The starting section of this script has already written and contains the following line:

```
#1/usr/bin/python3

import requests

Credentials = [  (“192.168.1.1”, “admin”, “s3cr3t”,),
                 (“192.168.1.2”, “netadmin:, “0th3rs3cr3t”) ]

Headers = { “Content-Type” :  “application/yang-data+json”,
            “Accept” :  “application/yang-data+json” }
```

2. This script needs to be completed by dragging the individual command lines below into their correct order to allow the script to correctly accomplish its purpose. Indicate the ends of the **for** block and of the **while** block by properly placing the “..**End of for**” and “..**End of while**” symbols.
3. Drag the lines into their correct order to complete the script as required. Make sure to also properly place the “..**End of for**” and “..**End of with**” symbols to indicate the end of the respective blocks in code

..End of with	
for IP.Login.Password in Credentials:	
File.write(Response.text)	
URL=I”https://[IP]:443/restconf/data/Cisco-IOS-XE-native:native	
with open(I[IP],conf’, ‘W’) as File	
..End of for	
Response = requests.get(URL auth=(Login.Password), headers=Headers,verify=False)	

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1. There are plans to extent the script to display a list of known IOS-XE devices by their IP addresses and allow the administration to select which devices to backup. Aside from other necessary changes in the script, which of the following storage options for the credentials would allow for the most straight-forward implementation?

a. Credentials = "192.168.1.1,admin,s3cr3t," \
192.168.1.2,netadmin,0th3rs3crt"
b. Credentials = ["192.168.1.1,admin,s3cr3t",
"192.168.1.2,netadmin,0th3rs3crt"]
c. Credentials = {"192.168.1.1" : ("admin", "s3cr3t"),
"192.168.1.2", : ("netadmin", "0th3rs3crt")}
d. Credentials = [("192.168.1.1", "admin", "s3cr3t"),
("192.168.1.2", "netadmin", "0th3rs3crt")]

Solution

Part 1

	for IP.Login.Password in Credentials:
	URL=I"https://[IP]:443/restconf/data/Cisco-IOS-XE-native:native
	Response = requests.get(URL auth=(Login.Password), headers=Headers,verify=False)
	with open(I[IP],conf', 'W') as File
	File.write(Response.text)
	..End of with
	..End of for

Part 2

Answer: d

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SECTION 3.4: The End

You reached the end of the exam!

Click the "End Exam Section" button to finish your exam.

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**********The End**********

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